

# Prime-Zeta Divisor Tomography: Local Zero Processes, Frozen-Edge Rigidity, and Derivative Caustics

Exact densities, phase recovery, small-target monodromy,  
and inverse prime spectroscopy

KEIO

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**Scope.** This paper concerns the absolute-convergence half-plane of prime-supported Dirichlet series. General almost-periodic projection and Jessen theories are used as foundations. The principal results are smooth and marked divisor laws, complete local point-process limits, deterministic hard-core criteria, an exact phase-fidelity frontier, quantized small-target escape with divisor monodromy, inverse support and phase recovery, an exact large-deviation profile at the outer edge, a universal rooted edge divisor, and the complete internal-band geometry of the derivative tower. Component masses and barycentres recover their boundary primes, while moving high derivatives produce exponentially accurate zero lattices. The exact projection edge is not claimed as a new general phenomenon. Numerical experiments are kept separate from the unconditional statements, and uncertified decimal thresholds are not used as hypotheses of the principal theorems.

# Abstract

Let

$$F_c(s) = \sum_p c_p p^{-s}, \quad \Re s > 1,$$

under the coefficient hypotheses stated below, and write  $F_\omega$  when  $|\omega_p| = 1$ . We identify vertical translation with a uniquely ergodic prime-phase hull. Its one-point Kac–Rice density  $G(\sigma, a)$  is smooth, radial, and phase blind. On every compact horizontal strip, multiplicity, distinct-point, and simple-point counts have the same exact linear main term; a marked law records the derivative at an  $a$ -point. For the untwisted prime zeta function the count extends to the full half-plane  $\Re s > 1$ . This proves the literal distinct-point height law of Belovas, Čepaitytė, and Sabaliauskas and strengthens their linear-counting conjecture to an exact asymptotic. The projection edge itself is used from classical almost-periodic theory and is not claimed as a new general phenomenon.

At every fixed derivative order, the empirical law of the vertically translated  $a$ -point divisor of any deterministic twist converges to the same stationary Haar divisor process. All factorial  $k$ -point correlations are therefore given by joint Kac–Rice densities, and compact-window zero bounds turn them into finite formulae for the Laplace functional, count laws, void probabilities, and Palm nearest-neighbour distributions. The self-inclusive smoothed two-point function is Bohr almost periodic; its factorial version is an almost-periodic function plus a  $C_0$  diagonal correction. An explicit variational lower bound for  $|F'_\omega(\rho)|$  produces a deterministic simple-zero region and a twist-independent planar hard core. In the complementary nondegenerate regime,

$$K_2^{(m,a)}(s, s+h) = \frac{|h|^2}{16} \mathcal{C}_m(\Re s, a) + O(|h|^3),$$

and the full confluent hierarchy has the corresponding Vandermonde repulsion.

Anchored edge data restore precisely the information erased by stationary averaging. If  $\rho = \sigma + it$  is a nonzero  $a$ -point and  $\varepsilon = A_c(\sigma) - |a|$ , then the exact phase defect is

$$\sum_p |c_p| p^{-\sigma} \left| 1 - \frac{c_p}{|c_p|} \frac{|a|}{a} p^{-it} \right|^2 = 2\varepsilon.$$

Hence a nonzero-target edge sequence recovers all coefficient phases, while a zero-edge sequence recovers their relative phases, with common rotation as the only ambiguity. Together with sparse samples of the right-edge curve, this is inverse complete for the coefficient system. More strongly, after recentering at any zero whose real part approaches the universal zero edge  $S = \sigma_e$ , the normalized analytic germ converges, uniformly over all twists, to

$$H_e(z) = 2^{1-(S+z)} - \mathcal{P}(S+z).$$

Thus the rooted divisor and every rooted Palm law collapse to one deterministic profile. Boundary-attaining twists and equal edge germs are classified exactly by common rotation and the vertical Kronecker flow; in shrinking edge layers, distinct zero ordinates separate at least logarithmically in the reciprocal defect. This gives a phase-readable, uniformly discrete edge divisor even though its entire stationary local law is phase blind.

The phase defect also converts the outer-edge large-deviation theorem into an exact density–fidelity frontier. If  $\mathcal{C}(d)$  is the vertical intensity of zeros whose weighted relative-phase distortion is at most  $d$ , then, with  $\beta_e = (\sigma_e - 1)^{-1}$  and  $\ell_d = \log(2/d)$ ,

$$-\log \mathcal{C}(d) = K_e (d/2)^{-\beta_e} \ell_d^{-(1+\beta_e)} \left( 1 + (1 + \beta_e) \frac{\log \ell_d}{\ell_d} + O(\ell_d^{-1}) \right).$$

This is a limiting-intensity theorem, not a first-occurrence estimate. Its finite-prime consequence is stated only as a sufficient reconstruction certificate.

Finally, the quantized  $a \rightarrow 0$  escape law is lifted from measures to divisors. On every compact subband of the hard-core region, the bounded  $a$ -point divisor consists of uniformly separated holomorphic continuations of the zero divisor; a zero-free corridor separates these persistent branches from a far-right logarithmic lattice. A loop around  $a = 0$  fixes every persistent branch and acts by a unit shift on the escaping lattice. The escape mass and reciprocal lattice spacing—equivalently, the physical displacement of one monodromy shift—recover the least support prime, while the bounded mass has an even  $C^\infty$  expansion in  $|a|$ .

The paper also classifies the internal dominance bands of the derivative tower and its moving-order zero lattices. It proves no statement across  $\Re s = 1$  and no new case of the Riemann hypothesis, Montgomery’s pair-correlation conjecture, or a prime-gap conjecture.

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**Keywords.** Prime zeta function; almost periodic Dirichlet series; local zero process; Palm law; Kac–Rice formula; hard core; large deviations; phase recovery; rate–distortion; divisor monodromy; derivative caustic; inverse spectral problem.

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# 1 Introduction and main conclusions

The prime zeta function

$$\mathcal{P}(s) = \sum_{p \in \mathbb{P}} p^{-s}$$

is absolutely convergent in  $\Re s > 1$  and is connected to the Riemann zeta function by Möbius inversion,

$$\mathcal{P}(s) = \sum_{k \geq 1} \frac{\mu(k)}{k} \log \zeta(ks),$$

on compatible branches. Its continuation has a natural boundary at  $\Re s = 0$  and a dense family of singularities generated by the zeros and pole of  $\zeta$ ; see Titchmarsh [20], Fröberg [9], and Landau–Walfisz [12]. The present paper asks a different question: how much of the zero geometry is already forced, and how much arithmetic can be recovered, inside the elementary half-plane  $\Re s > 1$ ?

Belovas, Čepaitytė, and Sabaliauskas [2] numerically studied the rightmost zeros and formulated conjectures concerning their edge, their linear number, and the uniform distribution of normalized heights. The right projection itself is a polygonal property of an almost-periodic Dirichlet series. General criteria are due to Jessen–Tornerhave [10] and, in a direct projection form, to Sepulcre–Vidal [18]. A recent preprint of Camargo [7] independently specializes equivalent-series methods to the prime zeta function and obtains the same projection edge. Accordingly, the new content claimed here is not the mere existence of that boundary, nor the classical existence of a multiplicity-weighted Jessen mean count. It is the explicit phase-blind density, the upgrade to distinct and simple divisors, marked slope laws, quantized escape, endpoint transfer, inverse theorems, and derivative geometry built on the boundary. In particular, we do not claim Conjecture 1 of [2] as a new result; the literal distinct-point forms and sharpenings of its metric Conjectures 2–3 are the relevant application of the new package.

The conclusions may be organized into fourteen layers. The first nine concern the horizontal divisor, edge, derivative, spectral, and arithmetic structures; the last five pass to the complete fixed-level local process, rooted edge rigidity, the exact phase-fidelity frontier, and pointwise small-target surgery.

(I) **Smooth density.** Every unimodular twist has the same smooth  $a$ -point density  $G(\sigma, a)$  in  $\Re s > 1$ . Its zero density is positive exactly before the polygon edge and is infinitely flat at that edge. We determine the flatness by an explicit Cramér large-deviation law whose power exponent is  $(\sigma_e - 1)^{-1}$  and whose logarithmic correction is forced by the prime number theorem.

(II) **Simple and marked divisors.** For  $1 < \gamma < \sigma_e$ ,

$$M_{\omega, \gamma}(T) = c_\gamma T + o(T), \quad c_\gamma = \int_\gamma^{\sigma_e} G(\sigma, 0) d\sigma > 0.$$

Multiplicity, distinct, and simple counts have this same main term. A marked Kac–Rice law records the derivative at each  $a$ -point and proves a quartic law for near-critical slopes. The normalized heights are uniformly distributed, giving the literal distinct-point versions of the 2025 metric conjectures.

(III) **The endpoint.** For the untwisted  $\mathcal{P}$ , the exponential envelope  $e^{\mathcal{P}} = \zeta H$  controls the layer  $1 < \Re \rho < 1 + \delta$ . It yields

$$\#\{\mathcal{P}(\rho) = a : \Re \rho > 1, 0 < \Im \rho < T\} = C_a T + o(T)$$

for every fixed  $a$  under each of the three counting conventions. Simple points in every occupied compact band recur with bounded vertical gaps. In particular, the weak endpoint prediction  $M_1(T) = O(T)$  is sharpened to  $M_1(T) \sim c_1 T$ .

- (IV) **Quantized small-target escape.** As  $a \rightarrow 0$  through nonzero values, exactly  $\log 2/(2\pi)$  units of horizontal  $a$ -point mass escape to infinity. The escaping points form a simple asymptotic lattice. For a general infinite prime support the jump is  $\log p_0/(2\pi)$  and therefore recovers its least prime  $p_0$ .
- (V) **Tomography and conservation.** All nonzero  $a$ -point Riesz measures determine the coefficient moduli of a bounded prime-supported Dirichlet series. A single vertical limiting value law already recovers the full prime sequence, and the nonzero  $a$ -point right-edge curve is an even smaller complete datum. Integrating over the target plane gives an exact prime-spectral mass identity and, for character supports, one scalar recovers the modulus radical. In the derivative tower, one bounded zero component recovers the two prime labels on its boundary from its mass and barycentre.
- (VI) **Derivative caustics.** The zero edge of  $(-1)^m \mathcal{P}^{(m)}$  is the pinch point at which the right-hand value gap of  $(-1)^{m-1} \mathcal{P}^{(m-1)}$  changes topology. The edge sequence is a lossless encoding of the primes. More generally every prime owns at most one dominance band, every prime is born at a finite derivative order, and the bands are disjoint, ordered, and persist through the tower under the explicit shift in (6.23).
- (VII) **Crystalline moving-order limit.** After the rescaling  $\sigma \mapsto \sigma/m$ , the horizontal zero-density measures converge vaguely to a weighted prime-indexed picket fence. Before vertical averaging, the actual zeros in each fixed crossover window form an exponentially accurate simple lattice. Its horizontal velocity, vertical spacing, and phase offset recover the two adjacent primes and their twist ratio.
- (VIII) **Equivalent spectral forms.** The same density constant is a Weyl coefficient, a heat coefficient, an Abelian spectral-zeta residue, a Dixmier trace, and the exponential type of a symmetric canonical product. These are Tauberian repackagings, not additional zero-distribution hypotheses.
- (IX) **Arithmetic transfer.** Reciprocal prime series have matching pole and convergence edges. Root-of-unity filters split ordered prime factorizations into equidistributed length classes while retaining a completely explicit family of off-axis poles.
- (X) **Complete fixed-level local divisor process.** For every deterministic unimodular twist, vertical translation of the  $a$ -point divisor has a unique stationary local limit, namely the divisor of the Haar prime-phase model. Its factorial moment densities are the joint Kac–Rice functions  $K_k$ , for every  $k \geq 1$ . Uniform compact-window zero bounds make the correlation hierarchy determinate: the Laplace functional, every finite-window count law, void probabilities, and Palm nearest-neighbour laws are finite inclusion–exclusion expressions in the  $K_k$ . Thus the local law is universal in the twist even though an individual anchored configuration is not. The self-inclusive smoothed pair function is Bohr almost periodic; the factorial pair function lies in  $AP + C_0$  and has product mean.
- (XI) **Hard-core transition and diagonal law.** Put

$$m(\sigma) = \sup_{\lambda > 0} \frac{2^{-\sigma} - \sum_{p \geq 3} p^{-\sigma} |1 - \lambda \log(p/2)|}{\lambda}.$$

Whenever  $\mathfrak{m}(\sigma) > 0$ , every zero of every unimodular twist on  $\Re s = \sigma$  is simple and satisfies a twist-independent derivative lower bound; Taylor's theorem gives an explicit positive separation radius. In the nondegenerate complementary regime the two-point density vanishes quadratically on the diagonal, with coefficient determined by the third jet. These are deterministic and Kac–Rice forms of fixed-level repulsion. Any quoted decimal transition value is kept separate from the theorem unless supported by an interval certificate. The same dual construction gives symbolic hard-core and ordinate-separation criteria at every fixed derivative level.

- (XII) **Rooted edge rigidity and ordinate phase recovery.** Horizontal densities, edge locations, and the entire stationary local law depend only on the coefficient moduli. Nevertheless, the defect identity at a single nonzero  $a$ -point gives

$$\left| \frac{c_p}{|c_p|} - \frac{a}{|a|} p^{it} \right| \leq \left( \frac{2\{A_c(\sigma) - |a|\}}{|c_p| p^{-\sigma}} \right)^{1/2}.$$

An edge-approaching sequence therefore recovers all phases. For zeros it recovers the ratios of the phases, modulo their unavoidable common rotation. Together with the right-edge curve, which recovers the moduli, this gives a hybrid inverse datum for the full coefficient system. At the zero edge, recentering at any near-edge root gives the universal analytic limit

$$H_e(z) = 2^{1-(\sigma_e+z)} - \mathcal{P}(\sigma_e + z),$$

uniformly over the twist. The rooted divisors and rooted Palm laws collapse to its deterministic divisor. Equality of edge germs, with or without an imaginary translation, is equivalent to equality of the twists modulo common rotation and the corresponding vertical flow. In a shrinking edge layer the minimum ordinate separation grows at least logarithmically in the reciprocal defect.

- (XIII) **Exact phase-fidelity frontier.** For a zero  $\rho = \sigma + it$ , the weighted chordal distortion of the natural relative-phase decoder is exactly

$$\mathcal{D}_\omega(\rho) = 2 \left( \sum_{p \geq 3} p^{-\sigma} - 2^{-\sigma} \right).$$

Consequently a distortion threshold is exactly a horizontal edge layer. The outer-edge Cramér law therefore gives the complete logarithmic asymptotic for its vertical intensity  $\mathcal{C}(d)$  and, after inversion, an exact density–fidelity frontier. The induced condition for learning all phases at primes  $p \leq Q$  is a sufficient certificate with explicitly computable intensity; neither optimality nor a deterministic first-hit height is claimed.

- (XIV) **Small-target divisor surgery and monodromy.** On compact subbands of the deterministic hard-core region, every zero continues to a unique holomorphic  $a$ -point branch, uniformly in the twist and height, and every bounded  $a$ -point arises this way after inserting fixed horizontal buffers. A growing zero-free corridor separates this persistent divisor from the far-right escaping lattice. When the target winds once around 0, every persistent branch has identity monodromy while the escaping lattice undergoes a bilateral unit shift. The escaping mass and reciprocal lattice step—equivalently, the physical displacement of one shift—encode the least prime of the support; the bounded contribution has an even  $C^\infty$  expansion in  $|a|$  to every finite order.

## 1.1 What is and is not settled

This revision settles, in the following fixed-scale sense, the questions left as Problems 3 and 6 in Version 2.0. For Problem 3, at every fixed derivative order the complete stationary local divisor process is now determined. Its factorial  $k$ -point functions, finite-window count and void laws, and Palm nearest-neighbour and consecutive-gap laws are given by exact finite Kac–Rice/inclusion–exclusion formulae. The corrected pair statement is that the self-inclusive smoothed pair function is Bohr almost periodic, whereas the factorial pair function is an almost-periodic term plus a compactly supported diagonal correction. The local theory also gives a symbolic deterministic hard core on one side of the support transition and Vandermonde, in particular quadratic pair, repulsion in the nondegenerate bulk. This is a qualitative limit theorem on each fixed compact window; it supplies neither a height-error term nor an elementary closed form for the gap density.

For Problem 6, one no longer needs the complete moving derivative tower in order to recover phases. At a single fixed level, the ordinates of supplied near-edge nonzero  $a$ -points recover all coefficient phases. Supplied near-edge zeros recover all relative phases; the remaining common rotation is unavoidable because it leaves the zero set unchanged. The rooted zero-edge profile is universal, and its anchored germ determines the twist modulo this common rotation; after quotienting the germ by imaginary translation it determines the corresponding vertical-flow orbit. Horizontal divisors alone remain phase blind, even when their entire stationary finite-order correlation hierarchy is retained.

Three further refinements should be separated from those two closures. First, the outer-edge large-deviation law has a deterministic rooted counterpart: every near-edge zero, after recentering and normalization, converges to the same analytic profile  $H_e$ , and the associated rooted divisors and Palm laws collapse to its divisor. Second, the exact zero phase defect turns the edge law into an exact asymptotic for the limiting intensity of any prescribed small distortion, together with its inverse density–fidelity frontier. This does not identify the first height at which that distortion occurs, and the finite-prime reconstruction condition extracted from it is sufficient rather than necessary. Third, the small-target measure limit is upgraded to a pointwise divisor theorem in the hard-core subbands: bounded zeros persist as holomorphic branches, a zero-free corridor separates them from the escaping logarithmic lattice, and the two families have respectively identity and shift monodromy. The corresponding bounded mass has an even smooth asymptotic jet.

The paper proves no new case of the Riemann hypothesis and no case of the twin prime, Polignac, Cramér, or Montgomery pair-correlation conjectures. The zeros studied here lie in the absolute-convergence half-plane and have fixed-scale almost-periodic statistics; they must not be conflated with the high zeros of the Riemann zeta function. Nor do the exact prime-gap dictionaries later in the paper estimate a prime gap: they are inverse identities only.

The exact projection edge is likewise not claimed as a new general theorem. It follows from classical analytic almost-periodic projection theory, and an independent prime-zeta specialization was given by Camargo. The distinct- and simple-divisor laws, local Kac–Rice hierarchy, deterministic hard core, rooted edge profile, ordinate phase recovery, phase-fidelity law, small-target divisor surgery, and inverse statements are the specialized contributions asserted here, subject to the priority qualification at the end of the paper. For arbitrary phases the universal zero-free statement remains the *open* half-plane  $\Re s > \sigma_e$ : an adversarial twist can attain its boundary, whereas the untwisted prime zeta function does not.



## 2 Almost-periodic hulls and exact projection sets

Let  $c = (c_p)_{p \in \mathbb{P}}$  be nonzero and suppose that  $|c_p| \ll (\log p)^M$  for some fixed  $M \geq 0$ . Put

$$F_c(s) = \sum_p c_p p^{-s}, \quad A_c(\sigma) = \sum_p |c_p| p^{-\sigma}, \quad \sigma > 1.$$

For fixed  $\sigma$ , define the inner polygonal radius

$$B_c(\sigma) = \max \left\{ 2 \max_p (|c_p| p^{-\sigma}) - A_c(\sigma), 0 \right\}.$$

**Proposition 2.1** (Phase annulus and real-part projection). *For every  $\sigma > 1$ ,*

$$\overline{\{F_c(\sigma + it) : t \in \mathbb{R}\}} = \{z \in \mathbb{C} : B_c(\sigma) \leq |z| \leq A_c(\sigma)\}.$$

Consequently, for every  $a \in \mathbb{C}$ ,

$$\boxed{\overline{\{\Re s : F_c(s) = a, \Re s > 1\}} = \{\sigma > 1 : B_c(\sigma) \leq |a| \leq A_c(\sigma)\}}. \quad (2.1)$$

The closure in the second formula is relative to  $(1, \infty)$ .

*Proof.* Unique factorization implies that the logarithms of distinct primes are linearly independent over  $\mathbb{Q}$ . Kronecker–Weyl approximation, first on a finite set of primes and then with the absolutely convergent tail restored, identifies the vertical value closure with

$$\left\{ \sum_p |c_p| p^{-\sigma} z_p : |z_p| = 1 \right\}.$$

The infinite polygon inequality gives precisely the stated closed annulus.

If a phase choice gives an equivalent Dirichlet series  $G$  with  $G(\sigma) = a$ , then arbitrarily large translates of  $F_c$  converge locally uniformly to  $G$  in  $\Re s > 1$ . Hurwitz’s theorem supplies  $a$ -points of  $F_c$  whose real parts tend to  $\sigma$ . Conversely, uniform differentiability on compact substrips shows that a limit of real parts of actual  $a$ -points must lie in the vertical value closure. This proves (2.1). This last projection step is a standard principle in analytic almost-periodic theory; compare [10, 18].  $\square$

For  $c_p = \omega_p$ ,  $|\omega_p| = 1$ , write  $F_\omega$ . At the zero target, the largest term is  $2^{-\sigma}$ , and the right edge is the unique solution

$$2^{-\sigma_e} = \sum_{p \geq 3} p^{-\sigma_e}, \quad \sigma_e = 1.7795446535469941164 \dots \quad (2.2)$$

**Corollary 2.2** (Exact zero projection and its boundary). *For every unimodular twist,*

$$\overline{\{\Re s : F_\omega(s) = 0, \Re s > 1\}} = (1, \sigma_e]. \quad (2.3)$$

*All such twists are zero-free in  $\Re s > \sigma_e$ . This cannot be changed to a universal closed-half-plane statement: the twist*

$$\omega_2 = 1, \quad \omega_p = -1 \quad (p \geq 3)$$

*satisfies  $F_\omega(\sigma_e) = 0$ .*

*For the untwisted prime zeta function,*

$$\mathcal{P}(s) \neq 0 \quad (\Re s \geq \sigma_e), \quad (2.4)$$

*although zero real parts approach  $\sigma_e$  from the left.*

*Proof.* Only the last assertion needs comment. Equality at the polygon boundary would force all phases  $p^{-it}$ ,  $p \geq 3$ , to oppose  $2^{-it}$ . In particular,  $t \log(5/3)$  and  $t \log(7/3)$  would both lie in  $2\pi\mathbb{Z}$ . If  $t \neq 0$ , their ratio would be rational, contradicting unique factorization;  $t = 0$  is impossible because all summands are positive.  $\square$

*Remark 2.3* (Priority boundary). The polygon equation (2.2) is the natural exact form of the numerically observed edge. The general mechanism predates the 2025 conjecture, and Camargo [7] subsequently obtained the phase-invariant projection closure by an equivalent-series argument. Later sections therefore use (2.2) as an input and concentrate on genuinely stronger density and inverse statements.

### 3 Smooth weighted densities and exact horizontal laws

Let  $(U_p)_{p \in \mathbb{P}}$  be independent Haar variables on the unit circle. For  $\sigma > 1$  define

$$X_\sigma = \sum_p U_p p^{-\sigma}, \quad Y_\sigma = - \sum_p (\log p) U_p p^{-\sigma}. \quad (3.1)$$

Both sums are then absolutely convergent; this random-Euler-product viewpoint goes back to Jessen–Wintner [11]. Write  $q_\sigma(x, y)$  for the joint density of  $(X_\sigma, Y_\sigma)$ , when it exists, and define the derivative-weighted value density

$$G(\sigma, a) = \int_{\mathbb{C}} |y|^2 q_\sigma(a, y) dA(y). \quad (3.2)$$

Here and throughout,  $dA$  is ordinary (unnormalized) planar Lebesgue measure.

**Lemma 3.1** (Bessel-product smoothing). *The pair  $(X_\sigma, Y_\sigma)$  has a  $C^\infty$  compactly supported density on  $\mathbb{C}^2$ . On every compact  $K \subseteq (1, \infty)$ , the function  $q_\sigma(x, y)$  and all its derivatives depend continuously and uniformly on  $\sigma \in K$ . Consequently,  $G \in C^\infty((1, \infty) \times \mathbb{C})$ .*

Moreover, with

$$R(\sigma) = \mathcal{P}(\sigma), \quad r(\sigma) = \max\{2^{1-\sigma} - \mathcal{P}(\sigma), 0\}, \quad (3.3)$$

one has

$$G(\sigma, a) > 0 \iff |a| < R(\sigma) \text{ and } (r(\sigma) = 0 \text{ or } |a| > r(\sigma)), \quad (3.4)$$

and  $G(\sigma, a) = 0$  outside the interior of the support. When  $r(\sigma) > 0$ , both boundary circles have zero weighted density; when  $r(\sigma) = 0$ , the origin is an interior point and has positive density. The function is radial in  $a$ .

*Proof.* For  $u, v \in \mathbb{C}$ , the joint characteristic function is

$$\hat{q}_\sigma(u, v) = \prod_p J_0(p^{-\sigma} |u - (\log p)v|), \quad (3.5)$$

where  $J_0$  is the Bessel function. Fix as many disjoint pairs of primes as desired. For every dual direction  $(u, v)$ , at least one of the two linear forms belonging to each pair is bounded below by a fixed multiple of  $|(u, v)|$ ; cancellation can occur for at most one logarithmic slope. Since

$$|J_0(x)| \ll (1 + x)^{-1/2}, \quad (3.6)$$

the factors from these fixed pairs give arbitrarily high polynomial decay. After any fixed number of differentiations in  $(u, v, \sigma)$ , unused pairs provide the same decay. Thus  $\widehat{q}_\sigma$  is a Schwartz function, uniformly for  $\sigma$  in a compact subinterval. Fourier inversion proves the first assertion.

The support of  $X_\sigma$  is the Minkowski sum of circles of radii  $p^{-\sigma}$ ; the polygon theorem identifies it with the closed annulus in (3.3), interpreted as a closed disc when  $r(\sigma) = 0$ . At a strict interior point, finitely many phase variables can be chosen so that the phase-to-value map has real rank two and its  $\sigma$ -derivative is nonzero. The coarea formula gives a positive local contribution to (3.2), and convolution with the unused variables preserves positivity. The genuine boundary circles are critical polygonal alignments and have zero weighted density. Common rotation of all  $U_p$  proves radially.  $\square$

For  $a \in \mathbb{C}$  and a unimodular twist  $F_\omega$ , let

$$J_a(\sigma) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \log |F_\omega(\sigma + it) - a| dt. \quad (3.7)$$

The following theorem is the metric step beyond the projection law.

**Theorem 3.2** (Phase-blind smooth  $a$ -point law). *For every  $a \in \mathbb{C}$ , the limit (3.7) exists, is independent of  $\omega$ , and is  $C^\infty$  on  $(1, \infty)$ . One has*

$$\boxed{J_a''(\sigma) = 2\pi G(\sigma, a).} \quad (3.8)$$

If multiplicities are denoted by  $m_\rho$ , then for  $1 < \alpha < \beta < \infty$ ,

$$\boxed{\sum_{\substack{F_\omega(\rho)=a \\ \alpha < \Re \rho < \beta \\ 0 < \Im \rho < T}} m_\rho = T \int_\alpha^\beta G(\sigma, a) d\sigma + o(T).} \quad (3.9)$$

The error may depend on  $\alpha, \beta, a$ , and  $\omega$ .

*Proof.* Kronecker–Weyl equidistribution identifies the vertical translation hull of every  $F_\omega$  with the Haar product model (3.1); the coefficient phases disappear under Haar rotation. Classical Jessen–Tornehave theory gives the existence and convexity of (3.7). The Poincaré–Lelong identity, averaged over the vertical flow, gives in the distributional sense

$$J_a''(\sigma) = 2\pi \mathbb{E}[|Y_\sigma|^2 \delta_a(X_\sigma)].$$

The right side is exactly (3.2). The logarithmic uniform-integrability and mean-convergence passage is the weighted-density theorem of Borchsenius–Jessen; the convenient modern formulation in Lee [13, Sections 2–3] applies after finite-prime approximation. Lemma 3.1 removes all exceptional differentiability points. Integrating (3.8) over  $(\alpha, \beta)$  gives (3.9).  $\square$

### 3.1 Asymptotic simplicity and marked divisor laws

The classical Jessen count is multiplicity weighted. For the present prime-supported family the loss incurred by that convention is sublinear.

**Theorem 3.3** (Asymptotic simplicity on compact strips). *Fix  $a \in \mathbb{C}$  and a compact interval  $I = [\alpha, \beta] \Subset (1, \infty)$ . If  $m(\rho)$  denotes the multiplicity of an  $a$ -point of  $F_\omega$ , then*

$$E_{\omega,a,I}(T) := \sum_{\substack{F_\omega(\rho)=a \\ \Re \rho \in I, 0 < \Im \rho < T}} (m(\rho) - 1) = o(T).$$

*Consequently the multiplicity count  $N^{\text{mult}}$ , the distinct-point count  $N^{\text{dist}}$ , and the simple-point count  $N^{\text{simp}}$  satisfy*

$$N_{\omega,a,I}^{\text{mult}}(T) = N_{\omega,a,I}^{\text{dist}}(T) + o(T) = N_{\omega,a,I}^{\text{simp}}(T) + o(T).$$

*The assertion is for fixed  $a$  and  $I$ ; no uniformity as  $\alpha \downarrow 1$  is claimed.*

*Proof.* Let  $\mathbb{H} = \mathbb{T}^{\mathbb{P}}$  with Haar probability and put

$$F_u(s) = \sum_p u_p p^{-s}, \quad (R_t u)_p = u_p p^{-it}.$$

Then  $F_\omega(s + it) = F_{R_t \omega}(s)$ . Unique factorization makes the finite subsets of  $(\log p)_p$  linearly independent over  $\mathbb{Q}$ , so this Kronecker flow is uniquely ergodic on  $\mathbb{H}$ .

For a compact  $K \Subset \{\Re s > 1\}$ , let  $\mathcal{B}_{a,K}$  be the set of  $u$  for which  $F_u - a$  has a multiple zero in  $K$ . It is closed and Haar-null. To see the second assertion, fix a prime  $q$  and condition on all phases except  $u_q$ . With  $H(s) = \sum_{p \neq q} u_p p^{-s}$ , a multiple root must satisfy

$$H'(s) + (\log q)(H(s) - a) = 0, \quad u_q = q^s(a - H(s)).$$

The first left-hand side is not identically zero, by uniqueness of absolutely convergent Dirichlet series. It has finitely many zeros in  $K$ , and each permits at most one value of  $u_q$ . Fubini's theorem proves Haar nullity.

There is also a uniform bound, over  $u \in \mathbb{H}$ , for the number of zeros in a fixed compact box, counted with multiplicity. Indeed the hull is compact in the compact-open topology and contains no function identically equal to  $a$ ; a failure of the bound would contradict Hurwitz's and Rouché's theorems on a slightly larger box. Choose  $K = I + i[-1, 1]$ . The excess-zero functional is bounded by a constant times  $\mathbf{1}_{\mathcal{B}_{a,K}}$ . Unique ergodicity and the Portmanteau inequality for this closed null set therefore give

$$\int_0^T Z_K^{\text{exc}}(R_t \omega) dt = o(T).$$

Every multiple point of  $F_\omega - a$  with ordinate in  $(1, T - 1)$  is seen by this moving box for two units of time. The two end windows contribute only  $O(1)$  by the uniform local bound. Hence  $E_{\omega,a,I}(T) = o(T)$ . Finally  $N^{\text{mult}} - N^{\text{dist}} = E$  and  $0 \leq N^{\text{dist}} - N^{\text{simp}} \leq E$ .  $\square$

**Theorem 3.4** (Marked simple  $a$ -point law). *Let  $I \Subset (1, \infty)$  and  $\Psi \in C_c(I \times \mathbb{C})$ . Then*

$$\frac{1}{T} \sum_{\substack{F_\omega(\rho)=a, F'_\omega(\rho) \neq 0 \\ 0 < \Im \rho < T}} \Psi(\Re \rho, F'_\omega(\rho)) \longrightarrow \int_I \int_{\mathbb{C}} \Psi(\sigma, y) |y|^2 q_\sigma(a, y) dA(y) d\sigma.$$

*In particular, the simple-point count has the same main term as (3.9).*

*Proof.* On a unit-height box, sum the displayed mark over the simple roots of  $F_u - a$ . This bounded zero functional is continuous except when a root lies on the boundary or is multiple. The second exceptional set is Haar-null by the conditioning argument in the preceding proof. The first is Haar-null as well: the expected number of roots in a shrinking collar is  $O(\delta)$  by the Kac–Rice formula, and one lets  $\delta \downarrow 0$ . Thus the functional is Haar–Riemann integrable, and unique ergodicity transfers its Haar mean to every twist. Complex Kac–Rice evaluates that mean as

$$\int_I \mathbb{E} \left[ \Psi(\sigma, Y_\sigma) |Y_\sigma|^2 \delta_a(X_\sigma) \right] d\sigma,$$

which is the asserted integral. The unmarked statement follows by a compact cutoff in the uniformly bounded derivative range and Theorem 3.3.  $\square$

**Corollary 3.5** (Near-critical slopes and large residues). *For fixed  $a$  and  $I \Subset (1, \infty)$ ,*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \# \{ \rho : F_\omega(\rho) = a, \Re \rho \in I, 0 < \Im \rho < T, 0 < |F'_\omega(\rho)| \leq \varepsilon \} = \int_I \int_{|y| \leq \varepsilon} |y|^2 q_\sigma(a, y) dA(y) d\sigma$$

and, as  $\varepsilon \downarrow 0$ , the right side equals

$$\frac{\pi \varepsilon^4}{2} \int_I q_\sigma(a, 0) d\sigma + O_{I,a}(\varepsilon^6).$$

If  $\mathcal{D}_z = (1 - zF_\omega)^{-1}$ , then in the same strip

$$\frac{1}{T} \# \{ \rho : \rho \text{ is a simple pole of } \mathcal{D}_z, |\operatorname{Res}_\rho \mathcal{D}_z| > R \} = \frac{\pi}{2|z|^4 R^4} \int_I q_\sigma(z^{-1}, 0) d\sigma + O_{I,z}(R^{-6}) + o_T(1).$$

The constants are not uniform as the target tends to zero.

**Corollary 3.6** (Complete simple valence in thin strips). *Let  $S$  be an infinite prime support and suppose, for a fixed  $m \geq 0$ , that*

$$\sum_{p \in S} (\log p)^m p^{-\sigma} \longrightarrow \infty \quad (\sigma \downarrow 1).$$

For every  $a \in \mathbb{C}$ , every  $\delta > 0$ , and every fixed unimodular twist, there is a compact interval  $I \Subset (1, 1 + \delta)$  and a constant  $\kappa > 0$  such that

$$N_{m,a}^{\text{simp}}(I; T) = \kappa T + o(T).$$

Hence every value is assumed infinitely often, at simple points, in every boundary strip. The hypothesis holds for all primes, cofinite prime supports, and positive Chebotarev-density supports.

*Proof.* Near 1, the total polygonal radius diverges whereas its largest summand remains bounded. Every fixed target therefore lies in the strict interior of the value annulus on some compact interval  $I \Subset (1, 1 + \delta)$ . The weighted density is positive there, and Theorems 3.2 and 3.3 give the result.  $\square$

## 4 The 2025 conjectures and the endpoint at $\Re s = 1$

Put

$$\rho(\sigma) = G(\sigma, 0). \tag{4.1}$$

By Lemma 3.1 and (2.2),

$$\rho(\sigma) > 0 \quad (1 < \sigma < \sigma_e), \quad \rho(\sigma) = 0 \quad (\sigma \geq \sigma_e). \tag{4.2}$$

Since  $G$  is smooth through the boundary, the edge is infinitely soft:

$$\rho^{(j)}(\sigma_e) = 0 \quad (j \geq 0). \tag{4.3}$$

### 4.1 Exact density, edge convergence, and height uniformity

For  $1 < \gamma < \sigma_e$  define

$$c_\gamma = \int_\gamma^{\sigma_e} \rho(\sigma) d\sigma. \quad (4.4)$$

**Theorem 4.1** (Exact simple form of the BCS metric conjectures). *For every unimodular twist and  $1 < \gamma < \sigma_e$ ,*

$$M_{\omega,\gamma}^{\text{mult}}(T) := \sum_{\substack{F_\omega(\rho)=0 \\ \Re \rho > \gamma \\ 0 < \Im \rho < T}} m_\rho = c_\gamma T + o(T), \quad (4.5)$$

and

$$M_{\omega,\gamma}^{\text{dist}}(T) = M_{\omega,\gamma}^{\text{simp}}(T) = c_\gamma T + o(T).$$

where  $c_\gamma > 0$ . Moreover, for  $0 \leq x \leq 1$ ,

$$\frac{M_{\omega,\gamma}^\bullet(xT)}{M_{\omega,\gamma}^\bullet(T)} \rightarrow x, \quad \bullet \in \{\text{mult}, \text{dist}, \text{simp}\}. \quad (4.6)$$

For the untwisted prime zeta function, if

$$\Sigma(T) = \sup\{\Re \rho : \mathcal{P}(\rho) = 0, 0 < \Im \rho < T\},$$

then

$$\Sigma(T) \rightarrow \sigma_e. \quad (4.7)$$

Thus (4.7) recovers BCS Conjecture 1 from the earlier general projection theory. Equations (4.5)–(4.6) prove the literal distinct-point Conjecture 2 and strengthen Conjecture 3 from two-sided linear bounds to an exact asymptotic, in their stated range  $1 < \gamma < \sigma_e$ .

*Proof.* Choose any  $\beta > \sigma_e$  and apply (3.9) to  $(\gamma, \beta)$ ; there are no zeros with real part in  $(\sigma_e, \beta)$ , and positivity follows from (4.2). Theorem 3.3 transfers the same main term to the distinct and simple counts. Replacing  $T$  by  $xT$  proves (4.6). For every  $\varepsilon > 0$ , (4.5) supplies zeros with real part in  $(\sigma_e - \varepsilon, \sigma_e)$  for all sufficiently large  $T$ , while Corollary 2.2 supplies the upper bound. This proves (4.7).  $\square$

The result has a useful two-variable form. For compactly supported continuous  $\Phi$  on  $(1, \sigma_e) \times (0, 1)$ ,

$$\frac{1}{T} \sum_{\substack{F_\omega(\rho)=0 \\ 0 < \Im \rho < T}} m_\rho \Phi\left(\Re \rho, \frac{\Im \rho}{T}\right) \rightarrow \int_1^{\sigma_e} \int_0^1 \Phi(\sigma, u) \rho(\sigma) du d\sigma. \quad (4.8)$$

Indeed, (4.8) holds first for rectangles by subtracting (4.5) at two heights and then for all such  $\Phi$  by approximation. Thus the real part and normalized height are asymptotically independent at the scale detected by the limiting divisor. Also, every fixed vertical line carries  $o(T)$  zeros because the limiting horizontal measure has no atoms. The same limit holds with distinct or simple points, by Theorem 3.3.

## 4.2 Syndetic recurrence of simple points

The density law gives many points on average. Almost periodicity upgrades existence in any occupied band to a bounded-window statement.

**Theorem 4.2** (Bounded-window recurrence). *Let  $I \Subset (1, \infty)$  and suppose  $\int_I G(\sigma, a) d\sigma > 0$ . For every fixed unimodular twist there is  $H = H(I, a, \omega) > 0$  such that every sufficiently high interval  $[u, u + H]$  contains the ordinate of a simple  $a$ -point with real part in  $I$ .*

*Proof.* Theorems 3.2 and 3.3 provide one simple point in the interior of the band. Choose a small disc about it on whose boundary  $F_\omega - a$  is nonzero. The set of uniform almost periods on this compact disc is relatively dense, because it is the return set of a minimal translation on the compact prime-phase hull. Rouché's theorem transports the simple zero to every corresponding translate. Shrinking the disc keeps the real part inside  $I$  and yields the asserted bound  $H$ .  $\square$

**Corollary 4.3** (Syndetic approach to the zero edge). *For every  $\eta > 0$  and every fixed unimodular twist, the strip*

$$\sigma_e - \eta < \Re s < \sigma_e$$

*contains simple zeros whose ordinates have bounded gaps. Thus the right edge is approached in every sufficiently high bounded-height window, not merely along a sparse sequence.*

*Proof.* Choose  $I \Subset (\max\{1, \sigma_e - \eta\}, \sigma_e)$ . Then  $G(\sigma, 0) > 0$  on  $I$ , and Theorem 4.2 applies.  $\square$

For  $\sigma \geq \sigma_e$ , the  $p = 2$  term dominates and

$$J_0(\sigma) = -\sigma \log 2. \tag{4.9}$$

Consequently,

$$c_\gamma = \frac{-\log 2 - J'_0(\gamma)}{2\pi}. \tag{4.10}$$

## 4.3 Exponential transfer across the abscissa of convergence

The remaining endpoint is special to  $\omega_p \equiv 1$ . In  $\Re s > 1$ ,

$$e^{\mathcal{P}(s)} = \zeta(s)H(s), \quad H(s) = \exp\left(-\sum_{k \geq 2} \frac{\mathcal{P}(ks)}{k}\right). \tag{4.11}$$

The function  $H$  is holomorphic and nonvanishing in  $\Re s > \frac{1}{2}$ .

**Lemma 4.4** (Exponential boundary majorant). *Fix  $a \in \mathbb{C}$  and set*

$$E(s) = \zeta(s)H(s), \quad Q_a(s) = E(s) - e^a.$$

*Then  $Q_a$  is meromorphic in  $\Re s > \frac{1}{2}$ , with only the pole at  $s = 1$ . Its multiplicity-weighted zero divisor has a locally bounded horizontal density near  $\sigma = 1$ . In particular, for sufficiently small fixed  $\delta > 0$ ,*

$$N_{Q_a}(1, 1 + \delta; T) \ll_a \delta T + o(T). \tag{4.12}$$

*Proof.* For  $\Re s > 1$ ,

$$E(s) = \prod_p e^{p^{-s}} = \sum_{n \geq 1} \frac{b_n}{n^s}, \quad b_n = \prod_p \frac{1}{\nu_p(n)!}. \quad (4.13)$$

For  $\sigma > \frac{1}{2}$ ,

$$\sum_{n \geq 1} \frac{b_n^2}{n^{2\sigma}} = \prod_p I_0(2p^{-\sigma}) < \infty, \quad (4.14)$$

and the same is true after inserting any fixed logarithmic power. Hence the Dirichlet polynomials of (4.13) converge to the meromorphic continuation in the required local mean-square sense. The generalized Jessen transfer theorem of Borchsenius–Jessen [6], in the form used by Lee [13, Theorem 3.1], applies after excising the isolated pole.

For completeness, the random series (3.1) are well defined in  $L^2$  for  $\sigma > \frac{1}{2}$ . The proof of Lemma 3.1, now using prime blocks rather than absolute support, gives a Schwartz joint density and a weighted density  $G_*(\sigma, z)$  smooth on this larger half-plane. The random envelope is  $e^{X_\sigma}$ , with derivative  $e^{X_\sigma} Y_\sigma$ . At a preimage  $X_\sigma = a + 2\pi i k$ , the Jacobian of the exponential cancels the factor  $|e^{X_\sigma}|^2$ . The zero density of  $Q_a$  is therefore

$$\tilde{G}_a(\sigma) = \sum_{k \in \mathbb{Z}} G_*(\sigma, a + 2\pi i k). \quad (4.15)$$

The series and all its derivatives converge locally uniformly because  $G_*$  is Schwartz in the value variable. It is thus bounded near  $\sigma = 1$ , and integration over  $(1, 1 + \delta)$  proves (4.12).  $\square$

**Theorem 4.5** (Exact endpoint  $a$ -point law). *For every fixed  $a \in \mathbb{C}$ ,*

$$M_a^{\text{mult}}(T) := \sum_{\substack{\mathcal{P}(\rho)=a \\ \Re \rho > 1 \\ 0 < \Im \rho < T}} m_\rho = C_a T + o(T), \quad C_a = \int_1^\infty G(\sigma, a) d\sigma, \quad (4.16)$$

where  $0 < C_a < \infty$ . The normalized ordinates are uniformly distributed:

$$M_a^{\text{dist}}(T) = M_a^{\text{simp}}(T) = C_a T + o(T), \quad \frac{M_a^\bullet(xT)}{M_a^\bullet(T)} \longrightarrow x \quad (\bullet \in \{\text{mult}, \text{dist}, \text{simp}\}). \quad (4.17)$$

In particular,

$$\boxed{M_1(T) := \sum_{\substack{\mathcal{P}(\rho)=0 \\ \Re \rho > 1 \\ 0 < \Im \rho < T}} m_\rho = c_1 T + o(T), \quad c_1 = \int_1^{\sigma_e} \rho(\sigma) d\sigma > 0.} \quad (4.18)$$

and the same asymptotic holds for the distinct and simple zero counts. This strengthens the separate endpoint estimate  $M_1(T) = O(T)$  in [2].

*Proof.* For fixed  $\delta > 0$ , Theorem 3.2 gives the multiplicity asymptotic in  $\Re s > 1 + \delta$ . Every zero of  $\mathcal{P} - a$  is a zero of  $Q_a = e^\mathcal{P} - e^a$  with the same multiplicity. Lemma 4.4 therefore bounds the omitted layer by  $O_a(\delta T) + o(T)$ . First let  $T \rightarrow \infty$  and then  $\delta \downarrow 0$ . This squeezes  $M_a^{\text{mult}}(T)/T$  to the integral in (4.16).

For  $a \neq 0$ , the integral is finite because no  $a$ -point is possible once  $\mathcal{P}(\sigma) < |a|$ ; for  $a = 0$ , its support ends at  $\sigma_e$ . Positivity follows because, as  $\sigma \downarrow 1$ , the strict polygon annulus contains every fixed target on some nonempty interval. On  $\Re s \geq 1 + \delta$ , Theorem 3.3 makes the excess multiplicity  $o(T)$ ; the omitted boundary layer is again at most  $O_a(\delta T) + o(T)$ . Letting  $\delta \downarrow 0$  proves the distinct and simple formulas. Height uniformity follows by replacing  $T$  by  $xT$ .  $\square$



*Remark 4.6* (The exponential is a majorant, not an equivalence). It is essential that

$$\mathcal{P}(s) = a \implies e^{\mathcal{P}(s)} = e^a$$

is only one direction. The reverse implication gives  $\mathcal{P}(s) = a + 2\pi ik$  for some  $k \in \mathbb{Z}$ ; indeed,

$$\operatorname{div}_0(E - e^a) = \sum_{k \in \mathbb{Z}} \operatorname{div}_0(\mathcal{P} - a - 2\pi ik) \quad (\Re s > 1). \quad (4.19)$$

Thus the envelope is used only for a boundary-layer upper bound. No analogous factorization is available for an arbitrary twist, so Theorem 4.5 is not asserted for  $F_\omega$ .

#### 4.4 Quantized small-target escape

The target  $a = 0$  is singular from the viewpoint of the far right. Nonzero targets have a finite right edge, whereas zero points of the hull retain a terminal contribution from the least supported prime. The discrepancy is a quantized atom at infinity.

**Theorem 4.7** (Quantized escape of  $a$ -point mass). *Let  $S$  be an infinite prime support for which the smooth density conclusion of Lemma 3.1 holds, let  $p_0 = \min S$ , and put*

$$F_{S,\omega}(s) = \sum_{p \in S} \omega_p p^{-s}.$$

For fixed  $\sigma_0 > 1$ , write

$$C_{S,a}(\sigma_0) = \int_{\sigma_0}^{\infty} G_S(\sigma, a) d\sigma.$$

Then

$$\lim_{a \rightarrow 0, a \neq 0} C_{S,a}(\sigma_0) = C_{S,0}(\sigma_0) + \frac{\log p_0}{2\pi}. \quad (4.20)$$

More precisely, on the one-point compactification  $[\sigma_0, \infty]$ ,

$$G_S(\sigma, a) d\sigma \implies G_S(\sigma, 0) d\sigma + \frac{\log p_0}{2\pi} \delta_\infty. \quad (4.21)$$

Thus the jump recovers the least support prime:

$$p_0 = \exp \left( 2\pi \left[ \lim_{a \rightarrow 0, a \neq 0} C_{S,a}(\sigma_0) - C_{S,0}(\sigma_0) \right] \right). \quad (4.22)$$

For the full prime zeta function the escaping mass is  $\log 2/(2\pi)$ .

*Proof.* Integrating  $J_a'' = 2\pi G_S$  to  $+\infty$  gives

$$2\pi C_{S,a}(\sigma_0) = -J_a'(\sigma_0) \quad (a \neq 0), \quad 2\pi C_{S,0}(\sigma_0) = -\log p_0 - J_0'(\sigma_0).$$

Indeed,  $J_a(\sigma) = \log |a|$  once the entire value support lies inside the circle of radius  $|a|$ , whereas eventual  $p_0$ -dominance gives  $J_0'(\sigma) = -\log p_0$ . If  $f_\sigma$  is the marginal density of  $X_\sigma$ , then

$$J_a'(\sigma_0) = \int_{\mathbb{C}} \log |x - a| \partial_\sigma f_{\sigma_0}(x) dA(x).$$

Smooth compact support of  $f_\sigma$  and local integrability of the logarithmic kernel give  $J_a'(\sigma_0) \rightarrow J_0'(\sigma_0)$ . This proves (4.20). Local uniform convergence  $G_S(\sigma, a) \rightarrow G_S(\sigma, 0)$  on finite  $\sigma$ -intervals, together with convergence of total masses, proves (4.21).  $\square$

**Proposition 4.8** (The escaping simple lattice). *Let  $p_1$  be the next prime in  $S$  and  $\eta = \log p_1 / \log p_0 - 1 > 0$ . As  $a \rightarrow 0$ , the remote  $a$ -points are simple and, uniformly in their integer label  $k$ ,*

$$\rho_{a,k} = \frac{\log(1/|a|)}{\log p_0} + i \frac{\arg \omega_{p_0} - \arg a + 2\pi k}{\log p_0} + O(|a|^\eta), \quad (4.23)$$

$$F'_{S,\omega}(\rho_{a,k}) = -(\log p_0)a + O(|a|^{1+\eta}). \quad (4.24)$$

Their positive-height density is  $\log p_0 / (2\pi)$ , exactly the atom in (4.21).

*Proof.* The displayed leading points are the simple solutions of  $\omega_{p_0} p_0^{-s} = a$ . On equal small circles around them, the sum over  $p \geq p_1$  is  $O(|a|^{1+\eta})$ . Rouché's theorem and one Newton step give (4.23)–(4.24). Consecutive leading ordinates differ by  $2\pi / \log p_0$ .  $\square$

## 5 Prime spectral tomography from nonzero $a$ -points

We now reverse the preceding construction. For a bounded coefficient system  $c$ , let

$$X_{c,\sigma} = \sum_p |c_p| p^{-\sigma} U_p, \quad \mu_{c,\sigma} = \text{Law}(X_{c,\sigma}), \quad (5.1)$$

and define

$$J_{c,a}(\sigma) = \mathbb{E} \log |X_{c,\sigma} - a|, \quad \nu_{c,a} = \frac{1}{2\pi} J''_{c,a} \quad (5.2)$$

on  $(1, \infty)$ . The measure  $\nu_{c,a}$  is the limiting horizontal multiplicity-weighted  $a$ -point measure.

**Theorem 5.1** (All- $a$  tomographic equivalence). *Let  $c = (c_p)$  and  $d = (d_p)$  be bounded coefficient systems. The following are equivalent.*

- (i)  $|c_p| = |d_p|$  for every prime  $p$ .
- (ii)  $\mu_{c,\sigma} = \mu_{d,\sigma}$  for every  $\sigma > 1$ .
- (iii)  $J_{c,a}(\sigma) = J_{d,a}(\sigma)$  for every  $\sigma > 1$  and every  $a \neq 0$ .
- (iv)  $\nu_{c,a} = \nu_{d,a}$  on  $(1, \infty)$  for every  $a \neq 0$ , with the common normalization

$$J_{c,a}(\sigma), J_{d,a}(\sigma) \rightarrow \log |a|, \quad J'_{c,a}(\sigma), J'_{d,a}(\sigma) \rightarrow 0 \quad (\sigma \rightarrow \infty).$$

Thus the family of all nonzero  $a$ -point horizontal measures is a complete invariant of the coefficient moduli and is maximally insensitive to their phases.

*Proof.* Condition (i) gives equality in law in (5.1), and hence (ii)  $\Rightarrow$  (iii) as well. Distributional differentiation gives (iii)  $\Rightarrow$  (iv).

Conversely, put  $r = |a|$ . Rotational invariance and Jensen's circle formula yield

$$J_{c,a}(\sigma) = \mathbb{E} \log \max\{r, |X_{c,\sigma}|\}. \quad (5.3)$$

Once  $A_c(\sigma) < r$ , this equals  $\log r$ . The normalization at infinity therefore removes both affine ambiguities in the second derivative and gives

$$J_{c,a}(\sigma) = \log r + 2\pi \int_{(\sigma, \infty)} (u - \sigma) \nu_{c,a}(du). \quad (5.4)$$

Thus (iv) $\Rightarrow$ (iii).

For fixed  $\sigma$ , logarithmic potential theory gives

$$\Delta_a J_{c,a}(\sigma) = 2\pi\mu_{c,\sigma} \quad (5.5)$$

in distributions. Equality for all  $a \neq 0$ , together with total mass one and the normalization at infinity, also determines any mass at the origin; hence (iii) $\Rightarrow$ (ii). Finally, equality of the laws gives equality of variances:

$$\sum_p |c_p|^2 p^{-2\sigma} = \mathbb{E}|X_{c,\sigma}|^2 = \mathbb{E}|X_{d,\sigma}|^2 = \sum_p |d_p|^2 p^{-2\sigma}. \quad (5.6)$$

Uniqueness of absolutely convergent Dirichlet series proves (i).  $\square$

**Corollary 5.2** (Explicit recovery of a vertical value law). *Writing  $R_{c,\sigma} = |X_{c,\sigma}|$ , the one-sided radial derivatives satisfy*

$$r \partial_r^- J_{c,r}(\sigma) = \mathbb{P}(R_{c,\sigma} < r), \quad r \partial_r^+ J_{c,r}(\sigma) = \mathbb{P}(R_{c,\sigma} \leq r). \quad (5.7)$$

Thus the all- $a$  horizontal divisor recovers the full vertical value law at every  $\sigma > 1$ .

*Proof.* Differentiate (5.3) under the Stieltjes integral. The two one-sided derivatives record a possible atom on the circle of radius  $r$ . Rotational invariance then reconstructs the planar law from its radial distribution.  $\square$

## 5.1 Target-plane conservation and compressed tomography

The horizontal density can also be integrated in the target variable. This loses most of the value distribution but leaves an exact spectral trace.

**Proposition 5.3** (Target-plane conservation). *For a prime support  $S$ , put  $P_S(s) = \sum_{p \in S} p^{-s}$  and let  $G_S$  be the corresponding weighted density. With ordinary planar Lebesgue measure  $dA$ ,*

$$\boxed{\int_{\mathbb{C}} G_S(\sigma, a) dA(a) = \sum_{p \in S} (\log p)^2 p^{-2\sigma} = P_S''(2\sigma).} \quad (5.7a)$$

More generally, for  $1 < \alpha < \beta \leq \infty$ ,

$$\int_{\mathbb{C}} \int_{\alpha}^{\beta} G_S(\sigma, a) d\sigma dA(a) = \frac{1}{2} \sum_{p \in S} (\log p) (p^{-2\alpha} - p^{-2\beta}). \quad (5.7b)$$

In particular, if  $C_S(a) = \int_1^{\infty} G_S(\sigma, a) d\sigma$ , then

$$\boxed{\int_{\mathbb{C}} C_S(a) dA(a) = \frac{1}{2} \sum_{p \in S} \frac{\log p}{p^2} = -\frac{1}{2} P_S'(2).} \quad (5.7c)$$

For the  $m$ th derivative tower the right sides become, respectively,  $P_S^{(2m+2)}(2\sigma)$  and  $-P_S^{(2m+1)}(2)/2$ .

*Proof.* Tonelli's theorem and Haar orthogonality give

$$\int_{\mathbb{C}} G_S(\sigma, a) dA(a) = \mathbb{E}|Y_{S,\sigma}|^2 = \sum_{p,q \in S} \frac{(\log p)(\log q)}{(pq)^\sigma} \mathbb{E}(U_p \overline{U_q}) = P_S''(2\sigma).$$

Integrating in  $\sigma$  proves the remaining formulas. In the  $m$ th derivative case,  $Y$  has coefficient  $(\log p)^{m+1} p^{-\sigma}$ .  $\square$

**Corollary 5.4** (One scalar hears the modulus radical). *For an integer  $q \geq 1$ , define*

$$\mathfrak{M}(q) = \frac{1}{2} \sum_{p \nmid q} \frac{\log p}{p^2}.$$

*Then*

$$\boxed{\mathfrak{M}(q_1) = \mathfrak{M}(q_2) \iff \text{rad}(q_1) = \text{rad}(q_2).} \quad (5.7d)$$

*Equivalently, for Dirichlet-character prime series, the complete horizontal density, the full right-edge curve, and this single target-integrated mass all classify exactly the set of primes dividing the modulus. They do not recover character phases or prime powers in the modulus.*

*Proof.* Only injectivity of the scalar needs proof. Equality gives a finite relation  $\sum_{p \in E} \varepsilon_p (\log p)/p^2 = 0$  with  $\varepsilon_p \in \{-1, 1\}$ . Multiplication by  $D = \prod_{p \in E} p^2$  and exponentiation give  $\prod_{p \in E} p^{\varepsilon_p D/p^2} = 1$ . Unique factorization forces  $E = \emptyset$ . The other equivalences follow from Theorem 5.1 and Theorem 5.7.  $\square$

## 5.2 One vertical line already contains every prime

The preceding theorem used all vertical lines because it allowed arbitrary weights. For the unit-coefficient prime series, one line suffices.

**Theorem 5.5** (One-line recovery of the prime spectrum). *Fix  $\sigma > 1$ . The vertical limiting value law of*

$$F_\omega(\sigma + it) = \sum_p \omega_p p^{-\sigma - it}$$

*determines the complete prime sequence  $2, 3, 5, \dots$*

*Proof.* For  $X_\sigma$  as in (3.1), the law determines

$$\phi_\sigma(t) = \mathbb{E} e^{it \Re X_\sigma} = \mathbb{E} J_0(t |X_\sigma|) = \prod_p J_0(tp^{-\sigma}). \quad (5.8)$$

If  $j_{0,\ell}$  are the positive zeros of  $J_0$ , then near the origin

$$\log J_0(z) = - \sum_{m \geq 1} b_m z^{2m}, \quad b_m = \frac{1}{m} \sum_{\ell \geq 1} j_{0,\ell}^{-2m} > 0. \quad (5.9)$$

It follows that

$$\log \phi_\sigma(t) = - \sum_{m \geq 1} b_m \mathcal{P}(2m\sigma) t^{2m}. \quad (5.10)$$

Hence the law determines every power sum

$$S_m = \sum_p (p^{-2\sigma})^m.$$

Arrange  $x_1 > x_2 > \cdots$  with  $x_k = p_k^{-2\sigma}$ . Successive root peeling gives

$$x_1 = \lim_{m \rightarrow \infty} S_m^{1/m}, \quad x_{k+1} = \lim_{m \rightarrow \infty} \left( S_m - \sum_{j \leq k} x_j^m \right)^{1/m}, \quad (5.11)$$

and finally  $p_k = x_k^{-1/(2\sigma)}$ .  $\square$

*Remark 5.6.* For general  $c_p$ , one fixed line recovers only the unordered multiset  $\{|c_p|p^{-\sigma} : p \in \mathbb{P}\}$ . Prime labels are recovered by using all  $\sigma$ , as in Theorem 5.1, or by the edge curve below.

### 5.3 Minimal right-edge tomography

For  $a \neq 0$  define

$$E_c(a) = \sup \{ \Re s : F_c(s) = a, \Re s > 1 \}. \quad (5.12)$$

**Theorem 5.7** (Right-edge tomography). *Assume that at least two coefficients  $c_p$  are nonzero. For every  $r = |a|$  in the range of  $A_c$ , one has*

$$\boxed{E_c(a) = A_c^{-1}(r)}. \quad (5.13)$$

Consequently, the following data determine one another:

$$\boxed{(|c_p|)_{p \in \mathbb{P}} \iff A_c(\sigma) \iff E_c(r)}. \quad (5.14)$$

The edge depends only on  $|a|$  and the coefficient moduli.

*Proof.* The triangle inequality forbids an  $a$ -point whenever  $A_c(\sigma) < r$ , proving the upper bound. At  $\sigma_0 = A_c^{-1}(r)$ , align every hull phase with  $a/r$ . The resulting equivalent series takes the value  $a$  at  $\sigma_0$ , and the zero of its difference from  $a$  is simple because

$$\frac{d}{d\sigma} \sum_p |c_p| p^{-\sigma} \neq 0.$$

Kronecker approximation and Hurwitz's theorem therefore produce actual  $a$ -points with real parts approaching  $\sigma_0$ . This proves (5.13). Inverting the edge gives  $A_c$ ; uniqueness of positive Dirichlet series then recovers  $|c_p|$  prime by prime.  $\square$

**Corollary 5.8** (Can one hear the primes?). *For every unimodular twist and every  $r > 0$ ,*

$$\boxed{E_\omega(r) = \mathcal{P}^{-1}(r)}. \quad (5.15)$$

Thus  $E_\omega^{-1}(\sigma) = \mathcal{P}(\sigma)$ , and the primes are recovered recursively by

$$p_1 = \lim_{\sigma \rightarrow \infty} \mathcal{P}(\sigma)^{-1/\sigma}, \quad (5.16)$$

$$p_{k+1} = \lim_{\sigma \rightarrow \infty} \left( \mathcal{P}(\sigma) - \sum_{j \leq k} p_j^{-\sigma} \right)^{-1/\sigma}. \quad (5.17)$$

For the untwisted function and positive target  $a = r$ , the right endpoint is the actual real point  $s = \mathcal{P}^{-1}(r)$ ; for a general twist it is only asserted as a supremum of a closure.

**Corollary 5.9** (Sparse edge sampling). *Suppose  $E_c(r_n) = E_d(r_n)$  for distinct positive  $r_n$  converging to an interior point of the common edge range. Then  $|c_p| = |d_p|$  for every prime  $p$ . Thus even an edge sample with one interior accumulation point is inverse complete.*

*Proof.* At the common edge values  $\sigma_n$  one has  $A_c(\sigma_n) = r_n = A_d(\sigma_n)$ . The  $\sigma_n$  accumulate in  $\Re s > 1$ ; the identity theorem gives  $A_c = A_d$ , followed by Dirichlet-series uniqueness.  $\square$

*Remark 5.10* (What zero data cannot hear). The zero-target measure alone is not inverse complete. Both  $2^{-s}$  and  $2^{-s} + 3^{-s}$  have no zeros in  $\Re s > 1$ , although their modulus spectra differ. Moreover, every limiting datum above is phase blind: it recovers  $|c_p|$  but never the individual phases  $c_p/|c_p|$ , and it does not identify the actual zero ordinates of two twists.

## 6 Derivative edges and value caustics

For  $m \geq 0$  and a unimodular sequence  $\omega$ , put

$$\mathcal{P}_{m,\omega}(s) = \sum_p \omega_p (\log p)^m p^{-s}, \quad \Re s > 1. \quad (6.1)$$

Thus  $\mathcal{P}_{m,1} = (-1)^m \mathcal{P}^{(m)}$ . Define

$$a_{m,p}(\sigma) = (\log p)^m p^{-\sigma}, \quad A_m(\sigma) = \sum_p a_{m,p}(\sigma), \quad (6.2)$$

and

$$R_m(\sigma) = \max \left\{ 0, \max_p \left( a_{m,p}(\sigma) - \sum_{q \neq p} a_{m,q}(\sigma) \right) \right\}. \quad (6.3)$$

**Lemma 6.1** (Fixed-order logarithmic-weight transfer). *Fix  $m \geq 0$  and define*

$$X_{m,\sigma} = \sum_p U_p (\log p)^m p^{-\sigma}, \quad Y_{m,\sigma} = - \sum_p U_p (\log p)^{m+1} p^{-\sigma}.$$

*The pair  $(X_{m,\sigma}, Y_{m,\sigma})$  has a  $C^\infty$  compactly supported density  $q_{m,\sigma}$ , locally uniformly with all derivatives for  $\sigma > 1$ . If*

$$G_m(\sigma, a) = \int_{\mathbb{C}} |y|^2 q_{m,\sigma}(a, y) dA(y),$$

*then the phase-annulus projection, the Jessen identity*

$$J''_{m,a}(\sigma) = 2\pi G_m(\sigma, a),$$

*the horizontal counting law, asymptotic simplicity, and the marked simple-point law of Sections 2–3 all hold for  $\mathcal{P}_{m,\omega}$ , for each fixed  $m$ . All assertions are locally uniform in  $\sigma$ , but no uniformity in  $m \rightarrow \infty$  is asserted.*

*Proof.* For every  $j \geq 0$  and  $\delta > 0$ ,

$$\sum_p (\log p)^{m+j} p^{-1-\delta} < \infty.$$

Hence the series and every fixed  $s$ -derivative converge locally uniformly in  $\Re s > 1$ . This is precisely the tail input used in the finite-prime Kronecker–Weyl argument and proves the phase-annulus projection for the present weights.

The joint characteristic function is

$$\prod_p J_0((\log p)^m p^{-\sigma} |u - (\log p)v|).$$

The fixed-pair argument of Lemma 3.1 supplies arbitrary polynomial decay. Differentiation in  $u, v$ , or  $\sigma$  inserts only further powers of  $\log p$ , which are absorbed by the displayed locally uniform summability. Fourier inversion therefore gives the asserted smooth density. The Jessen and Kac–Rice arguments then apply without change.

For completeness, the simplicity proof also survives verbatim. The weighted translation hull is compact in the compact-open topology. After conditioning on all phases except the phase at a prime  $q$ , a multiple  $a$ -point must satisfy

$$H'(s) + (\log q)(H(s) - a) = 0,$$

and uniqueness of absolutely convergent Dirichlet series shows that this function is not identically zero. The conditional exceptional phase set is therefore finite on compact boxes and Haar-null. The compact-hull zero bound, unique ergodicity, and the marked Kac–Rice argument from Section 3 complete the transfer.  $\square$

Lemma 6.1 and Proposition 2.1 give the exact global law

$$\overline{\{\Re s : \mathcal{P}_{m,\omega}(s) = a\}} = \{\sigma > 1 : R_m(\sigma) \leq |a| \leq A_m(\sigma)\}. \quad (6.4)$$

For  $m \geq 1$ , the largest term can change with  $\sigma$ ; (6.4), rather than a single interval slogan, is the safe global statement.

Set

$$D_m(\sigma) = (\log 2)^m 2^{-\sigma} - \sum_{p \geq 3} (\log p)^m p^{-\sigma}. \quad (6.5)$$

**Theorem 6.2** (Derivative zero-edge tower). *For every  $m \geq 0$ , there is a unique  $\sigma_m > 1$  such that*

$$\boxed{\sum_{p \geq 3} \left(\frac{\log p}{\log 2}\right)^m \left(\frac{2}{p}\right)^{\sigma_m} = 1.} \quad (6.6)$$

*It is the exact right endpoint of every unimodular derivative twist:*

$$\sup \overline{\{\Re s : \mathcal{P}_{m,\omega}(s) = 0, \Re s > 1\}} = \sigma_m. \quad (6.7)$$

*The untwisted  $\mathcal{P}_{m,1}$  has no actual zero on  $\Re s = \sigma_m$ , although zero real parts approach it. An adversarial twist can attain the boundary, so the universal zero-free half-plane is  $\Re s > \sigma_m$ .*

*The dominance functions satisfy*

$$\boxed{D'_m(\sigma) = -D_{m+1}(\sigma),} \quad (6.8)$$

and

$$D_m(\sigma) < 0 \quad (1 < \sigma < \sigma_m), \quad D_m(\sigma) > 0 \quad (\sigma > \sigma_m). \quad (6.9)$$

*Proof.* After division by the  $p = 2$  term, the left side of (6.6) decreases strictly from  $+\infty$  to 0 as  $\sigma$  runs from  $1^+$  to infinity. This proves existence, uniqueness, and (6.9). On  $[\sigma_m, \infty)$  the  $p = 2$  term is the largest one, so  $R_m = D_m$  and (6.4) gives (6.7). Equality at the edge is attained in the hull by

putting the  $p = 2$  term opposite to every other term, and Hurwitz approximation puts the edge in the closure of actual zero real parts.

For the untwisted series, an actual boundary zero would force equality in the triangle inequality. The phases at  $p = 3, 5, 7$  would then impose rational relations between  $\log(5/3)$  and  $\log(7/3)$ , which unique factorization forbids. Termwise differentiation gives (6.8).  $\square$

The differential recursion turns a list of separate boundaries into a single geometric mechanism.

**Theorem 6.3** (Derivative–caustic correspondence). *Define the right-hand 2-dominance caustic radius*

$$\boxed{r_m^* = D_m(\sigma_{m+1})}. \quad (6.10)$$

Then  $r_m^* > 0$ , and  $D_m$  increases strictly on  $(\sigma_m, \sigma_{m+1})$ , decreases strictly on  $(\sigma_{m+1}, \infty)$ , and tends to 0 at infinity.

If  $0 < r < r_m^*$ , there are unique

$$\sigma_m < \ell_m(r) < \sigma_{m+1} < u_m(r)$$

with  $D_m(\ell_m(r)) = D_m(u_m(r)) = r$ . Let  $\alpha_m(r)$  solve  $A_m(\alpha_m(r)) = r$ . Then, for every  $|a| = r$ ,

$$\boxed{\{\Re s : \mathcal{P}_{m,\omega}(s) = a\} \cap [\sigma_m, \infty) = [\sigma_m, \ell_m(r)] \cup [u_m(r), \alpha_m(r)]}. \quad (6.11)$$

At  $r = r_m^*$  the two positive-density bands pinch at  $\sigma_{m+1}$ ; the density vanishes at the pinch. If  $r_m^* < r < A_m(\sigma_m)$ , the terminal projection is the single interval  $[\sigma_m, \alpha_m(r)]$ . Equality  $r = A_m(\sigma_m)$  leaves only  $\{\sigma_m\}$ , and larger  $r$  leaves no point in  $[\sigma_m, \infty)$ .

The pinch is quadratic. For  $a = r_m^* e^{i\phi}$ , the hull function

$$H_{m,\phi}(s) = e^{i\phi} \left( (\log 2)^m 2^{-s} - \sum_{p \geq 3} (\log p)^m p^{-s} \right)$$

satisfies

$$H_{m,\phi}(\sigma_{m+1}) = a, \quad H'_{m,\phi}(\sigma_{m+1}) = 0, \quad H''_{m,\phi}(\sigma_{m+1}) \neq 0. \quad (6.12)$$

Equivalently,

$$\boxed{\text{the next derivative zero edge is the current derivative's right-hand } a\text{-point caustic.}} \quad (6.13)$$

*Proof.* Equations (6.8)–(6.9) give the asserted monotonicity and unique maximum. On the terminal branch, (6.4) reduces the  $a$ -point condition to  $D_m(\sigma) \leq r \leq A_m(\sigma)$ , which yields (6.11) and all endpoint cases. Finally,

$$H'_{m,\phi} = -e^{i\phi} D_{m+1}, \quad H''_{m,\phi} = e^{i\phi} D_{m+2}.$$

At  $\sigma_{m+1}$  the first expression is zero, whereas the second is nonzero because  $\sigma_{m+1} < \sigma_{m+2}$ .  $\square$

## 6.1 Internal dominance bands and the crystalline limit

The terminal 2-band is only one part of the derivative geometry. We now classify all internal bands. Besides closing the internal-band problem posed in Version 2.0, the classification will give exact masses and barycentres for every bounded zero component. For a prime  $p$ , put

$$g_{m,p}(\sigma) := 2a_{m,p}(\sigma) - A_m(\sigma) = a_{m,p}(\sigma) - \sum_{q \neq p} a_{m,q}(\sigma), \quad (6.14)$$



and, for  $r \geq 0$ , define

$$B_{m,p}(r) := \{\sigma > 1 : g_{m,p}(\sigma) > r\}, \quad B_{m,p} := B_{m,p}(0), \quad (6.15)$$

$$\mathfrak{D}_m := \{p : B_{m,p} \neq \emptyset\}, \quad m_p := \min\{m \geq 0 : p \in \mathfrak{D}_m\}. \quad (6.16)$$

Thus  $R_m(\sigma) = \max\{0, \max_p g_{m,p}(\sigma)\}$ . We use the phase-independent density  $G_m(\sigma, a)$  from Lemma 6.1. For  $p < q$  write

$$\tau_{p,q} := \frac{\log(\log q / \log p)}{\log(q/p)}, \quad \sigma_{p,q}^{(m)} := m\tau_{p,q}. \quad (6.17)$$

If  $p \geq 3$  has neighbouring primes  $p^- < p < p^+$ , let

$$W_p = (\tau_{p,p^+}, \tau_{p^-,p}), \quad W_2 = (\tau_{2,3}, \infty). \quad (6.18)$$

Strict concavity of  $u \mapsto \log u$  shows that these windows are nonempty and tile  $(0, \infty)$  up to their endpoints. For later use put

$$\phi_{p,q}(\tau) = \log \frac{\log q}{\log p} - \tau \log \frac{q}{p}.$$

**Theorem 6.4** (Complete dominance-band classification). *The following assertions hold.*

- (i) *For every  $m \geq 0$ , prime  $p$ , and  $r \geq 0$ , the set  $B_{m,p}(r)$  is an open interval, possibly empty.*
- (ii) *If  $m \geq 1$ , then  $B_{m,p} \subset mW_p$ . If  $p < q$  both belong to  $\mathfrak{D}_m$ , then*

$$\sup B_{m,q} < \sigma_{p,q}^{(m)} < \inf B_{m,p}. \quad (6.19)$$

*Consequently the nonempty bands are pairwise disjoint and are ordered, from left to right, by decreasing prime.*

- (iii) *The set  $\mathfrak{D}_m$  is finite, contains 2, and*

$$B_{m,2} = (\sigma_m, \infty).$$

*Every  $p \in \mathfrak{D}_m \setminus \{2\}$  satisfies*

$$\frac{p}{(\log p)^m} < \frac{2}{(\log 2)^m}. \quad (6.20)$$

- (iv) *For each unimodular twist define the twist-dependent closed projection*

$$Z_{m,\omega} := \overline{\{\Re s : \mathcal{P}_{m,\omega}(s) = 0, \Re s > 1\}}.$$

*Then the closure, although not the underlying set of actual zero real parts, is independent of  $\omega$  and is given exactly by*

$$\boxed{Z_{m,\omega} = (1, \sigma_m] \setminus \bigcup_{p \in \mathfrak{D}_m \setminus \{2\}} B_{m,p}.} \quad (6.21)$$

*Each  $B_{m,p}$  is a zero-free vertical strip for every unimodular twist. For the untwisted derivative its closed band  $\overline{B_{m,p}}$  is zero-free. At every interior point of  $Z_{m,\omega}$  the zero density  $G_m(\sigma, 0)$  is positive.*

*Proof.* Multiplication by  $p^\sigma$  gives

$$p^\sigma (g_{m,p}(\sigma) - r) = (\log p)^m - \sum_{q \neq p} (\log q)^m e^{-\sigma \log(q/p)} - r e^{\sigma \log p}. \quad (6.22)$$

The right side is strictly concave; hence its positive set is an open interval. Dominance by  $p$  implies dominance over each individual term. Comparison with the two neighbouring terms and monotonicity of secant slopes of  $\log$  give  $B_{m,p} \subset mW_p$  for  $m \geq 1$  and (6.19). Comparison with the  $p = 2$  term gives (6.20), hence finiteness. The assertion for  $p = 2$  is Theorem 6.2.

Finally, (6.4) at  $a = 0$  says that the projection closure is  $\{\sigma > 1 : \max_p g_{m,p}(\sigma) \leq 0\}$ , which is (6.21). The triangle inequality gives zero-freeness in an open band. Equality on a band endpoint for the untwisted series would force simultaneous phase alignment for at least three distinct primes, contradicting unique factorization. The smooth-density theorem, applied to the weights  $(\log p)^m$ , gives positivity at every strict interior point.  $\square$

The internal bands persist once born, and every prime eventually appears.

The following explicit estimate is useful both for proof and for certified computation. For  $p \geq 3$  set

$$\begin{aligned} r_+ &= \log \frac{p^+}{p}, & r_- &= \log \frac{p}{p^-}, & c_+ &= \log \frac{\log p^+}{\log p}, & c_- &= \log \frac{\log p}{\log p^-}, \\ \tau_p^* &= \frac{c_+ + c_-}{r_+ + r_-}, & \phi_p^* &= \frac{c_+ r_- - c_- r_+}{r_+ + r_-} < 0, & \rho_p &= \frac{r_+}{r_-}, \end{aligned}$$

and

$$\mathcal{K}(\rho) = (1 + \rho) \rho^{-\rho/(1+\rho)} \in (1, 2].$$

Also put

$$\begin{aligned} \delta_+ &= \tau_p^* - \frac{1}{\log p^+} > 0, & \delta_- &= \frac{1}{\log p^-} - \tau_p^* > 0, \\ T_+(m) &= \frac{p^+}{m\delta_+ - 1} \left( \frac{p^+}{p^+ + 1} \right)^{m\delta_+ - 1} \quad (m\delta_+ > 1), & T_-(m) &= \frac{p^-}{m\delta_- + 1}, \end{aligned}$$

with  $T_+(m) = \infty$  when  $m\delta_+ \leq 1$ .

**Lemma 6.5** (Explicit birth bounds). *For every prime  $p \geq 3$ ,*

$$\boxed{\frac{\log \mathcal{K}(\rho_p)}{|\phi_p^*|} < m_p \leq \min \left\{ m \geq 1 : m|\phi_p^*| > \log(2 + T_+(m) + T_-(m)) \right\}.} \quad (6.22a)$$

*The lower bound remains valid if  $(p^-, p^+)$  is replaced by any pair of primes  $q_1 < p < q_2$ , with the corresponding parameters.*

*Proof.* At  $\tau = \tau_p^* + s$  the two neighbouring exponentials equal

$$e^{m\phi_p^*} (e^{-mr+s} + e^{mr-s}).$$

Their minimum over  $s \in \mathbb{R}$  is  $\mathcal{K}(\rho_p) e^{m\phi_p^*}$ . Dominance forces this number to be  $< 1$ , proving the lower bound.

For the converse, evaluate (6.26) at  $\tau_p^*$ . Concavity of  $u \mapsto \log u$  gives

$$\sum_{q > p^+} e^{m\phi_{p,q}(\tau_p^*)} \leq e^{m\phi_p^*} T_+(m), \quad \sum_{q < p^-} e^{m\phi_{p,q}(\tau_p^*)} \leq e^{m\phi_p^*} T_-(m),$$

where the displayed expressions are elementary integral bounds for the two tails. Adding the two neighbouring terms gives

$$\sum_{q \neq p} e^{m\phi_{p,q}(\tau_p^*)} \leq e^{m\phi_p^*} (2 + T_+(m) + T_-(m)).$$

The right side is  $< 1$  under the condition in (6.22a), and hence  $p \in \mathfrak{D}_m$ .  $\square$

**Theorem 6.6** (Monotonicity, birth, and the prime-gap scale). *For every  $m \geq 0$  and every prime  $p$ ,*

$$\boxed{B_{m,p} + \frac{1}{\log p} \subset B_{m+1,p}.} \quad (6.23)$$

Thus  $\mathfrak{D}_m \subset \mathfrak{D}_{m+1}$ , every  $m_p$  is finite, and  $p \in \mathfrak{D}_m$  if and only if  $m \geq m_p$ .

For  $p \geq 3$  let  $g_- = p - p^-$ ,  $g_+ = p^+ - p$ , and  $g_{\max} = \max(g_-, g_+)$ . As  $p \rightarrow \infty$ ,

$$\left(\frac{p \log p}{g_{\max}}\right)^2 (1 - o(1)) \leq m_p \ll \log(2 + 2g_{\max}) \frac{p^2 \log^2 p}{g_- g_+}. \quad (6.24)$$

In particular, Cramér's gap conjecture would imply  $m_p = p^{2+o(1)}$ . Unconditionally the normalized quantity

$$Y_p := \frac{m_p}{p^2 \log^2 p}$$

satisfies, along every sequence  $p_j \rightarrow \infty$ ,

$$Y_{p_j} \rightarrow 0 \iff \max\{g_-(p_j), g_+(p_j)\} \rightarrow \infty.$$

In particular,

$$\liminf_{p \rightarrow \infty} Y_p = 0 < \limsup_{p \rightarrow \infty} Y_p < \infty. \quad (6.25)$$

More precisely,  $\limsup Y_p > 0$  is equivalent to  $\liminf_n (p_{n+2} - p_n) < \infty$ . The latter holds by Maynard's theorem on infinitely many bounded clusters containing three primes [16]. No assertion that  $m_p g_- g_+ / (p^2 \log^2 p)$  is unbounded is made.

*Proof.* Let  $\sigma' = \sigma + 1/\log p$ . The function  $u \mapsto u e^{-u/\log p}$  is uniquely maximized at  $u = \log p$ . Termwise comparison at  $u = \log q$  therefore proves (6.23). For  $\tau \in W_p$ , write

$$\frac{\sum_{q \neq p} a_{m,q}(m\tau)}{a_{m,p}(m\tau)} = \sum_{q \neq p} \exp\{m\phi_{p,q}(\tau)\}. \quad (6.26)$$

Every exponent is negative on  $W_p$ , and dominated convergence makes the sum tend to zero; this proves finite birth.

For completeness, put  $L = \log p$ . Uniformly for the two neighbouring gaps, the prime number theorem and Taylor expansion give

$$\phi_p^* = -\frac{r_+ r_-}{2L^2} (1 + o(1)), \quad \delta_{\pm} = \frac{r_+ + r_-}{2L^2} (1 + o(1)), \quad r_{\pm} = \frac{g_{\pm}}{p} (1 + o(1)).$$

The elementary inequality

$$2 \log \mathcal{K}(\rho) \geq \min\{\rho, \rho^{-1}\} \quad (\rho > 0)$$

and the lower half of (6.22a) therefore yield

$$m_p > \frac{\log \mathcal{K}(\rho_p)}{|\phi_p^*|} \geq (1 - o(1)) \left( \frac{pL}{g_{\max}} \right)^2.$$

For the converse choose, with a sufficiently large absolute  $C$ ,

$$m = \left\lceil C \log(2 + 2g_{\max}) \frac{p^2 L^2}{g_- g_+} \right\rceil.$$

Then

$$m|\phi_p^*| = \left( \frac{C}{2} + o(1) \right) \log(2 + 2g_{\max}), \quad T_+(m) + T_-(m) \ll \frac{g_- g_+}{(g_- + g_+) \log(2 + 2g_{\max})}.$$

Thus the upper condition in (6.22a) holds and proves (6.24). If  $g_{\max} \rightarrow \infty$ , the upper bound is  $O(\log(2 + 2g_{\max})/g_{\max})$ , because the other neighbouring gap is at least 2; hence  $Y_p \rightarrow 0$  along such a sequence. On the other hand, within each bounded three-prime cluster choose its middle prime. Both neighbouring gaps are then bounded, and the lower bound in (6.24) keeps  $Y_p$  away from zero. The upper bound in (6.24), split according as  $g_{\max}$  is bounded or tends to infinity, also bounds  $Y_p$  globally and proves the displayed sequential equivalence. A positive limsup is equivalent to bounded  $g_{\max}$  along a subsequence, which is the same as  $\liminf_n (p_{n+2} - p_n) < \infty$ . This proves (6.25).  $\square$

The differential recursion is prime-by-prime, not merely terminal. When  $B_{m,p} = (\ell_{m,p}, u_{m,p})$ , we allow  $u_{m,2} = \infty$ .

**Theorem 6.7** (Prime-by-prime internal caustics). *If  $p \in \mathfrak{D}_m$ , then  $g_{m,p}$  has a unique global maximum at  $\ell_{m+1,p}$ . With*

$$r_{m,p}^* := g_{m,p}(\ell_{m+1,p}), \quad (6.27)$$

one has  $r_{m,p}^* > 0$  and

$$g'_{m,p}(\ell_{m+1,p}) = 0, \quad g''_{m,p}(\ell_{m+1,p}) < 0. \quad (6.28)$$

For  $0 \leq r < r_{m,p}^*$  the set  $B_{m,p}(r)$  is a nonempty open interval containing  $\ell_{m+1,p}$ ; its closure shrinks to that point as  $r \uparrow r_{m,p}^*$ , and it is empty for  $r \geq r_{m,p}^*$ . Moreover,

$$\ell_{m,p} < \ell_{m+1,p} \leq \ell_{m,p} + \frac{1}{\log p}. \quad (6.29)$$

For  $p \geq 3$  one also has

$$u_{m+1,p} \geq u_{m,p} + \frac{1}{\log p}.$$

Thus every dominant prime has its own derivative-caustic correspondence: the pinch of its level- $m$   $a$ -point band is the left edge of its level- $(m+1)$  zero-free band.

*Proof.* Termwise differentiation gives

$$g'_{m,p} = -g_{m+1,p}, \quad g''_{m,p} = g_{m+2,p}. \quad (6.30)$$

Theorem 6.6 ensures that the next two  $p$ -bands exist. For  $p \geq 3$ , since  $g_{m+1,p}$  is nonpositive outside its band and positive inside,  $g_{m,p}$  increases on  $(1, \ell_{m+1,p})$ , decreases on  $(\ell_{m+1,p}, u_{m+1,p})$ , and increases again after  $u_{m+1,p}$  toward 0 from below. Because  $B_{m,p}$  is nonempty, its positive interval must therefore contain  $\ell_{m+1,p}$ , where the unique global maximum is attained; beyond  $u_{m+1,p}$  the

function cannot rival this positive value. For  $p = 2$ , the same conclusion follows from the sign change of  $D_{m+1}$  at  $\sigma_{m+1}$  and the fact that  $D_m(\sigma) \rightarrow 0^+$ . At the left endpoint of the positive set of  $g_{m+1,p}$  one has  $g'_{m+1,p}(\ell_{m+1,p}) > 0$ , so

$$g_{m+2,p}(\ell_{m+1,p}) = -g'_{m+1,p}(\ell_{m+1,p}) < 0.$$

This proves the transverse quadratic pinch. The remaining assertions follow from the interval property and (6.23).  $\square$

We next describe the high-derivative limit. Fix consecutive primes  $p < q = p^+$ , set  $d_{p,q} = \log(q/p)$  and  $\tau = \tau_{p,q}$ , and put

$$\eta_{m,p} := \sum_{r \notin \{p,q\}} \frac{a_{m,r}(m\tau)}{a_{m,p}(m\tau)}. \quad (6.31)$$

At  $m\tau$  the  $p$ - and  $q$ -terms are equal. Furthermore

$$\eta_{m,p} = \Theta_p(e^{-m\mu_p}), \quad \mu_p := \min_{r \notin \{p,q\}} \left[ -\log \frac{\log r}{\log p} + \tau \log \frac{r}{p} \right] > 0. \quad (6.32)$$

The leading constant in (6.32) is the number, one or two, of minimizers in the definition of  $\mu_p$ .

**Theorem 6.8** (Thin zero windows and the picket-fence limit). *For each fixed consecutive pair  $p < q = p^+$ , as  $m \rightarrow \infty$ ,*

$$\ell_{m,p} - m\tau_{p,q} = (1 + o(1)) \frac{\eta_{m,p}}{d_{p,q}}, \quad m\tau_{p,q} - u_{m,q} = (1 + o(1)) \frac{\eta_{m,p}}{d_{p,q}}. \quad (6.33)$$

*In particular every band fills its window:*

$$\frac{\ell_{m,p}}{m} \longrightarrow \tau_{p,p^+}, \quad \frac{u_{m,p}}{m} \longrightarrow \tau_{p^-,p} \quad (p \geq 3). \quad (6.34)$$

*Locally on  $(0, \infty)$  the closed sets  $m^{-1}Z_{m,\omega}$  converge in the Fell topology to the crossover set. Equivalently, for every compact interval  $K \Subset (0, \infty)$  whose boundary contains no crossover point,*

$$(m^{-1}Z_{m,\omega}) \cap K \longrightarrow \{\tau_{p,p^+} : p \in \mathbb{P}\} \cap K \quad (6.35)$$

*in Hausdorff distance, uniformly in the choice of unimodular twist. The component near  $m\tau_{p,p^+}$  has width*

$$(2 + o(1)) \frac{\eta_{m,p}}{\log(p^+/p)}. \quad (6.36)$$

*Proof.* Divide the dominance inequality by the  $p$ -term and expand at  $\sigma = m\tau_{p,q}$ . The  $q/p$  ratio contributes  $e^{-d_{p,q}s} = 1 - d_{p,q}s + O(s^2)$ , while the sum of all remaining ratios is  $\eta_{m,p}(1 + o(1))$  uniformly for  $|s| = O(\eta_{m,p})$ . Solving the two endpoint equations proves (6.33). Formula (6.34) follows from the window inclusion. On a compact  $K \Subset (0, \infty)$  only finitely many windows occur; every associated prime has been born for all sufficiently large  $m$ . Equations (6.21) and (6.33) then give (6.35)–(6.36).  $\square$

The limiting set alone forgets how much divisor mass lies at each picket. The Jessen law restores that information exactly. Let  $J_{m,a}$  denote the analogue of (3.7) for  $\mathcal{P}_{m,\omega}$ , retaining the notation  $G_m$  fixed above. If  $q > p$  are adjacent members of  $\mathfrak{D}_m$ , write

$$C_{m;q,p} := [u_{m,q}, \ell_{m,p}] \quad (6.37)$$

for the bounded zero-projection component between their bands.

**Theorem 6.9** (Exact component mass, barycentre, and local recovery). *For every adjacent dominant pair  $p < q$ ,*

$$\boxed{\int_{C_{m;q,p}} G_m(\sigma, 0) d\sigma = \frac{1}{2\pi} \log \frac{q}{p}}, \quad (6.38)$$

$$\boxed{\int_{C_{m;q,p}} \sigma G_m(\sigma, 0) d\sigma = \frac{m}{2\pi} \log \frac{\log q}{\log p}}. \quad (6.39)$$

*Consequently its density barycentre is exactly  $m\tau_{p,q}$ , and for each of the multiplicity, distinct, and simple counting conventions, with  $m$  and the component fixed before  $T \rightarrow \infty$ ,*

$$N_{m;q,p}^\bullet(T) = \frac{\log(q/p)}{2\pi} T + o(T). \quad (6.40)$$

*Conversely, this one component recovers its two boundary primes. If*

$$D = 2\pi \int_C G_m(\sigma, 0) d\sigma, \quad E = \frac{2\pi}{m} \int_C \sigma G_m(\sigma, 0) d\sigma,$$

*then*

$$\boxed{p = \exp\left(\frac{D}{e^E - 1}\right), \quad q = pe^D}. \quad (6.41)$$

*Proof.* On a  $p$ -dominance band, Jensen's formula gives

$$J_{m,0}(\sigma) = \log a_{m,p}(\sigma) = m \log \log p - \sigma \log p,$$

so  $J'_{m,0} = -\log p$ . Integrating  $J''_{m,0} = 2\pi G_m(\sigma, 0)$  across (6.37) proves (6.38). One integration by parts gives (6.39). Asymptotic simplicity gives (6.40). Finally  $D = \log(q/p)$  and  $E = \log(\log q / \log p)$ , from which (6.41) follows algebraically.  $\square$

**Theorem 6.10** (All-target topology and caustic conservation). *Fix  $m \geq 1$  and  $r = |a| > 0$ , and let  $\alpha_m(r)$  be the unique solution of  $A_m(\alpha_m(r)) = r$ . Put*

$$\mathfrak{D}_m(r) = \{p \in \mathfrak{D}_m : r < r_{m,p}^*\}.$$

*Then*

$$\boxed{\{\Re s : \overline{\mathcal{P}_{m,\omega}(s) = a, \Re s > 1}\} = (1, \alpha_m(r)] \setminus \bigcup_{p \in \mathfrak{D}_m(r)} B_{m,p}(r)}, \quad (6.42)$$

*and the number of connected components of this closed support is*

$$\boxed{b_0(m, r) = 1 + \#\mathfrak{D}_m(r)}. \quad (6.43)$$

*At  $r = 0$  the corresponding number is  $\#\mathfrak{D}_m$ .*

*If  $q > p$  label consecutive active bands from left to right and  $C$  is the bounded component between them, then the two formulas (6.38)–(6.39), and hence the recovery formula (6.41), remain valid with  $G_m(\sigma, 0)$  replaced by  $G_m(\sigma, a)$ . If the 2-band is active, the terminal component between its right endpoint and  $\alpha_m(r)$  has*

$$\int_C G_m(\sigma, a) d\sigma = \frac{\log 2}{2\pi}, \quad \int_C \sigma G_m(\sigma, a) d\sigma = \frac{m \log \log 2 - \log r}{2\pi}. \quad (6.44)$$

*As  $r$  crosses a caustic value  $r_{m,p}^*$ , the  $p$ -band disappears and the two neighbouring components merge. Their zeroth and first moments add exactly to those of the merged component. The same telescoping law holds for simultaneous caustics.*

*Proof.* Equation (6.42) is the phase-annulus criterion (6.4): the outer inequality is  $r \leq A_m(\sigma)$  and every failure of the inner inequality is precisely one of the disjoint sets  $B_{m,p}(r)$ . Removing  $k$  disjoint open intervals from  $(1, \alpha_m(r)]$  gives  $k + 1$  components, proving (6.43). At the critical radius the strict superlevel set is empty and its two endpoints have coalesced; the support closure has therefore merged through a density-zero pinch.

On  $B_{m,p}(r)$ , conditioning on all phases except  $U_p$  and using Jensen's circle formula gives the same affine plateau as at target zero:

$$J_{m,a}(\sigma) = m \log \log p - \sigma \log p, \quad J'_{m,a}(\sigma) = -\log p.$$

For  $A_m(\sigma) < r$ , the same formula after a common Haar rotation gives  $J_{m,a}(\sigma) = \log r$  and  $J'_{m,a}(\sigma) = 0$ . Integrating  $J''_{m,a} = 2\pi G_m(\sigma, a)$  once and then by parts proves all moment formulas. The merger laws reduce to  $\log(q/p) + \log(p/h) = \log(q/h)$  and the analogous identity for logarithms of logarithms.  $\square$

The component masses have a useful scaling limit. Define

$$\nu_m := (\sigma \mapsto \sigma/m)_*(G_m(\sigma, 0) d\sigma).$$

Then, vaguely on  $(0, \infty)$ ,

$$\nu_m \Rightarrow \frac{1}{2\pi} \sum_p \log \frac{p^+}{p} \delta_{\tau_{p,p^+}}. \quad (6.45)$$

An atom  $(\tau, w)$  of the limiting measure recovers its consecutive prime pair by setting  $D = 2\pi w$  and using

$$p = \exp\left(\frac{D}{e^{\tau D} - 1}\right), \quad p^+ = pe^D. \quad (6.46)$$

Moreover, if  $w_p = (2\pi)^{-1} \log(p^+/p)$ , then

$$\sum_{p \leq p_N} w_p = \frac{\log p_{N+1} - \log 2}{2\pi}, \quad \sum_{p \leq p_N} \tau_{p,p^+} w_p = \frac{\log \log p_{N+1} - \log \log 2}{2\pi}. \quad (6.47)$$

These exact telescoping laws show that the picket measure is a locally finite, lossless encoding of the consecutive-prime chain, not a source of information unavailable from all other edge data.

Indeed, on a compact subset of  $(0, \infty)$  only finitely many prime windows occur. Every associated prime has been born for all sufficiently large  $m$ ; (6.33) collapses the intervening component to its crossover, while (6.38) keeps its total mass exactly equal to the displayed atomic weight. Testing against a compactly supported continuous function proves the vague convergence in (6.45).

*Remark 6.11* (Nonrigorous numerical orientation). Floating-point minimization suggests, for example, the initial birth pattern  $m_3 = 10, m_5 = 22, m_7 = 44, m_{11} = 103, \dots$ . Such values are not used above. Ordinary double-precision optimization, including floating-point tail estimates, is not an exact certificate. Any numerical table promoted to theorem status must instead be verified by directed interval arithmetic with archived code and data.

## 7 Convex edge interpolation and inverse prime spectroscopy

Interpolate the integer derivative order by the unique solution  $\sigma(u)$ ,  $u \geq 0$ , of

$$\sum_{p \geq 3} \left(\frac{\log p}{\log 2}\right)^u \left(\frac{2}{p}\right)^{\sigma(u)} = 1. \quad (7.1)$$

Thus  $\sigma(m) = \sigma_m$ .

**Proposition 7.1** (Strictly convex edge curve). *The function  $\sigma(u)$  is real analytic, strictly increasing, and strictly convex. If*

$$x_p = \log \frac{\log p}{\log 2}, \quad y_p = \log \frac{p}{2}, \quad w_p = e^{ux_p - \sigma(u)y_p},$$

then  $\sum_{p \geq 3} w_p = 1$  and

$$\sigma'(u) = \frac{\sum_p w_p x_p}{\sum_p w_p y_p} > 0, \quad (7.2)$$

$$\sigma''(u) = \frac{\sum_p w_p (x_p - \sigma'(u)y_p)^2}{\sum_p w_p y_p} > 0. \quad (7.3)$$

*Proof.* The implicit-function theorem applies to (7.1). First and second implicit differentiation give (7.2)–(7.3). Strictness follows because  $x_p/y_p$  is not constant in  $p$ .  $\square$

Set

$$L = \log \frac{3}{2}, \quad \kappa = \frac{\log(\log 3 / \log 2)}{\log(3/2)} = 1.1358825679163036 \dots, \quad (7.4)$$

and, for  $p \geq 3$ ,

$$q_p = \frac{\log p}{\log 3} \left( \frac{3}{p} \right)^\kappa. \quad (7.5)$$

Then  $q_3 = 1$ ,  $q_p$  decreases strictly with  $p$ , and

$$q_5 = 0.8200411528584832 \dots, \quad q_7 = 0.6765500255651028 \dots. \quad (7.6)$$

For orientation, direct prime summation gives the following initial values; the digits shown are stable under enlargement of the prime cutoff.

| $m$ | $\sigma_m$ | $r_m^*$    |
|-----|------------|------------|
| 0   | 1.77954465 | 0.08048045 |
| 1   | 2.60762384 | 0.02861372 |
| 2   | 3.50440655 | 0.00995714 |
| 3   | 4.44616892 | 0.00340929 |
| 4   | 5.42063867 | 0.00115243 |
| 5   | 6.42040617 | 0.00038547 |

(7.6a)

**Theorem 7.2** (Asymptotic edge law). *As  $u \rightarrow \infty$ ,*

$$\sigma(u) = \kappa u + \frac{q_5^u}{L} + O(q_7^u). \quad (7.7)$$

*In particular,*

$$0 < \sigma_{m+1} - \sigma_m < \sigma_{m+2} - \sigma_{m+1} < \kappa, \quad \sigma_{m+1} - \sigma_m \longrightarrow \kappa. \quad (7.8)$$

*Furthermore,*

$$r_m^* = CB^m(1 + O(q_5^m)), \quad (7.9)$$

where

$$B = (\log 2)2^{-\kappa} = 0.3154210960683259 \dots, \quad C = 2^{-\kappa} \left( 1 - \frac{\log 2}{\log 3} \right) = 0.1679477965425128 \dots. \quad (7.10)$$



*Proof.* The function

$$q(x) = \frac{\log x}{\log 3} \left( \frac{3}{x} \right)^\kappa$$

is strictly decreasing for  $x \geq 3$ , because  $(\log q)' = (1/\log x - \kappa)/x < 0$ . Put  $h(u) = \sigma(u) - \kappa u$ . Separating the  $p = 3$  term in (7.1) gives the exact identity

$$e^{Lh(u)} - 1 = \sum_{p \geq 5} q_p^u \left( \frac{3}{p} \right)^{h(u)}. \quad (7.11)$$

It follows first that  $h(u) > 0$  and  $h(u) = O(q_5^u)$ . Separating  $p = 5$  gives

$$e^{Lh(u)} - 1 = q_5^u + O(q_5^{2u} + q_7^u).$$

Since  $q_5^2 < q_7$ , expansion of the logarithm proves (7.7). Formula (7.2) is a weighted average of  $x_p/y_p$ , whose strict maximum is the  $p = 3$  value  $\kappa$ . Together with strict convexity this proves (7.8). Substitution of  $\sigma_{m+1} = \kappa(m+1) + O(q_5^m)$  into  $D_m(\sigma_{m+1})$  gives (7.9)–(7.10).  $\square$

The small error in (7.7) successively records 5, 7, 11,  $\dots$ , not merely the first correction.

**Theorem 7.3** (All primes from all derivative zero edges). *Given the exact sequence  $(\sigma_m)_{m \geq 0}$  and the anchors 2, 3, every prime is recovered recursively. Conversely, the prime sequence determines every  $\sigma_m$ . Thus*

$$\boxed{\{\text{complete derivative zero-edge tower}\}} \iff \boxed{\{\text{complete prime spectrum}\}}. \quad (7.12)$$

*Proof.* The forward construction from primes to edges is (6.6). For the inverse direction, put

$$h_m = \sigma_m - \kappa m, \quad E_m = e^{Lh_m} - 1.$$

Equation (7.11) is exact:

$$E_m = \sum_{p \geq 5} q_p^m \left( \frac{3}{p} \right)^{h_m}. \quad (7.13)$$

Since  $h_m \rightarrow 0$  and  $q_p$  decreases strictly,

$$q_5 = \lim_{m \rightarrow \infty} E_m^{1/m}. \quad (7.14)$$

The strictly decreasing real function  $q(x)$  then identifies 5. If the primes 5, 7,  $\dots$ ,  $p_j$  have been recovered, subtract their exact terms and put

$$E_m^{(j)} = E_m - \sum_{5 \leq p \leq p_j} q_p^m \left( \frac{3}{p} \right)^{h_m}.$$

Then

$$q_{p_{j+1}} = \lim_{m \rightarrow \infty} (E_m^{(j)})^{1/m}. \quad (7.15)$$

The upper bound follows by fixing  $m_0 \geq 1$  and using

$$E_m^{(j)} \leq q_{p_{j+1}}^{m-m_0} \sum_{p > p_j} q_p^{m_0},$$

whose last sum is finite. Induction recovers all primes.  $\square$

## 8 Equivalent encodings of the zero-density law

This section records what else closes once the exact density theorem is known. The statements are not new zero estimates; they are exact Tauberian, moment, and operator encodings of the same measure.

### 8.1 Regular variation and the logical strength of the BCS statements

For fixed  $\gamma$ , the BCS height assertion

$$\frac{M_\gamma(xT)}{M_\gamma(T)} \longrightarrow x \quad (0 < x < 1) \quad (8.1)$$

is exactly regular variation of index one:  $M_\gamma(T) = T\ell_\gamma(T)$  with  $\ell_\gamma$  slowly varying. Their separate two-sided linear bound says  $M_\gamma(T) = \Theta(T)$ . Even together these assertions do not force  $\ell_\gamma(T)$  to converge. Theorem 4.1 supplies the strictly stronger conclusion

$$\ell_\gamma(T) \longrightarrow c_\gamma. \quad (8.2)$$

Now fix a compact interval  $I \Subset (1, \sigma_e)$  and put

$$\kappa_I = \int_I \rho(\sigma) d\sigma > 0. \quad (8.3)$$

List the positive ordinates of the simple zeros whose real parts lie in  $I$ :

$$0 < \tau_1 \leq \tau_2 \leq \cdots,$$

discarding a finite initial set if convenient. Theorem 3.3 shows that this choice has the same Weyl coefficient as either other counting convention. Let  $D_I e_n = \tau_n e_n$ .

**Theorem 8.1** (Weyl–heat equivalence and its consequences). *The following three assertions are equivalent:*

$$N_I(T) \sim \kappa_I T, \quad (8.4)$$

$$\tau_n \sim \frac{n}{\kappa_I}, \quad (8.5)$$

$$\mathrm{Tr}(e^{-tD_I}) \sim \frac{\kappa_I}{t} \quad (t \downarrow 0). \quad (8.6)$$

They hold for the prime-zeta divisor by Theorem 4.1. They imply

$$\lim_{s \downarrow 1} (s-1) \mathrm{Tr}(D_I^{-s}) = \kappa_I, \quad (8.7)$$

$$\sum_{\tau_n \leq T} \frac{1}{\tau_n} = \kappa_I \log T + o(\log T), \quad (8.8)$$

and

$$D_I^{-1} \in \mathcal{L}^{1,\infty} \setminus \mathcal{S}_1, \quad \mathrm{Tr}_\Omega(D_I^{-1}) = \kappa_I \quad (8.9)$$

for every Dixmier trace  $\mathrm{Tr}_\Omega$ . Furthermore,

$$\mathcal{S}_I(z) = \det(I - z^2 D_I^{-2}) = \prod_n \left(1 - \frac{z^2}{\tau_n^2}\right) \quad (8.10)$$

is an entire function of order one and exponential type  $\pi\kappa_I$ , and

$$\log \det(I + x^2 D_I^{-2}) \sim \pi\kappa_I x. \quad (8.11)$$

Thus

$$\boxed{\kappa_I = \frac{1}{2\pi} \int_I J_0''(\sigma) d\sigma = \lim_{T \rightarrow \infty} \frac{N_I(T)}{T} = \lim_{t \downarrow 0} t \operatorname{Tr}(e^{-tD_I}) = \operatorname{Tr}_\Omega(D_I^{-1}) = \frac{1}{\pi} \operatorname{type} \mathcal{S}_I.} \quad (8.12)$$

*Proof.* The equivalence of (8.4) and (8.5) follows by inversion of a monotone counting function. The equivalence of (8.4) and (8.6) is Karamata's Laplace–Stieltjes theorem [3]. Equations (8.7)–(8.8) follow by Abelian partial summation. Relation (8.5) gives  $s_n(D_I^{-1}) \sim \kappa_I/n$ , proving measurability and (8.9); see [8, 19] for the operator-ideal background.

Finally,  $D_I^{-2}$  is trace class. Partial summation of  $\sum_n \log(1 + x^2/\tau_n^2)$  against  $N_I$  gives

$$\kappa_I \int_0^\infty \log(1 + x^2/t^2) dt = \pi\kappa_I x + o(x),$$

which proves (8.11); the standard canonical-product estimate [14] gives the stated type.  $\square$

*Remark 8.2* (No false converses). A real-axis limit such as (8.7), a logarithmic harmonic asymptotic, or a single Dixmier trace generally records only an averaged density and does not by itself imply the pointwise Weyl law (8.4). Only the Weyl–heat equivalence is invoked in both directions without extra regular-variation assumptions.

The two-variable limit (4.8) has the operator form

$$\frac{1}{\Lambda} \operatorname{Tr}[g(B)f(D/\Lambda)] \longrightarrow \left( \int g(\sigma)\rho(\sigma) d\sigma \right) \left( \int_0^\infty f(u) du \right), \quad (8.13)$$

where  $B$  is the diagonal operator of zero real parts,  $D$  that of positive heights,  $g$  is continuous with compact support in  $(1, \sigma_e)$ , and  $f \in C_c([0, \infty))$ . Hence the Jessen measure is also a heat-trace tomography:

$$\lim_{t \downarrow 0} t \operatorname{Tr}[g(B)e^{-tD}] = \int g(\sigma)\rho(\sigma) d\sigma. \quad (8.14)$$

## 8.2 Moment tomography and the infinitely soft edge

Put

$$\mu_n = \int_1^{\sigma_e} \sigma^n \rho(\sigma) d\sigma. \quad (8.15)$$

Since  $\rho$  is positive on an interval, its Hankel matrices are strictly positive:

$$\sum_{i,j=0}^N c_i \overline{c_j} \mu_{i+j} > 0 \quad ((c_0, \dots, c_N) \neq 0). \quad (8.16)$$

In particular,

$$\mu_n^2 < \mu_{n-1} \mu_{n+1}, \quad \lim_{n \rightarrow \infty} \mu_n^{1/n} = \lim_{n \rightarrow \infty} \frac{\mu_{n+1}}{\mu_n} = \sigma_e. \quad (8.17)$$

Thus the edge can be recovered from the moment sequence alone.

The centered Jessen excess

$$K(\sigma) = J_0(\sigma) + \sigma \log 2$$

obeys the exact Green formula

$$K(\sigma) = 2\pi \int_{\sigma}^{\sigma_e} (u - \sigma) \rho(u) \, du. \quad (8.18)$$

It is positive, strictly decreasing, and strictly convex before the edge, but every derivative vanishes at  $\sigma_e$ . Hence  $J_0$  is smooth but not real analytic at the edge. Equivalently, the top-edge Laplace profile

$$\Lambda(t) = \int_1^{\sigma_e} e^{-t(\sigma_e - \sigma)} \rho(\sigma) \, d\sigma \quad (8.19)$$

satisfies

$$\Lambda(t) = O_A(t^{-A}) \quad \text{for every } A > 0, \quad (8.20)$$

but is not  $O(e^{-ct})$  for any  $c > 0$ . Spectrally, the dimension in every fixed nonempty edge window remains one while its Weyl coefficient  $\kappa_{(\sigma_e - \delta, \sigma_e)}$  vanishes faster than every power of  $\delta$  as  $\delta \downarrow 0$ .

## 9 Consequences for factorization series and prime subsets

### 9.1 Fröberg-type ordered factorization series

For  $z \neq 0$ , consider

$$\mathcal{D}_z(s) = \frac{1}{1 - z\mathcal{P}(s)}. \quad (9.1)$$

In the half-plane where the geometric expansion is absolute,

$$\mathcal{D}_z(s) = \sum_{n \geq 1} \frac{A_z(n)}{n^s}, \quad A_z(n) = z^{\Omega(n)} \frac{\Omega(n)!}{\prod_p \nu_p(n)!}. \quad (9.2)$$

Thus  $A_z(n)$  counts ordered words in the prime factors of  $n$ , with weight  $z$  per letter. For  $z = 1$ , the identity, its Tauberian main term, and the resulting oscillation problem were studied by Montgomery and Tenenbaum [15]. The poles of  $\mathcal{D}_z$  are the  $1/z$ -points of  $\mathcal{P}$ .

**Corollary 9.1** (Exact pole density and sign blindness). *Put*

$$\beta_z = \mathcal{P}^{-1}(|z|^{-1}).$$

For  $1 < \gamma < \beta_z$ ,

$$\sum_{\substack{\mathcal{D}_z \text{ has a pole at } \rho \\ \Re \rho > \gamma \\ 0 < \Im \rho < T}} m_{\rho}^{\text{pole}} = T \int_{\gamma}^{\beta_z} G(\sigma, 1/z) \, d\sigma + o(T). \quad (9.3)$$

The density depends on  $z$  only through  $|z|$ . In particular, the pole divisors of  $(1 - \mathcal{P})^{-1}$  and  $(1 + \mathcal{P})^{-1}$  have identical limiting horizontal measures. Their common right edge is

$$\beta_1 = \mathcal{P}^{-1}(1) = 1.3994333287 \dots \quad (9.4)$$

*Proof.* Apply Theorem 3.2 to  $a = 1/z$  and use radially. The right edge is Theorem 5.7.  $\square$

For  $z > 0$ , the real pole at  $\beta_z$  is simple and has residue  $-1/(z\mathcal{P}'(\beta_z))$ . A Tauberian main term, when applicable, therefore has constant

$$C_z = -\frac{1}{\beta_z z \mathcal{P}'(\beta_z)}. \quad (9.5)$$

There can be no fixed power-saving remainder

$$\sum_{n \leq x} A_z(n) = C_z x^{\beta_z} + O(x^{\beta_z - \delta}) \quad (9.6)$$

for any  $\delta > 0$ : such a remainder would continue  $\mathcal{D}_z$  meromorphically to  $\Re s > \beta_z - \delta$  with no pole there except  $\beta_z$ , whereas (9.3) supplies linearly many further poles in every strip  $(\beta_z - \varepsilon, \beta_z)$ . For  $z = 1$  this recovers the obstruction in [15]; (9.3) adds the exact phase-blind pole density and asymptotic simplicity.

## 9.2 Reciprocal abscissae and character phases

Let  $\chi$  be a Dirichlet character modulo  $q$  and write

$$\mathcal{P}_\chi(s) = \sum_p \chi(p) p^{-s}, \quad \mathcal{P}_q(\sigma) = \sum_{p \nmid q} p^{-\sigma}.$$

For  $a \neq 0$ , let  $\beta_{q,a} > 1$  be the unique solution of  $\mathcal{P}_q(\beta_{q,a}) = |a|$ .

**Theorem 9.2** (Reciprocal–abscissa equivalence). *Put  $c(n) = \Omega(n)! / \prod_p \nu_p(n)!$ . In its half-plane of absolute convergence,*

$$\boxed{\frac{1}{a - \mathcal{P}_\chi(s)} = \frac{1}{a} \sum_{\substack{n \geq 1 \\ (n,q)=1}} \frac{\chi(n)c(n)}{a^{\Omega(n)} n^s}.} \quad (9.6a)$$

*The abscissa of ordinary convergence, the abscissa of absolute convergence, and the right edge of its pole set all coincide:*

$$\boxed{\sigma_c = \sigma_a = \sup\{\Re \rho : \mathcal{P}_\chi(\rho) = a\} = \beta_{q,a}.} \quad (9.6b)$$

*On each compact strip to the left of this edge, all but  $o(T)$  of the poles are simple.*

*Proof.* The multinomial theorem gives  $\mathcal{P}_\chi(s)^k = \sum_{\Omega(n)=k} \chi(n)c(n)n^{-s}$ , and the geometric series gives (9.6a). Taking absolute values gives exactly

$$\frac{1}{|a|} \sum_{k \geq 0} \left( \frac{\mathcal{P}_q(\sigma)}{|a|} \right)^k,$$

so  $\sigma_a = \beta_{q,a}$ . The triangle inequality excludes poles to its right. Immediately to its left, the target  $a$  lies in the strict interior of the value annulus: its outer radius exceeds  $|a|$ , while its inner radius is strictly smaller. The smooth  $a$ -point law supplies actual poles with real parts arbitrarily close to  $\beta_{q,a}$  from the left. If  $\sigma_c < \beta_{q,a}$ , the Dirichlet series would be holomorphic in a half-plane containing one of those poles, a contradiction. Thus  $\sigma_c = \sigma_a$ , and Theorem 3.3 gives the final claim.  $\square$

### 9.3 Ordered factorization length modulo $m$

Let  $\beta = \mathcal{P}^{-1}(1)$  and, for  $m \geq 2$  and  $0 \leq r < m$ , put

$$\mathcal{C}_{m,r}(s) = \sum_{\substack{n \geq 1 \\ \Omega(n) \equiv r \pmod{m}}} \frac{c(n)}{n^s}, \quad \Omega(1) = 0, \quad c(1) = 1.$$

**Theorem 9.3** (Modular prime-word decomposition). *For  $\omega = e^{2\pi i/m}$ ,*

$$\boxed{\mathcal{C}_{m,r}(s) = \frac{1}{m} \sum_{j=0}^{m-1} \frac{\omega^{-jr}}{1 - \omega^j \mathcal{P}(s)} = \frac{\mathcal{P}(s)^r}{1 - \mathcal{P}(s)^m}.} \quad (9.6c)$$

*All  $m$  functions have exactly the same pole divisor in  $\Re s > 1$ . For every  $I \in (1, \beta)$ , their simple, distinct, and multiplicity pole counts satisfy*

$$N_{m,r}^\bullet(I; T) = mT \int_I G(\sigma, 1) d\sigma + o(T), \quad \bullet \in \{\text{simp}, \text{dist}, \text{mult}\}. \quad (9.6d)$$

*Their summatory functions are equidistributed:*

$$\boxed{\sum_{\substack{n \leq x \\ \Omega(n) \equiv r \pmod{m}}} c(n) \sim -\frac{x^\beta}{m\beta \mathcal{P}'(\beta)}.} \quad (9.6e)$$

*Proof.* Root-of-unity filtering of  $\sum_{\Omega(n)=k} c(n)n^{-s} = \mathcal{P}(s)^k$  gives (9.6c). At a pole,  $\mathcal{P}(s)^m = 1$ , so the numerator  $\mathcal{P}(s)^r$  never vanishes; the pole divisors are therefore identical. The  $m$  distinct unit targets have the same radial density, which gives (9.6d) and asymptotic simplicity. At  $s = \beta$  the residue is  $-1/(m\mathcal{P}'(\beta))$ . Triangle equality and rational independence of the prime logarithms show that no other point on  $\Re s = \beta$  is singular. Wiener–Ikehara then proves (9.6e).  $\square$

**Corollary 9.4** (Exact cancellation and parity overtaking). *For  $r \neq r' \pmod{m}$ , exactly  $\gcd(m, r - r')$  of the  $m$  unit-target pole families cancel from  $\mathcal{C}_{m,r} - \mathcal{C}_{m,r'}$ ; the other  $m - \gcd(m, r - r')$  remain. Consequently, for every  $\varepsilon > 0$ , the difference of the two summatory functions is*

$$\Omega_\pm(x^{\beta-\varepsilon}), \quad (9.6f)$$

*and changes sign infinitely often. In particular, ordered prime words have asymptotically even and odd lengths in equal proportions, but the leading class overtakes the other infinitely many times and the error admits no fixed power saving.*

*Proof.* At a unit target  $\xi$ , cancellation is equivalent to  $\xi^{r-r'} = 1$ , giving the gcd count. The remaining nonreal poles approach  $\Re s = \beta$  arbitrarily closely by the  $a$ -point law. The Phragmén–Landau oscillation argument used in [15] now applies verbatim to the real coefficient difference and gives (9.6f).  $\square$

### 9.4 Arithmetic progressions and Dirichlet phases

Let

$$\mathcal{P}_{q,a}(s) = \sum_{p \equiv a \pmod{q}} p^{-s}, \quad (a, q) = 1. \quad (9.7)$$

The prime number theorem in arithmetic progressions supplies enough prime blocks for the smoothing proof. Consequently, the projection, density, and tomography theorems hold with the support restricted to this residue class. In particular, the nonzero  $a$ -point right-edge curve of  $\mathcal{P}_{q,a}$  reconstructs the complete prime subset  $\{p : p \equiv a \pmod{q}\}$ .

For a Dirichlet character  $\chi$ ,

$$\mathcal{P}_\chi(s) = \sum_p \chi(p) p^{-s} \quad (9.8)$$

has coefficient modulus one away from primes dividing its modulus and zero at those primes. Hence all characters omitting the same finite prime set have identical limiting  $a$ -point measures in  $\Re s > 1$ , although their individual zero ordinates can differ. All- $a$  tomography recovers precisely the omitted primes and no phase data. This gives a concrete classification of what the horizontal law can and cannot see.

There is also a field-theoretic consequence. For a number field  $K$ , let  $S_K^{\text{spl}}$  be the rational primes that split completely in  $K$  and form the prime-supported series  $\sum_{p \in S_K^{\text{spl}}} p^{-s}$ .

**Corollary 9.5** (Complete-split tomography). *The full  $a$ -point density, or equivalently the full right-edge curve, determines  $S_K^{\text{spl}}$ . It therefore determines the Galois closure of  $K$  up to  $\mathbb{Q}$ -isomorphism. If  $K/\mathbb{Q}$  is Galois, it determines  $K$  itself.*

*Proof.* Target-plane conservation for all  $\sigma$  gives the Dirichlet series  $\sum_{p \in S_K^{\text{spl}}} (\log p)^2 p^{-2\sigma}$ , and uniqueness recovers its support. The same follows directly from edge tomography. Bauer's theorem [5] says that the completely split primes, up to finitely many exceptions, determine the normal closure; the final assertion is immediate for a Galois field.  $\square$

Finally, in  $\Re s > 1$ ,

$$\log L(s, \chi) = \sum_{k \geq 1} \frac{1}{k} \mathcal{P}_{\chi^k}(ks). \quad (9.9)$$

Every Euler-logarithm layer therefore has a scaled phase-blind density and edge law. The layers are not independent, so (9.9) is a structural decomposition rather than a claim about zeros of  $L(s, \chi)$  or about GRH.

## 10 The exact logarithmic profile at the outer edge

The smoothness argument proves that the zero density is flat at  $\sigma_e$ , but smoothness alone does not determine the scale of that flatness. We now compute it. This section solves the edge-profile problem left open in the preceding version of the paper. All constants below are defined by convergent integrals; decimal values are included only for orientation.

Put

$$A_{\geq 3}(\sigma) = \sum_{p \geq 3} p^{-\sigma}, \quad \varepsilon(\sigma) = A_{\geq 3}(\sigma) - 2^{-\sigma} = -D_0(\sigma), \quad (10.1)$$

and write

$$S_\sigma = \sum_{p \geq 3} p^{-\sigma} U_p, \quad x_0(\sigma) = 2^{-\sigma}.$$

Thus  $\varepsilon > 0$  for  $1 < \sigma < \sigma_e$  and

$$\varepsilon(\sigma) = |D_1(\sigma_e)|(\sigma_e - \sigma)(1 + O(\sigma_e - \sigma)). \quad (10.2)$$

Let  $I_0$  be the modified Bessel function and define

$$h(x) = x - \log I_0(x), \quad H_j(\sigma) = \int_0^\infty h(x)(\log x)^j x^{-1/\sigma-1} dx, \quad j \in \mathbb{Z}_{\geq 0}. \quad (10.3)$$

The integrals converge because  $h(x) = x - x^2/4 + O(x^4)$  at zero and  $h(x) = \frac{1}{2} \log(2\pi x) + O(x^{-1})$  at infinity. For  $\lambda \geq 0$  set

$$\Lambda_\sigma(\lambda) = \sum_{p \geq 3} \log I_0(\lambda p^{-\sigma}), \quad \Psi_\sigma(\lambda) = \lambda A_{\geq 3}(\sigma) - \Lambda_\sigma(\lambda) = \sum_{p \geq 3} h(\lambda p^{-\sigma}). \quad (10.4)$$

For  $1 < \sigma < \sigma_e$  there is a unique  $\lambda_\sigma > 0$  satisfying

$$\Lambda'_\sigma(\lambda_\sigma) = x_0(\sigma), \quad \text{equivalently} \quad \Psi'_\sigma(\lambda_\sigma) = \varepsilon(\sigma). \quad (10.5)$$

Moreover,  $\lambda_\sigma \rightarrow \infty$  as  $\sigma \uparrow \sigma_e$ . We use the nonconflicting notation

$$\mathcal{J}_e(\sigma) = \Lambda_\sigma^*(x_0) = \lambda_\sigma x_0 - \Lambda_\sigma(\lambda_\sigma) = \Psi_\sigma(\lambda_\sigma) - \lambda_\sigma \varepsilon, \quad \mathcal{L}_\sigma = \log \lambda_\sigma. \quad (10.6)$$

**Lemma 10.1** (Radial reduction and derivative weight). *Let  $f_{S,\sigma}$  be the planar density of  $S_\sigma$ . As  $\sigma \uparrow \sigma_e$ ,*

$$\boxed{\rho(\sigma) = D_1(\sigma_e)^2 (1 + O(\sqrt{\varepsilon})) f_{S,\sigma}(x_0)}. \quad (10.7)$$

*In particular*  $-\log \rho(\sigma) = -\log f_{S,\sigma}(x_0) + O(1)$ .

*Proof.* Rotation invariance gives  $f_{X,\sigma}(0) = f_{S,\sigma}(x_0)$ . On the fibre  $X_\sigma = 0$ , write  $S_\sigma = |S_\sigma|e^{i\psi}$ ; then

$$\sum_{p \geq 3} p^{-\sigma} |1 - U_p e^{-i\psi}|^2 = 2\varepsilon.$$

After substituting  $2^{-\sigma} U_2 = -S_\sigma$  into  $Y_\sigma$  and applying Cauchy–Schwarz,

$$||Y_\sigma| - (|D_1(\sigma)| - (\log 2)\varepsilon)| \leq \left( 2\varepsilon \sum_{p \geq 3} \log^2(p/2) p^{-\sigma} \right)^{1/2}.$$

The last prime sum is locally bounded at  $\sigma_e$ . Integrating  $|Y_\sigma|^2$  over the conditional density on the fibre proves (10.7).  $\square$

**Lemma 10.2** (Mellin–PNT expansion). *Uniformly for  $\sigma$  in a compact subset of  $(1, \infty)$ , for every fixed  $K \in \mathbb{Z}_{\geq 0}$  and  $\lambda \rightarrow \infty$ ,*

$$\Psi_\sigma(\lambda) = \lambda^{1/\sigma} \left( \sum_{j=0}^K \frac{H_j(\sigma)}{(\log \lambda)^{j+1}} + O_K((\log \lambda)^{-K-2}) \right). \quad (10.8)$$

*The differentiated expansion is obtained by termwise differentiation; in particular its first two coefficients are  $H_0/\sigma$  and  $H_1/\sigma - H_0$ .*

*Proof.* Summation by parts and the prime number theorem with de la Vallée Poussin remainder reduce the prime sum, up to  $O(\lambda^{1/\sigma} e^{-c\sqrt{\log \lambda}})$ , to

$$M_\sigma(\lambda) = \int_2^\infty \frac{h(\lambda t^{-\sigma})}{\log t} dt.$$



The substitution  $x = \lambda t^{-\sigma}$  gives

$$M_\sigma(\lambda) = \lambda^{1/\sigma} \int_0^{\lambda^{2-\sigma}} \frac{h(x)x^{-1/\sigma-1}}{\log \lambda - \log x} dx.$$

On  $\lambda^{-1/2} \leq x \leq \lambda^{1/2}$  expand the denominator in powers of  $(\log x)/(\log \lambda)$ . The two tails are negligible by the endpoint estimates for  $h$ , and extending each coefficient integral to  $(0, \infty)$  gives (10.8). Repeating the argument with  $\tilde{h}(x) = xh'(x)$  gives the derivative formula. Here  $\tilde{h}(x) = x + O(x^2)$  at zero and  $\tilde{h}(x) = \frac{1}{2} + O(x^{-1})$  at infinity, so the same tail split is valid. Integration by parts yields, for  $j \geq 1$ ,

$$\int_0^\infty \tilde{h}(x)(\log x)^j x^{-1/\sigma-1} dx = \frac{H_j(\sigma)}{\sigma} - jH_{j-1}(\sigma).$$

For  $j = 0$  the corresponding coefficient is  $H_0(\sigma)/\sigma$ .  $\square$

**Lemma 10.3** (Exponential-tilt density bound). *For  $\sigma < \sigma_e$  sufficiently close to  $\sigma_e$ , with  $\mathcal{J}_e$  as in (10.6),*

$$\mathcal{J}_e(\sigma) - 2\mathcal{L}_\sigma - O(1) \leq -\log f_{S,\sigma}(x_0) \leq \mathcal{J}_e(\sigma) + O(\mathcal{J}_e(\sigma)^{1/2}). \quad (10.9)$$

*Proof.* Tilt the independent phases by

$$\frac{d\mathbb{P}_\lambda}{d\mathbb{P}} = \exp(\lambda \Re S_\sigma - \Lambda_\sigma(\lambda)).$$

Under this law the phase of the  $p$ th summand has von Mises density  $e^{\lambda p^{-\sigma} \cos \theta} / (2\pi I_0(\lambda p^{-\sigma}))$ . Five primes of size comparable with  $\lambda^{1/\sigma}$ , together with the van der Corput estimate for their circle integrals, give

$$\|f_{S,\sigma}^{(\lambda)}\|_\infty \ll \lambda^2.$$

Indeed, Bertrand's postulate supplies five primes  $q_j$  with  $\lambda q_j^{-\sigma} \asymp 1$ . The characteristic function of each corresponding tilted circle is bounded by  $\min\{1, C(q_j^{-\sigma}|u|)^{-1/2}\}$ , uniformly in the direction of  $u \in \mathbb{R}^2$ . The product of five such bounds is integrable, and the change of scale  $u = \lambda v$  gives the displayed  $L^1$  estimate by Fourier inversion. Convolution with all remaining summands cannot increase the  $L^\infty$  norm. The exact change-of-measure identity

$$f_{S,\sigma}(z) = e^{-\lambda \Re z + \Lambda_\sigma(\lambda)} f_{S,\sigma}^{(\lambda)}(z)$$

at  $z = x_0$  and  $\lambda = \lambda_\sigma$  proves the left inequality.

For the reverse inequality put  $a_p = p^{-\sigma}$  and isolate the two circles belonging to 3 and 5. Their convolution has density

$$\frac{\mathbf{1}_{|a_3 - a_5| < |z| < a_3 + a_5}}{\pi^2 \sqrt{(|z|^2 - (a_3 - a_5)^2)((a_3 + a_5)^2 - |z|^2)}} \geq \frac{1}{2\pi^2 a_3 a_5}$$

on its open annulus. Put

$$S' = \sum_{p \geq 7} a_p U_p, \quad A_7 = \sum_{p \geq 7} a_p, \quad \xi = A_7 - \Re S', \quad \eta = \Im S'.$$

Tilt  $S'$  with parameter  $\lambda_7$  chosen so that  $\mathbb{E}_{\lambda_7} \xi = (1 - 2\vartheta)\varepsilon$ , where  $\vartheta = C(\log \lambda_\sigma)^{1/2} \lambda_\sigma^{-1/(2\sigma)}$  and  $C$  is a sufficiently large absolute constant. Consider the event

$$\mathcal{E} = \left\{ (1 - 3\vartheta)\varepsilon \leq \xi \leq (1 - \vartheta)\varepsilon, \quad \eta^2 < \vartheta \varepsilon (a_3 + a_5) \right\}.$$

For  $\sigma$  sufficiently close to  $\sigma_e$ , elementary geometry shows that  $x_0 - S'$  lies in the open annulus on  $\mathcal{E}$ .

The standard von Mises moment estimates, summed over  $p \geq 7$ , give

$$\mathrm{Var}_{\lambda_7}(\xi) \ll \frac{\lambda_7^{1/\sigma-2}}{\log \lambda_7}, \quad \mathbb{E}_{\lambda_7} \eta^2 \ll \lambda_7^{-1}.$$

Lemma 10.2, differentiated once, implies  $\lambda_7 \asymp \lambda_\sigma$  and  $\varepsilon \asymp \lambda_\sigma^{1/\sigma-1} / \log \lambda_\sigma$ . In particular,

$$\frac{\mathrm{Var}_{\lambda_7}(\xi)}{\vartheta^2 \varepsilon^2} \ll C^{-2}, \quad \frac{\mathbb{E}_{\lambda_7} \eta^2}{\vartheta \varepsilon (a_3 + a_5)} = o(1).$$

Chebyshev's inequality therefore gives  $\mathbb{P}_{\lambda_7}(\mathcal{E}) \geq c > 0$  when  $C$  is chosen large. Define the deficit rate functions

$$\mathcal{R}_\sigma(u) = \sup_{\lambda \geq 0} \{ \Psi_\sigma(\lambda) - \lambda u \}, \quad \mathcal{R}_{\sigma, \geq 7}(u) = \sup_{\lambda \geq 0} \left\{ \sum_{p \geq 7} h(\lambda p^{-\sigma}) - \lambda u \right\}.$$

Thus  $\mathcal{R}_\sigma(\varepsilon) = \mathcal{I}_e(\sigma)$ . The inverse change of measure yields

$$\mathbb{P}(\mathcal{E}) \geq c \exp\{-\mathcal{R}_{\sigma, \geq 7}((1-2\vartheta)\varepsilon) - \lambda_7 \vartheta \varepsilon\}.$$

Deleting the two fixed primes changes the deficit function  $\Psi_\sigma$  by  $h(\lambda a_3) + h(\lambda a_5) = O(\log \lambda)$ . The envelope identity  $\mathcal{R}'_\sigma(u) = -\lambda_\sigma(u)$ , where  $\lambda_\sigma(u)$  is the unique maximizing parameter in the definition of  $\mathcal{R}_\sigma(u)$ , and the differentiated form of Lemma 10.2 consequently give

$$\mathcal{R}_{\sigma, \geq 7}((1-2\vartheta)\varepsilon) = \mathcal{I}_e(\sigma) + O(\vartheta \lambda_\sigma \varepsilon + \log \lambda_\sigma).$$

Since  $\lambda_\sigma \varepsilon \asymp \mathcal{I}_e$  and  $\vartheta \mathcal{I}_e \asymp \mathcal{I}_e^{1/2}$ , integrating the two-circle density over  $\mathcal{E}$  proves the right inequality.  $\square$

Define

$$\beta_e = \frac{1}{\sigma_e - 1}, \quad h_e = \frac{H_1(\sigma_e)}{H_0(\sigma_e)}, \quad (10.10)$$

$$C_e = \frac{\sigma_e - 1}{\sigma_e} H_0(\sigma_e)^{1+\beta_e} \sigma_e^{-\beta_e}, \quad K_e = C_e (1 + \beta_e)^{-(1+\beta_e)}. \quad (10.11)$$

**Theorem 10.4** (Exact logarithmic edge profile). *As  $\sigma \uparrow \sigma_e$ ,*

$$\boxed{-\log \rho(\sigma) = \mathcal{I}_e(\sigma) + O(\mathcal{I}_e(\sigma)^{1/2})}. \quad (10.12)$$

*More precisely, with  $\mathcal{L} = \mathcal{L}_\sigma$ ,*

$$\mathcal{I}_e(\sigma) = C_e \varepsilon^{-\beta_e} \mathcal{L}^{-(1+\beta_e)} \left( 1 + \frac{(1+\beta_e)h_e}{\mathcal{L}} + O(\mathcal{L}^{-2}) \right), \quad (10.13)$$

*where*

$$\mathcal{L} = (1 + \beta_e) \left( \log \frac{1}{\varepsilon} - \log \mathcal{L} + \log \frac{H_0(\sigma_e)}{\sigma_e} \right) + O(\mathcal{L}^{-1}). \quad (10.14)$$

*Consequently, if  $\ell = \log(1/\varepsilon)$ , then*

$$\boxed{-\log \rho(\sigma) = K_e \varepsilon^{-\beta_e} \ell^{-(1+\beta_e)} \left( 1 + (1 + \beta_e) \frac{\log \ell}{\ell} + O(\ell^{-1}) \right)}. \quad (10.15)$$

Equivalently, for  $\delta = \sigma_e - \sigma$  and  $\ell_\delta = \log(1/\delta)$ ,

$$-\log \rho(\sigma) = K'_e \delta^{-\beta_e} \ell_\delta^{-(1+\beta_e)} \left( 1 + (1 + \beta_e) \frac{\log \ell_\delta}{\ell_\delta} + O(\ell_\delta^{-1}) \right), \quad (10.16)$$

where  $K'_e = K_e |D_1(\sigma_e)|^{-\beta_e}$ .

*Proof.* Lemma 10.1 and Lemma 10.3 prove (10.12). Apply Lemma 10.2 and its derivative at the saddle point  $\Psi'_\sigma(\lambda) = \varepsilon$ . Solving the derivative equation gives

$$\lambda^{1-1/\sigma} = \frac{H_0(\sigma)}{\sigma \varepsilon \log \lambda} \left( 1 + \frac{H_1(\sigma)/H_0(\sigma) - \sigma}{\log \lambda} + O((\log \lambda)^{-2}) \right).$$

Substitution into  $\Psi_\sigma(\lambda) - \lambda \varepsilon$  yields (10.13)–(10.14), first with  $\sigma$  in the constants. Since  $\sigma_e - \sigma \asymp \varepsilon$ , replacing  $\sigma$  by  $\sigma_e$  changes the expression only within the stated error. Iterating (10.14) once and using (10.12) gives (10.15). Finally (10.2) gives (10.16).  $\square$

Numerically,

$$\beta_e \doteq 1.28280015, \quad K_e \doteq 1.38465461, \quad K'_e \doteq 6.35167631. \quad (10.17)$$

These decimals are high-precision evaluations, not interval certificates and not part of the logical statement. The theorem explains the source of the infinite softness: the edge position is fixed by the initial prime spectrum, while the cost of approaching it is paid by aligning primes of size about  $\lambda_\sigma^{1/\sigma_e}$ .

**Conjecture 10.5** (Gaussian prefactor). As  $\sigma \uparrow \sigma_e$ , let  $v_1(\sigma)$  and  $v_2(\sigma)$  be the two coordinate variances of  $S_\sigma$  under the saddle-point tilt. Then

$$\rho(\sigma) = \frac{D_1(\sigma_e)^2}{2\pi \sqrt{v_1(\sigma)v_2(\sigma)}} e^{-\mathcal{J}_e(\sigma)} (1 + o(1)).$$

This requires a local central limit theorem for the tilted triangular array and is not used elsewhere in the paper.

## 11 Picket transforms and an exact prime-gap dictionary

The weighted limit (6.45) admits a useful transform. Write  $p_n$  for the  $n$ th prime and set

$$w_n = \frac{1}{2\pi} \log \frac{p_{n+1}}{p_n}, \quad \tau_n = \frac{\log(\log p_{n+1}/\log p_n)}{\log(p_{n+1}/p_n)}. \quad (11.1)$$

**Theorem 11.1** (Mellin abscissa of the weighted picket fence). *The ordered series*

$$Z_{\text{pf}}(s) = \sum_{n \geq 1} w_n \tau_n^s \quad (11.2)$$

converges, indeed absolutely, exactly in the half-plane  $\Re s > 1$ . For  $\Re s \leq 1$  its ordered partial sums are not Cauchy.

*Proof.* Put  $x_n = \log p_n$  and  $\Delta_n = x_{n+1} - x_n$ . Then  $w_n = \Delta_n/(2\pi)$  and the mean-value theorem gives

$$\tau_n = \frac{\log x_{n+1} - \log x_n}{x_{n+1} - x_n} = \frac{1}{\xi_n}, \quad x_n < \xi_n < x_{n+1}. \quad (11.3)$$

For  $\sigma = \Re s > 1$ , the sum  $\sum \Delta_n \xi_n^{-\sigma}$  is a Riemann-sum comparison with  $\int_{\log 2}^{\infty} x^{-\sigma} dx$ , proving absolute convergence.

Conversely fix  $s = \sigma + it$  with  $\sigma \leq 1$  and choose  $c > 1$  so close to 1 that  $|t| \log c < \pi/4$ . In each sufficiently large block  $X \leq x_n < x_{n+1} \leq cX$ , the phases  $\tau_n^{it}$  lie in one arc of length less than  $\pi/3$ . After a common rotation, the real part of the block sum is bounded below by a constant multiple of

$$X^{-\sigma} \sum_{X \leq x_n < x_{n+1} \leq cX} \Delta_n \asymp X^{1-\sigma}.$$

Here the omitted boundary interval is  $O(1)$  because Bertrand's postulate gives  $\Delta_n \leq \log 2$ . Thus the block tails fail to tend to zero for  $\sigma = 1$  and grow for  $\sigma < 1$ .  $\square$

**Theorem 11.2** (Topology growth in the derivative tower). *Let  $b_m$  be the number of connected components of the closed zero projection  $Z_{m,\omega}$ . Then*

$$b_m = \#\mathfrak{D}_m, \quad b_m - b_{m-1} = \#\{p : m_p = m\} \quad (m \geq 1), \quad b_m \longrightarrow \infty. \quad (11.4)$$

Unconditionally,

$$\boxed{\frac{\sqrt{m}}{(\log m)^2} \ll b_m \ll \sqrt{m}.} \quad (11.5)$$

Under Cramér's prime-gap conjecture,

$$\boxed{b_m = m^{1/2+o(1)}}. \quad (11.6)$$

*Proof.* The first line follows from (6.21), monotone birth, and finite birth for every prime. Since  $g_- g_+ \geq 2g_{\max}$  and  $\log(2 + 2g_{\max})/g_{\max}$  is bounded, the upper bound in (6.24) gives  $m_p \ll p^2 \log^2 p$ . Hence all primes  $p \ll \sqrt{m}/\log m$  have appeared by level  $m$ , and the prime number theorem gives the lower half of (11.5). For the upper half, the lower birth estimate shows that a prime  $p \in \mathfrak{D}_m$  must have an adjacent gap of length

$$\gg \frac{p \log p}{\sqrt{m}}.$$

In a dyadic interval  $x < p \leq 2x$ , the disjoint consecutive-prime gaps have total length  $O(x)$  and each gap can be charged to at most two endpoints. Hence that interval contains  $O(\sqrt{m}/\log x)$  dominant primes. The Baker–Harman–Pintz estimate  $g_{\max} \ll p^{0.525}$  [1] first restricts every such prime to  $p \leq m^{20/19+o(1)}$ ; summing the dyadic bounds above  $\sqrt{m}$ , and using  $\pi(\sqrt{m}) \ll \sqrt{m}/\log m$  below it, gives  $b_m \ll \sqrt{m}$ . Under Cramér, the two sides of (6.24) give  $m_p = p^{2+o(1)}$ , and inversion with the prime number theorem gives (11.6).  $\square$

The same atoms provide exact coordinates for standard prime-gap questions. The next corollary is a dictionary, not a proof of any of the conjectures named in it.

**Corollary 11.3** (Exact prime-gap dictionary). *Let  $g_n = p_{n+1} - p_n$ . Then*

$$2\pi p_n w_n = p_n \log \left( 1 + \frac{g_n}{p_n} \right) = g_n(1 + o(1)). \quad (11.7)$$

Consequently:

(i) *The twin-prime conjecture is equivalent to*

$$\liminf_{n \rightarrow \infty} p_n w_n = \frac{1}{\pi},$$

*and also to  $1/\pi$  being a subsequential limit of  $(p_n w_n)$ .*

(ii) *For every  $h \geq 1$ , the Polignac assertion that  $g_n = 2h$  infinitely often is equivalent to  $h/\pi$  being a subsequential limit of  $(p_n w_n)$ .*

(iii) *Cramér's conjecture  $g_n = O((\log p_n)^2)$  is equivalent to*

$$w_n = O\left(\frac{(\log p_n)^2}{p_n}\right), \quad (11.8)$$

*and also to  $|W_p| = O(1/p)$  over the primes.*

(iv) *For every fixed  $k \geq 1$ ,*

$$\liminf_n (p_{n+k} - p_n) = \liminf_n 2\pi p_n \sum_{j=0}^{k-1} w_{n+j}. \quad (11.9)$$

*Proof.* Formula (11.7) is the definition of  $w_n$  and Taylor expansion, using  $p_{n+1}/p_n \rightarrow 1$ . A bounded subsequence of  $p_n w_n$  forces  $g_n = O(1)$ ; parity and integrality then provide a constant-gap subsequence. This proves (i)–(ii). The first equivalence in (iii) follows again from (11.7). Expansion of the two secant slopes defining  $W_p$  gives

$$|W_p| = \frac{g_- + g_+}{2p(\log p)^2} (1 + o(1)), \quad (11.10)$$

which proves the second. Finally the sum in (11.9) telescopes to  $p_n \log(p_{n+k}/p_n)$ , differing from  $p_{n+k} - p_n$  by  $o(1)$  along every subsequence on which either has finite limit.  $\square$

For reference, (11.7) also gives exact, purely reformulatory versions of two other classical inequalities:

$$\sqrt{p_{n+1}} - \sqrt{p_n} < 1 \iff w_n < \frac{1}{\pi} \log(1 + p_n^{-1/2}), \quad (11.11)$$

and Firoozbakht's inequality is equivalent to

$$w_n < \frac{\log p_n}{2\pi n}. \quad (11.12)$$

No new case of either conjecture follows from these identities.

### 11.1 Crystalline zero lattices at consecutive-prime crossovers

The picket limit describes horizontal divisor mass after vertical averaging. At a fixed consecutive-prime crossover one can retain the ordinates and obtain an actual zero lattice. Fix consecutive primes  $p < q = p^+$  and put

$$d = d_{p,q} = \log \frac{q}{p}, \quad \tau = \tau_{p,q} = \frac{\log(\log q / \log p)}{\log(q/p)}. \quad (11.13)$$

For a prime  $r \notin \{p, q\}$  define

$$\psi_{p,q}(r) = \log \frac{\log r}{\log p} - \tau \log \frac{r}{p}, \quad \mu_{p,q} = - \max_{r \notin \{p,q\}} \psi_{p,q}(r) > 0. \quad (11.14)$$

Strict concavity shows that the maximum is attained among  $p^-$  and  $q^+$ , with only  $q^+$  present when  $p = 2$ . Choose a real lift  $\theta_{p,q} = \arg(\omega_q/\omega_p)$  and set

$$t_k = \frac{\theta_{p,q} + (2k+1)\pi}{d}, \quad s_{m,k}^0 = m\tau + it_k. \quad (11.15)$$

Changing the lift only reindexes  $k$ .

**Theorem 11.4** (Uniform two-prime zero lattice). *Fix  $p < q = p^+$  and  $A > 0$ . For all sufficiently large integers  $m$ , so that  $m\tau - A > 1$ , uniformly over all unimodular twists and all  $k \in \mathbb{Z}$ , the rectangle*

$$\mathcal{R}_{m,k}(A) = \left\{ s : |\Re s - m\tau| < A, \quad |\Im s - t_k| < \frac{\pi}{d} \right\} \quad (11.16)$$

*contains exactly one zero  $\rho_{m,k}$  of  $\mathcal{P}_{m,\omega}$ , counted with multiplicity. It is simple; there are no zeros on  $\partial \mathcal{R}_{m,k}(A)$ ; and these points exhaust the zeros in  $|\Re s - m\tau| \leq A$ . Moreover,*

$$\boxed{\rho_{m,k} = m\tau + i \frac{\theta_{p,q} + (2k+1)\pi}{d} + O_{p,q,A}(e^{-\mu_{p,q}m}).} \quad (11.17)$$

If

$$E_{m,\omega}(z) = \sum_{r \notin \{p,q\}} \frac{\omega_r}{\omega_p} e^{m\psi_{p,q}(r)} e^{-z \log(r/p)}, \quad (11.18)$$

then the first correction is

$$\boxed{\rho_{m,k} = s_{m,k}^0 - \frac{1}{d} E_{m,\omega}(it_k) + O_{p,q,A}(e^{-2\mu_{p,q}m}).} \quad (11.19)$$

*Proof.* Write  $s = m\tau + z$  and divide by the nonvanishing  $p$ -term. The defining identity for  $\tau$  gives

$$\frac{\mathcal{P}_{m,\omega}(m\tau + z)}{\omega_p(\log p)^m p^{-m\tau - z}} = 1 + \frac{\omega_q}{\omega_p} e^{-dz} + E_{m,\omega}(z). \quad (11.20)$$

The strictly concave function  $x \mapsto \log x - \tau x$  takes the same value at  $x = \log p$  and  $x = \log q$ , and its maximum lies strictly between them. Since  $p, q$  are consecutive,  $\psi_{p,q}(r) < 0$  for  $r \notin \{p, q\}$  and  $\mu_{p,q} > 0$ . For every fixed  $j \geq 0$ ,

$$\sup_{|\Re z| \leq A} |\partial_z^j E_{m,\omega}(z)| \ll_{p,q,A,j} e^{-\mu_{p,q}m}. \quad (11.21)$$

Indeed, factor out  $e^{-(m-m_0)\mu_{p,q}}$  with  $m_0\tau - A > 1$ ; the remaining majorant is a convergent log-weighted prime Dirichlet series. This also proves uniformity in  $k$  and in the twist.

Writing  $z = it_k + w$ , the two-term part becomes  $1 - e^{-dw}$ . On the vertical sides of (11.16) its modulus is at least  $1 - e^{-dA}$ , while on the horizontal sides it is at least  $1 + e^{-dA}$ . Equation (11.21) and Rouché's theorem give one zero, counted with multiplicity, in every rectangle and no boundary zero. The rectangles tile the fixed-width strip, so they exhaust it. Rouché on  $|w| = Ce^{-\mu_{p,q}m}$  gives (11.17). Finally

$$1 - e^{-dw} = dw + O(w^2), \quad E_{m,\omega}(it_k + w) = E_{m,\omega}(it_k) + O(e^{-\mu_{p,q}m}|w|),$$

which yields (11.19).  $\square$

**Corollary 11.5** (Exact strip count and crystalline spacing). *For fixed  $p, q, A$  and all sufficiently large  $m$ ,*

$$N_{m;p,q,A}(T) = \frac{\log(q/p)}{2\pi} T + O_{p,q,A}(1), \quad (11.22)$$

*uniformly over the twists, where the count is taken in  $|\Re s - m\tau| < A$  and  $0 < \Im s \leq T$ . Furthermore,*

$$\rho_{m,k+1} - \rho_{m,k} = \frac{2\pi i}{\log(q/p)} + O_{p,q,A}(e^{-\mu_{p,q}m}), \quad (11.23)$$

*and, for a fixed twist and a consistent indexing,*

$$\rho_{m+1,k} - \rho_{m,k} = \tau_{p,q} + O_{p,q,A}(e^{-\mu_{p,q}m}). \quad (11.24)$$

**Corollary 11.6** (Inverse recovery of the primes and phases). *Suppose the moving lattice belonging to a consecutive pair is known and put*

$$H = \lim_{m \rightarrow \infty} \Im(\rho_{m,k+1} - \rho_{m,k}), \quad \tau = \lim_{m \rightarrow \infty} \frac{\Re \rho_{m,k}}{m}, \quad d = \frac{2\pi}{H}.$$

*Then*

$$\log p = \frac{d}{e^{\tau d} - 1}, \quad q = pe^d. \quad (11.25)$$

*The ordinate offset recovers the relative phase:*

$$-e^{id\Im \rho_{m,k}} = \frac{\omega_q}{\omega_p} + O_{p,q,A}(e^{-\mu_{p,q}m}). \quad (11.26)$$

*Thus all consecutive-prime crossover lattices recover the whole twist up to one common unimodular factor.*

*Proof.* Equations (11.17), (11.23), and the algebra used in (6.41) give (11.25). Equation (11.15) gives (11.26), and consecutive ratios recover all phases after one phase is chosen as a normalization.  $\square$

The leading defect also detects which external neighbour is closest in exponential scale. Let

$$\mathcal{R}_{p,q} = \{r \notin \{p, q\} : -\psi_{p,q}(r) = \mu_{p,q}\}$$

If  $p = 2$ , then  $\mathcal{R}_{p,q} \subseteq \{q^+\}$ ; if  $p \geq 3$ , then  $\mathcal{R}_{p,q} \subseteq \{p^-, q^+\}$ .

**Corollary 11.7** (Second-neighbour defect profile). *Uniformly in  $k$ ,*

$$e^{\mu_{p,q}m}(\rho_{m,k} - s_{m,k}^0) = -\frac{1}{d} \sum_{r \in \mathcal{R}_{p,q}} \frac{\omega_r}{\omega_p} e^{-it_k \log(r/p)} + o(1). \quad (11.27)$$

*Define*

$$\mathcal{K}_m = \overline{\{e^{\mu_{p,q}m}(\rho_{m,k} - s_{m,k}^0) : k \in \mathbb{Z}\}}.$$

*If  $|\mathcal{R}_{p,q}| = 1$ , then*

$$d_{\mathbb{H}}(\mathcal{K}_m, \{z : |z| = 1/d\}) \longrightarrow 0;$$

*if  $|\mathcal{R}_{p,q}| = 2$ , then*

$$d_{\mathbb{H}}(\mathcal{K}_m, \{z : |z| \leq 2/d\}) \longrightarrow 0.$$

*Proof.* Multiply (11.19) by  $e^{\mu_{p,q}m}$ ; every nonmaximal tail term decays exponentially. With one surviving neighbour, a rational relation between  $\log(r/p)$  and  $\log(q/p)$  would exponentiate to a nontrivial equality between products of distinct primes, contradicting unique factorization. Thus Kronecker's theorem gives the circle. With two neighbours, any rational relation among  $1$ ,  $\log(p^-/p)/d$ , and  $\log(q^+/p)/d$  likewise exponentiates to a nontrivial prime-product identity. Unique factorization makes the three numbers rationally independent; their torus orbit is dense, and sums of two unit vectors fill the closed disc. The error in (11.27) is uniform in  $k$ , which proves the stated Hausdorff convergence.  $\square$

*Remark 11.8* (Scope and prior geometry). The theorem fixes  $p < q$  and then lets  $m \rightarrow \infty$ ; it is not uniform for a prime pair growing with  $m$ . Its phase-recovery datum is the complete ordinate-resolved moving tower, not phase-blind data at one derivative level. The distinct anchored fixed-level recovery mechanism is developed below. Binder, Pauli, and Saidak proved an analogous two-term rectangle theorem for high derivatives of the Riemann zeta function [4]; hence the Rouché lattice mechanism itself is not claimed as new. Prime-specific additions here are the decay scale  $\mu_{p,q}$ , the uniform first correction, arbitrary unimodular twists, inverse recovery, and the second-neighbour defect. A 2017 lecture of Nakamura was entitled “On zeros of derivatives of the prime zeta-function” [17]; priority for the bare prime-zeta specialization is therefore left open.

## 12 Universal local zero processes, Palm laws, and hard cores

The horizontal laws proved above determine more than one-point densities. At every fixed derivative level they determine the complete local zero process, including all finite-window counting laws and all nearest-neighbour laws. The point-process formulation is also the cleanest way to separate two genuinely different small-scale regimes: quartic planar repulsion in the bulk and an exact deterministic hard core in a dominance region.

Throughout this section, fix an integer  $m \geq 0$  and write

$$F_{m,u}(s) := \sum_p u_p (\log p)^m p^{-s}, \quad u = (u_p)_p \in \mathbb{H} := \mathbb{T}^{\mathbb{P}}, \quad \Re s > 1. \quad (1)$$

Thus  $F_{m,1} = (-1)^m \mathcal{P}^{(m)}$ . We also fix  $a \in \mathbb{C}$ . The zero divisor of  $F_{m,u} - a$ , with algebraic multiplicity, is denoted by

$$\Xi_{m,u,a} := \sum_{F_{m,u}(\rho)=a} m_\rho \delta_\rho. \quad (2)$$

It is a locally finite integer-valued measure on  $D := \{s \in \mathbb{C} : \Re s > 1\}$ . The vertical translation of a divisor is  $(\vartheta_t \Xi)(B) = \Xi(B + it)$ , and the Kronecker flow on  $\mathbb{H}$  is

$$(R_t u)_p = u_p p^{-it}. \quad (3)$$

The elementary but decisive covariance identity is

$$F_{m,R_t u}(s) = F_{m,u}(s + it), \quad \Xi_{m,R_t u,a} = \vartheta_t \Xi_{m,u,a}. \quad (4)$$

### 12.1 Compactness of the hull and smooth joint jets

**Lemma 12.1** (Continuous divisor map and a uniform local zero bound). *The map*

$$u \longmapsto F_{m,u}$$



from  $\mathbb{H}$  to the holomorphic functions on  $D$ , equipped with the compact-open topology, is continuous. The induced divisor map

$$\mathbb{H} \longrightarrow \mathcal{N}_{\text{lf}}(D), \quad u \longmapsto \Xi_{m,u,a}, \quad (5)$$

is continuous for the vague topology on locally finite divisors, where zeros are counted with multiplicity. Moreover, for every compact  $K \Subset D$ ,

$$M_{m,a}(K) := \sup_{u \in \mathbb{H}} \Xi_{m,u,a}(K) < \infty. \quad (6)$$

*Proof.* If  $K \Subset D$  and  $\Re s \geq 1 + \eta$  on  $K$ , then for every  $j \geq 0$ ,

$$\sum_p (\log p)^{m+j} p^{-1-\eta} < \infty.$$

The tails of the series for  $F_{m,u}$  and all its fixed derivatives are therefore uniform in  $u$ . Coordinatewise convergence in  $\mathbb{H}$  implies compact-open convergence of the corresponding functions.

No member of the hull is identically equal to  $a$ . For  $a \neq 0$  this follows by letting  $\Re s \rightarrow \infty$ , and for  $a = 0$  it follows from uniqueness of absolutely convergent Dirichlet series. Hurwitz's theorem, equivalently the argument principle on contours avoiding the limiting divisor, now gives continuity of the divisor map with multiplicities.

Suppose that (6) failed. There would be  $u_n \in \mathbb{H}$  for which the number of zeros in  $K$  tends to infinity. After passing to a subsequence, compactness of  $\mathbb{H}$  gives  $u_n \rightarrow u$ . Choose a bounded domain  $V \Subset D$  containing  $K$  whose boundary contains no zero of  $F_{m,u} - a$ . Rouché's theorem makes the number of zeros of  $F_{m,u_n} - a$  in  $V$  equal to the finite number for  $F_{m,u} - a$  for all large  $n$ , a contradiction.  $\square$

For pairwise distinct  $s_1, \dots, s_k \in D$  and an integer  $r \geq 0$ , set

$$\mathcal{J}_{k,r}(u; \mathbf{s}) := (F_{m,u}^{(\nu)}(s_j))_{1 \leq j \leq k, 0 \leq \nu \leq r} \in \mathbb{C}^{k(r+1)}. \quad (7)$$

**Lemma 12.2** (Smooth joint jet densities). *Every vector (7) has a  $C^\infty$  density with bounded derivatives of all orders. The density depends continuously, in every  $C^r$  norm, on the points  $s_j$  away from their diagonals. If points coalesce, the same assertion holds uniformly after replacing their values by Newton divided differences; the limiting variables are the corresponding normalized jets.*

*Proof.* Put  $N = k(r+1)$ . The vector (7) is a sum of independent Haar phases multiplied by coefficient vectors

$$c_p = ((-\log p)^\nu (\log p)^m p^{-s_j})_{j,\nu} \in \mathbb{C}^N.$$

The functions  $x \mapsto x^{m+\nu} e^{-s_j x}$  are linearly independent. Consequently the vectors  $c_p$  span  $\mathbb{C}^N$ . Indeed, otherwise a nonzero exponential polynomial would vanish at  $x = \log p$  for every sufficiently large prime. Such a function has only  $O(T)$  real zeros in  $[0, T]$ , whereas  $\#\{p : \log p \leq T\} \asymp e^T/T$ . The same argument after removing any finite set of primes supplies arbitrarily many disjoint spanning  $N$ -tuples.

The characteristic function of (7) is a product of Bessel factors

$$\prod_p J_0(|\langle \xi, c_p \rangle|).$$

On each spanning tuple at least one scalar product has size  $\gg_{\mathcal{T}} |\xi|$ . Since  $|J_0(x)| \ll (1+x)^{-1/2}$ , using arbitrarily many disjoint tuples gives decay faster than any prescribed negative power of  $|\xi|$ .

Differentiating in  $\xi$  or in the parameters inserts only fixed powers of  $\log p$ , which remain summable on compact subsets of  $D$ . Fourier inversion proves the smoothness, boundedness, and parameter continuity.

Newton divided differences are linear combinations of the values at distinct nodes and converge to normalized derivatives as the nodes merge. Their coefficient vectors vary continuously through this confluent limit. Applying the preceding characteristic-function argument to these vectors proves the last assertion.  $\square$

## 12.2 All correlation functions and the complete local process

Let  $q_s^{(m)}$  denote the joint density of  $(F_{m,U}(s_j), F'_{m,U}(s_j))_{j=1}^k$  for Haar-distributed  $U$ . For distinct  $s_j$  define

$$\begin{aligned} K_k^{(m,a)}(s_1, \dots, s_k) &:= \mathbb{E} \left[ \prod_{j=1}^k |F'_{m,U}(s_j)|^2 \prod_{j=1}^k \delta_a(F_{m,U}(s_j)) \right] \\ &= \int_{\mathbb{C}^k} \prod_{j=1}^k |y_j|^2 q_s^{(m)}(a, y_1, \dots, a, y_k) \prod_{j=1}^k dA(y_j). \end{aligned} \quad (8)$$

Here and below the delta notation is shorthand for evaluation of the smooth density. The multi-point complex Kac–Rice formula identifies (8) with the density of the  $k$ th factorial moment measure of the Haar zero divisor. Haar-nullity of multiple  $a$ -points, already proved for the fixed-order family, is exactly the nondegeneracy needed here. For  $k = 1$  this agrees with the one-point density used above:

$$K_1^{(m,a)}(s) = G_m(\Re s, a). \quad (9)$$

**Theorem 12.3** (Full local point-process universality). *For every  $m \geq 0$ ,  $a \in \mathbb{C}$ , and unimodular twist  $\omega$ , one has weak convergence on  $\mathcal{N}_{\text{lf}}(D)$ :*

$$\boxed{\frac{1}{T} \int_0^T \delta_{\vartheta_t \Xi_{m,\omega,a}} dt \implies \text{Law}(\Xi_{m,U,a}), \quad U \sim \text{Haar}(\mathbb{H}).} \quad (10)$$

In particular, for every  $k \geq 1$  and every compactly supported continuous test function  $\Psi$  on  $D^k$ , the vertical average of the  $k$ -fold factorial sum converges to

$$\int_{D^k} \Psi(s_1, \dots, s_k) K_k^{(m,a)}(s_1, \dots, s_k) \prod_{j=1}^k dA(s_j). \quad (11)$$

After anchoring the first ordinate, this is the relative-coordinate all- $k$  correlation law.

*Proof.* Unique factorization makes every nontrivial finite integer combination of the  $\log p$  nonzero. Hence the flow  $R_t$  is uniquely ergodic on the compact group  $\mathbb{H}$ , with Haar measure as its unique invariant probability. Thus

$$\frac{1}{T} \int_0^T \delta_{R_t \omega} dt \implies \text{Haar}(\mathbb{H})$$

for every starting point  $\omega$ . Push this convergence forward under the continuous divisor map of Lemma 12.1, and use (4); this proves (10).

For a compactly supported  $\Psi$ , the associated factorial sum is a continuous functional of a divisor with multiplicities, including when several simple zeros coalesce. It is bounded by Lemma 12.1. Taking its expectation in (10) and applying (8) proves (11).  $\square$

The preceding theorem admits a useful exact converse: the complete process is recovered from its correlation functions by a finite, rather than merely formal, expansion.

**Theorem 12.4** (Exact Laplace functional and finite-window laws). *Let  $f \geq 0$  be continuous with compact support  $K \Subset D$ . Then*

$$\mathbb{E} \exp\{-\langle f, \Xi_{m,U,a} \rangle\} = \sum_{k=0}^{M_{m,a}(K)} \frac{1}{k!} \int_{K^k} \prod_{j=1}^k (e^{-f(s_j)} - 1) K_k^{(m,a)}(\mathbf{s}) \prod_j dA(s_j). \quad (12)$$

Let  $B \Subset D$  be Borel with planar-null boundary and put

$$\alpha_k(B) := \int_{B^k} K_k^{(m,a)}(\mathbf{s}) \prod_j dA(s_j), \quad \alpha_0(B) := 1.$$

Then, for  $0 \leq j \leq M_{m,a}(\overline{B})$ ,

$$\mathbb{P}\{\Xi_{m,U,a}(B) = j\} = \sum_{k=j}^{M_{m,a}(\overline{B})} \frac{(-1)^{k-j}}{j!(k-j)!} \alpha_k(B). \quad (13)$$

In particular,

$$\mathbb{P}\{\Xi_{m,U,a}(B) = 0\} = \sum_{k=0}^{M_{m,a}(\overline{B})} \frac{(-1)^k}{k!} \alpha_k(B). \quad (14)$$

The same right sides are the limiting vertical frequencies for every fixed twist  $\omega$ .

*Proof.* For a divisor  $\xi$  having at most  $M$  points in  $K$ , with labels repeated according to multiplicity,

$$e^{-\langle f, \xi \rangle} = \prod_{x \in \xi} [1 + (e^{-f(x)} - 1)].$$

Expanding the product into ordered factorial sums and dividing the  $k$ th sum by  $k!$  gives (12). Likewise

$$z^{\xi(B)} = \sum_{k=0}^M \frac{(z-1)^k}{k!} (\xi(B))_k.$$

Expectation and coefficient extraction give (13); setting  $j = 0$  gives (14). The boundary condition makes the event  $\{\xi(B) = j\}$  a continuity event for the Haar law: its only possible boundary configurations have a zero on  $\partial B$ , and the expected number of those is zero because the one-point measure has a density. The final assertion is therefore an application of (10) and the Portmanteau theorem.  $\square$

### 12.3 The corrected almost-periodic pair structure

For  $\psi \in C_c(D)$  let

$$L_\psi(u) := \langle \psi, \Xi_{m,u,a} \rangle.$$

This is continuous and bounded on  $\mathbb{H}$ . For  $\psi_1, \psi_2 \in C_c(D)$  put

$$A_{\psi_1, \psi_2}(h) := \int_{\mathbb{H}} L_{\psi_1}(u) L_{\psi_2}(R_h u) \, du, \quad (15)$$

$$D_{\psi_1, \psi_2}(h) := \int_D \psi_1(s) \psi_2(s - ih) G_m(\Re s, a) \, dA(s), \quad (16)$$

and let  $C_{\psi_1, \psi_2}(h)$  denote the corresponding factorial, or distinct-label, pair correlation.

**Proposition 12.5** (Almost periodic plus a local diagonal correction). *The all-pairs function  $A_{\psi_1, \psi_2}$  is Bohr almost periodic,  $D_{\psi_1, \psi_2} \in C_c(\mathbb{R})$ , and*

$$C_{\psi_1, \psi_2}(h) = A_{\psi_1, \psi_2}(h) - D_{\psi_1, \psi_2}(h) \in AP(\mathbb{R}) + C_c(\mathbb{R}) \subset AP(\mathbb{R}) + C_0(\mathbb{R}). \quad (17)$$

Its Bohr mean is

$$\mathcal{M}(C_{\psi_1, \psi_2}) = \left( \int_D \psi_1(s) G_m(\Re s, a) \, dA(s) \right) \left( \int_D \psi_2(s) G_m(\Re s, a) \, dA(s) \right). \quad (18)$$

In general  $C_{\psi_1, \psi_2}$  itself is not Bohr almost periodic.

*Proof.* The first expression in (15) is a convolution of continuous functions on the compact abelian group  $\mathbb{H}$ , evaluated on the one-parameter subgroup  $R_h$ . Peter–Weyl approximation by finite character sums makes it a uniform limit of trigonometric polynomials in  $h$ , hence Bohr almost periodic. The repeated-label terms in the product  $L_{\psi_1}(u) L_{\psi_2}(R_h u)$  are exactly (16), so (17) follows. Compact support of the two test functions makes  $D$  compactly supported in  $h$ , and its Cesàro mean is zero. Unique ergodicity computes the mean of  $A$  as the product of the two Haar means, which are the displayed one-point integrals. Finally, a nonzero compactly supported continuous function cannot be Bohr almost periodic; thus subtracting the diagonal term does not in general preserve  $AP(\mathbb{R})$ .  $\square$

## 12.4 Projected processes and exact Palm gap laws

Fix a compact interval  $I \Subset (1, \infty)$  and restrict the divisor to the strip  $S_I = I + i\mathbb{R}$ . Its ordinate projection is

$$\eta_{m, u, a, I} := \sum_{\substack{F_{m, u}(\rho) = a \\ \Re \rho \in I}} m_\rho \delta_{\Im \rho}. \quad (19)$$

Under Haar measure this is a stationary point process. Its factorial correlation densities are

$$g_{I, k}^{(m, a)}(t_1, \dots, t_k) := \int_{I^k} K_k^{(m, a)}(\sigma_1 + it_1, \dots, \sigma_k + it_k) \prod_j d\sigma_j, \quad (20)$$

and depend only on the differences  $t_j - t_1$ . Its intensity is

$$c_I^{(m, a)} = g_{I, 1}^{(m, a)} = \int_I G_m(\sigma, a) \, d\sigma. \quad (21)$$

Assume in what follows that  $c_I^{(m, a)} > 0$ .

The projected Haar process is simple almost surely. Planar multiple roots are Haar-null, and the expected number of two distinct planar roots having exactly the same ordinate is zero: the latter pairs lie on a Lebesgue-null hypersurface while their factorial moment measure has the locally integrable density  $K_2^{(m, a)}$ .

Let  $\mathbb{P}_I^0$  denote the reduced Palm law obtained by anchoring a typical ordinate at the origin and deleting that anchored point. Under this law define

$$G_+ := \inf\{t > 0 : t \in \eta\}, \quad R_{\text{ord}} := \inf\{|t| : t \in \eta\}.$$

**Theorem 12.6** (Exact Palm nearest-neighbour and consecutive-gap laws). *For every  $u > 0$ ,*

$$\mathbb{P}_I^0\{G_+ > u\} = \frac{1}{c_I^{(m,a)}} \sum_{k \geq 0} \frac{(-1)^k}{k!} \int_{(0,u)^k} g_{I,k+1}^{(m,a)}(0, t_1, \dots, t_k) \prod_j dt_j, \quad (22)$$

$$\mathbb{P}_I^0\{R_{\text{ord}} > u\} = \frac{1}{c_I^{(m,a)}} \sum_{k \geq 0} \frac{(-1)^k}{k!} \int_{(-u,u)^k} g_{I,k+1}^{(m,a)}(0, t_1, \dots, t_k) \prod_j dt_j. \quad (23)$$

*Both sums terminate locally. At every  $u$  at which the boundary has zero Palm probability, the same laws are obtained by choosing an  $a$ -point of any fixed twist uniformly among those with  $\Re \rho \in I$  and  $0 < \Im \rho < T$ , and then letting  $T \rightarrow \infty$ .*

*The forward-gap distribution is absolutely continuous away from a possible hard-core endpoint, with density*

$$f_{+,I}^{(m,a)}(u) = \frac{1}{c_I^{(m,a)}} \sum_{k \geq 0} \frac{(-1)^k}{k!} \int_{(0,u)^k} g_{I,k+2}^{(m,a)}(0, t_1, \dots, t_k, u) \prod_j dt_j. \quad (24)$$

*Proof.* Campbell's formula says that the  $k$ th factorial moment density under the reduced Palm law is

$$\frac{g_{I,k+1}^{(m,a)}(0, t_1, \dots, t_k)}{c_I^{(m,a)}}.$$

Apply the finite void expansion (14) under this Palm law to  $(0, u)$  and  $(-u, u)$ ; this gives (22) and (23). Differentiating the finite sum in (22) and using symmetry among the  $k$  integration variables gives (24).

For the deterministic assertion, apply Theorem 12.3 to the bounded local Campbell functional which sums a bounded continuous mark over the roots in a unit-height window. The exceptional configurations having a root on a boundary or a multiple root form a Haar-null set. The numerator of the empirical Palm average converges to its Campbell expectation, while the denominator is  $c_I^{(m,a)}T + o(T)$ . Approximation from above and below handles the gap indicators at every Palm continuity point.  $\square$

The unconditioned projected void law is, similarly,

$$V_I(u) := \mathbb{P}\{\eta_{m,U,a,I}(0, u) = 0\} = \sum_{k \geq 0} \frac{(-1)^k}{k!} \int_{(0,u)^k} g_{I,k}^{(m,a)}(\mathbf{t}) \prod_j dt_j. \quad (25)$$

## 12.5 Confluent repulsion and the planar nearest-neighbour law

The following estimate both controls all diagonal singularities and strengthens the pair-repulsion statement to arbitrary order.

**Theorem 12.7** (Vandermonde  $k$ -repulsion). *Let  $K \Subset D$  and  $k \geq 2$ . There are  $r_K > 0$  and  $C_{K,k} < \infty$  such that whenever  $z_1, \dots, z_k \in K$  lie in a disc of radius  $r_K$  and are pairwise distinct,*

$$K_k^{(m,a)}(z_1, \dots, z_k) \leq C_{K,k} \prod_{1 \leq i < j \leq k} |z_i - z_j|^2. \quad (26)$$

*Thus  $K_k^{(m,a)}$  extends continuously by zero to the complete diagonal. The same argument applied separately to each collision cluster gives a locally bounded continuous extension across every partial*

diagonal. In particular, if  $k$  points are constrained to a disc of radius  $r$ , their  $k$ th factorial mass is  $O(r^{k(k+1)})$ .

*Proof.* Let

$$\Delta(\mathbf{z}) = \prod_{i < j} (z_j - z_i).$$

Replace the value vector  $(F(z_1), \dots, F(z_k))$  by its Newton divided differences. The complex determinant of this linear transformation is  $\Delta(\mathbf{z})^{-1}$ . Lemma 12.2 shows that the density of the divided-difference vector remains uniformly bounded as the nodes merge. Consequently the density of the value vector at  $(a, \dots, a)$  is  $O(|\Delta(\mathbf{z})|^{-2})$ .

On the event  $F(z_j) = a$  for every  $j$ , factor

$$F(z) - a = \prod_{j=1}^k (z - z_j) H(z)$$

in a fixed disc slightly larger than the cluster. The family  $F_{m,u} - a$  is uniformly bounded on that larger disc. Since its boundary stays a fixed distance from every  $z_j$ , the maximum principle applied to the quotient gives  $|H(z_j)| \leq C_{K,k}$ . Hence

$$\prod_{j=1}^k |F'(z_j)|^2 = \prod_{j=1}^k \left| H(z_j) \prod_{\ell \neq j} (z_j - z_\ell) \right|^2 \ll_{K,k} |\Delta(\mathbf{z})|^4.$$

Multiplying this conditional derivative bound by the value-density bound and using Kac–Rice proves (26). If only a proper subset of the points coalesces, perform the divided-difference change only inside that subset; the remaining value coordinates stay uniformly nondegenerate. This proves the assertion concerning partial diagonals. Scaling  $z_j = z_0 + rw_j$  gives  $r^{k(k-1)}$  from the squared Vandermonde and  $r^{2k}$  from planar volume, which proves the final estimate.  $\square$

For the exact leading pair coefficient define

$$\mathcal{C}_m(\sigma, a) := \mathbb{E} \left[ |F''_{m,U}(\sigma)|^4 \delta_a(F_{m,U}(\sigma)) \delta_0(F'_{m,U}(\sigma)) \right]. \quad (27)$$

**Theorem 12.8** (Confluent pair law). *Uniformly for  $s$  in compact subsets of  $D$ , as  $h \rightarrow 0$  in  $\mathbb{C}$ ,*

$$\boxed{K_2^{(m,a)}(s, s+h) = \frac{|h|^2}{16} \mathcal{C}_m(\Re s, a) + O(|h|^3).} \quad (28)$$

Moreover  $\mathcal{C}_m(\sigma, a) > 0$  if and only if  $(a, 0)$  is an interior point of the support of  $(F_{m,U}(\sigma), F'_{m,U}(\sigma))$ .

*Proof.* By vertical invariance take  $s = \sigma$ . Put

$$X = F_{m,U}(s), \quad Y = F'_{m,U}(s),$$

and

$$W_h = \frac{2}{h^2} (F_{m,U}(s+h) - F_{m,U}(s) - hF'_{m,U}(s)).$$

Lemma 12.2 gives a smooth density  $q_h(x, y, w)$  for  $(X, Y, W_h)$ , with  $q_h \rightarrow q_0$ , together with its first derivatives, where  $q_0$  is the density of  $(F, F', F'')$  at  $s$ . Taylor's formula gives, uniformly in  $U$ ,

$$F(s+h) = X + hY + \frac{1}{2}h^2W_h,$$

$$F'(s+h) = Y + hW_h + O(|h|^2).$$

On the two-point zero slice  $F(s) = F(s+h) = a$ , one therefore has

$$Y = -\frac{1}{2}hW_h, \quad F'(s+h) = \frac{1}{2}hW_h + O(|h|^2).$$

The complex delta transformation

$$\delta_a(F(s+h)) = |h|^{-2}\delta_0(Y + \frac{1}{2}hW_h)$$

on the slice  $X = a$  yields

$$\begin{aligned} K_2^{(m,a)}(s, s+h) &= |h|^{-2} \int_{\mathbb{C}} \left| \frac{hw}{2} \right|^2 \left| \frac{hw}{2} + O(|h|^2) \right|^2 q_h \left( a, -\frac{hw}{2}, w \right) dA(w) \\ &= \frac{|h|^2}{16} \int_{\mathbb{C}} |w|^4 q_0(a, 0, w) dA(w) + O(|h|^3), \end{aligned}$$

which is (28). The marginal density of  $(F, F')$  at  $(a, 0)$  is  $\int_{\mathbb{C}} q_0(a, 0, w) dA(w)$ . Smoothness and nonnegativity show that this marginal is positive exactly when the weighted integral with  $|w|^4$  is positive. The standard coarea description of the phase map identifies positivity of the smooth density with the interior of its support.  $\square$

Let  $\mathbb{P}_{I,\text{pl}}^0$  be the Palm law obtained by choosing a typical zero per unit height with real part in  $I$ , translating its ordinate to zero, and retaining only zeros whose real parts also lie in  $I$ . Let  $R_{\text{pl}}$  be the Euclidean distance from the anchor to the nearest remaining zero.

**Corollary 12.9** (Quartic planar nearest-neighbour law). *If  $c_I^{(m,a)} > 0$ , then*

$$\boxed{\mathbb{P}_{I,\text{pl}}^0\{R_{\text{pl}} \leq r\} = \frac{\pi r^4}{32c_I^{(m,a)}} \int_I \mathcal{C}_m(\sigma, a) d\sigma + O_I(r^5).} \quad (29)$$

*When the integral is positive, the planar nearest-neighbour repulsion exponent is therefore exactly four.*

*Proof.* The Palm expected number of other zeros within distance  $r$  of the anchor is

$$\frac{1}{c_I^{(m,a)}} \int_I \int_{\substack{|h| < r \\ \sigma + \Re h \in I}} K_2^{(m,a)}(\sigma, \sigma+h) dA(h) d\sigma.$$

Theorem 12.8 and  $\int_{|h| < r} |h|^2 dA(h) = \pi r^4/2$  give the displayed main term; anchors within  $r$  of an endpoint of  $I$  and the cubic remainder contribute  $O(r^5)$ . The difference between the expected number and the probability of at least one neighbour is bounded by the Palm second factorial moment. By Theorem 12.7, the latter is

$$O\left(\int_{|h_1|, |h_2| < r} |h_1 h_2 (h_1 - h_2)|^2 dA(h_1) dA(h_2)\right) = O(r^{10}),$$

which does not alter the stated error term.  $\square$

## 12.6 Symbolic deterministic hard cores

We now specialize to the unweighted zero problem  $m = 0$ ,  $a = 0$ . The following statements contain no numerical threshold and require no computer-assisted claim. Put

$$B_j(\sigma) := \sum_p p^{-\sigma} (\log p)^j, \quad j = 1, 2, \quad (30)$$

and define

$$f_\sigma(\lambda) := \sum_{p \geq 3} p^{-\sigma} \left| 1 - \lambda \log \frac{p}{2} \right|, \quad \lambda > 0, \quad (31)$$

$$\mathbf{m}(\sigma) := \sup_{\lambda > 0} \frac{2^{-\sigma} - f_\sigma(\lambda)}{\lambda}. \quad (32)$$

Whenever  $\mathbf{m}(\sigma) > 0$ , let  $d(\sigma)$  be the unique solution in  $(0, \sigma - 1)$  of

$$d(\sigma) B_2(\sigma - d(\sigma)) = 2\mathbf{m}(\sigma). \quad (33)$$

**Theorem 12.10** (Derivative and planar hard core). *Let  $\omega$  be any unimodular twist and let  $\rho = \sigma + it$  be a zero of  $F_{0,\omega}$ . Then*

$$\boxed{|F'_{0,\omega}(\rho)| \geq \mathbf{m}(\sigma)}. \quad (34)$$

*If  $\mathbf{m}(\sigma) > 0$ , the zero is simple and every other zero  $\rho'$  of the same twist satisfies*

$$\boxed{|\rho' - \rho| \geq d(\sigma)}. \quad (35)$$

*Consequently  $K_2^{(0,0)}(s, s') = 0$  throughout the corresponding forbidden region.*

*Proof.* Write  $U_p = \omega_p p^{-it}$ ,  $S = \sum_{p \geq 3} p^{-\sigma} U_p$ , and  $Y = F'_{0,\omega}(\rho)$ . The zero equation gives  $|S| = 2^{-\sigma}$ . Since

$$Y = - \sum_{p \geq 3} \log(p/2) p^{-\sigma} U_p,$$

for every  $\lambda > 0$ ,

$$2^{-\sigma} = |S| \leq \sum_{p \geq 3} p^{-\sigma} \left| 1 - \lambda \log \frac{p}{2} \right| + \lambda |Y|.$$

Taking the supremum proves (34).

Let  $r = |\rho' - \rho| < \sigma - 1$ ; otherwise (35) is automatic. Taylor's integral formula along the segment from  $\rho$  to  $\rho'$  and the bound  $|F''_{0,\omega}(z)| \leq B_2(\Re z)$  give

$$\mathbf{m}(\sigma) r \leq \frac{1}{2} r^2 B_2(\sigma - r).$$

The function  $r \mapsto r B_2(\sigma - r)$  is strictly increasing, so  $r \geq d(\sigma)$ . Since no twist can realize two zeros in the open forbidden set, its Haar factorial moment measure is zero there; continuity of  $K_2$  off the diagonal makes the density identically zero there.  $\square$

For unequal real parts define

$$\Lambda(\sigma_1, \sigma_2) := \sup_{\lambda \geq 0} \left\{ \lambda 2^{-\sigma_2} - \sum_{p \geq 3} |p^{-\sigma_1} - \lambda p^{-\sigma_2}| \right\}, \quad (36)$$

$$H(\sigma_1, \sigma_2) := \frac{[\Lambda(\sigma_1, \sigma_2) - 2^{-\sigma_1}]_+}{B_1(\sigma_1)}. \quad (37)$$



**Theorem 12.11** (Symbolic ordinate separation). *If  $\rho_j = \sigma_j + it_j$  are two zeros of the same unimodular twist and  $\sigma_1 \neq \sigma_2$ , then*

$$\boxed{|t_2 - t_1| \geq H(\sigma_1, \sigma_2)}. \quad (38)$$

*The same assertion with the indices interchanged may be used, so the maximum of the two directed bounds is valid.*

*Proof.* Set  $h = t_2 - t_1$ ,  $U_p = \omega_p p^{-it_1}$ , and  $V_p = U_p p^{-ih}$ . Rotate the second zero equation so that, for suitable  $x_p \in [-1, 1]$ ,

$$\sum_{p \geq 3} p^{-\sigma_2} x_p = 2^{-\sigma_2}.$$

Since  $|e^{ih \log p} - 1| \leq |h| \log p$ , the first zero equation implies

$$\sum_{p \geq 3} p^{-\sigma_1} x_p - |h| B_1(\sigma_1) \leq 2^{-\sigma_1}.$$

For every  $\lambda \geq 0$ ,

$$\sum_{p \geq 3} p^{-\sigma_1} x_p \geq \lambda 2^{-\sigma_2} - \sum_{p \geq 3} |p^{-\sigma_1} - \lambda p^{-\sigma_2}|.$$

Optimizing in  $\lambda$  and combining the last two inequalities proves (38).  $\square$

**Corollary 12.12** (A symbolic strip hard core). *Let  $I \Subset (1, \infty)$  be an interval on which*

$$d_I := \inf_{\sigma \in I} d(\sigma) > 0.$$

*Choose  $0 < \delta < d_I$ , and suppose*

$$H_{I,\delta} := \inf_{\substack{\sigma_1, \sigma_2 \in I \\ |\sigma_1 - \sigma_2| \geq \delta}} \max\{H(\sigma_1, \sigma_2), H(\sigma_2, \sigma_1)\} > 0.$$

*Then every two zeros in  $I + i\mathbb{R}$  have ordinate difference at least*

$$h_I := \min \left\{ \sqrt{d_I^2 - \delta^2}, H_{I,\delta} \right\} > 0. \quad (39)$$

*For  $0 < u < h_I$  the projected Haar process has at most one point in every interval of length  $u$ , and hence*

$$\boxed{\mathbb{P}_I^0\{G_+ > u\} = \mathbb{P}_I^0\{R_{\text{ord}} > u\} = 1, \quad V_I(u) = 1 - c_I^{(0,0)} u.} \quad (40)$$

*Proof.* If  $|\sigma_1 - \sigma_2| < \delta$ , the planar bound gives  $|t_1 - t_2| \geq \sqrt{d_I^2 - \delta^2}$ . Otherwise apply Theorem 12.11. Thus an interval shorter than  $h_I$  contains at most one projected point. Its expected count is  $c_I^{(0,0)} u$ , so its void probability is exactly one minus that expectation; the Palm assertions are immediate.  $\square$

There is a complementary no-core statement which requires no numerical construction.

**Corollary 12.13** (Exact small-gap alternative). *For a fixed-order projected process, suppose that*

$$g_{I,2}^{(m,a)}(0) > 0.$$

Then, as  $u \downarrow 0$ ,

$$\mathbb{P}_I^0\{G_+ \leq u\} = \frac{g_{I,2}^{(m,a)}(0)}{c_I^{(m,a)}}u + o(u), \quad (41)$$

$$\mathbb{P}_I^0\{R_{\text{ord}} \leq u\} = \frac{2g_{I,2}^{(m,a)}(0)}{c_I^{(m,a)}}u + o(u), \quad (42)$$

$$V_I(u) = 1 - c_I^{(m,a)}u + \frac{1}{2}g_{I,2}^{(m,a)}(0)u^2 + o(u^2). \quad (43)$$

In particular the ordinate process has no exact hard core.

*Proof.* Insert the continuous density  $g_{I,2}^{(m,a)}$  into (22), (23), and (25). Terms involving  $g_{I,k}$  with  $k \geq 3$  are  $O(u^2)$  in the Palm expansions and  $O(u^3)$  in the unconditioned expansion, because the confluent bounds above make these densities locally bounded. The asserted leading terms follow.  $\square$

*Remark 12.14* (Dominant-prime version at every fixed derivative level). The deterministic argument is not restricted to  $m = 0$  or to the prime 2. Put  $a_{m,p}(\sigma) = (\log p)^m p^{-\sigma}$  and fix a candidate dominant prime  $q$ . Then every zero of  $F_{m,\omega}$  on  $\Re s = \sigma$  satisfies

$$|F'_{m,\omega}(s)| \geq \sup_{\lambda \in \mathbb{R} \setminus \{0\}} \frac{a_{m,q}(\sigma) - \sum_{p \neq q} a_{m,p}(\sigma) |1 - \lambda \log(p/q)|}{|\lambda|}.$$

Whenever the right side is positive, the Taylor argument gives a planar hard core, with  $B_2$  replaced by  $\sum_p (\log p)^{m+2} p^{-\sigma}$ . This supplies symbolic hard-core neighbourhoods at any internal dominance edge for which the displayed dual margin is positive. A similarly directed linear program, replacing  $p^{-\sigma_j}$  in (36) by  $a_{m,p}(\sigma_j)$  and replacing  $B_1$  by  $\sum_{p \neq q} a_{m,p}(\sigma_1) |\log(p/q)|$ , gives ordinate separation. No claim of uniformity as  $m \rightarrow \infty$  is made.

## 12.7 What is closed and what remains open

Theorems 12.3–12.6 close the qualitative fixed- order local-statistics problem: for every fixed  $m$ , every target  $a$ , and every unimodular twist, the full local process, every factorial correlation, every finite-window count law, every void law, and every Palm nearest-neighbour law exist and are universal. Theorem 12.7 and Corollary 12.9 give the corresponding bulk collision law, while Theorems 12.10 and 12.11 give the opposite deterministic hard-core regime.

What remains open is effective rather than qualitative: a rate in the vertical limit, estimates uniform in a derivative order that grows with height, and closed-form evaluation of the Kac–Rice integrals. Numerical candidate values for hard-core transition points and computer-located multiple zeros are useful evidence, but are not used in any theorem of this section unless accompanied by independent interval-arithmetic certificates.

## 13 Ordinate phase tomography and universal edge rigidity

The horizontal divisor laws developed above deliberately average out the coefficient phases. This section records the complementary fact that the *anchored ordinates* of points approaching a right edge retain those phases with no loss. Combining this elementary rigidity with the deterministic hard core gives a universal rooted edge profile, an edge-layer spacing theorem, and a stable inverse problem. None of the results below interprets a limiting density as a first-occurrence height.

Throughout, let

$$F_c(s) = \sum_p c_p p^{-s}, \quad A_c(\sigma) = \sum_p |c_p| p^{-\sigma}, \quad \Re s > 1,$$

where  $c = (c_p)_p$  is bounded. The same symbol  $A_c$  will denote the holomorphic Dirichlet series  $\sum_p |c_p| p^{-s}$  when its argument is complex. We write  $\mathcal{S}_c = \{p : c_p \neq 0\}$ .

### 13.1 The exact phase-defect identities

The following identity is a quantitative equality case of the triangle inequality. It is the basic input for every inverse statement in this section.

**Theorem 13.1** (Exact phase defect at a nonzero target). *Let  $a \neq 0$ , put  $\alpha = a/|a|$ , and suppose that  $\rho = \sigma + it$  is an  $a$ -point of  $F_c$ . Define*

$$\varepsilon_a(\rho) = A_c(\sigma) - |a| \geq 0, \quad \zeta_p(\rho) = \frac{c_p}{|c_p|} \bar{\alpha} p^{-it} \quad (p \in \mathcal{S}_c).$$

Then

$$\sum_{p \in \mathcal{S}_c} |c_p| p^{-\sigma} |1 - \zeta_p(\rho)|^2 = 2\varepsilon_a(\rho). \quad (44)$$

Consequently, for every  $p \in \mathcal{S}_c$ ,

$$\left| \frac{c_p}{|c_p|} - \alpha p^{it} \right| \leq \left( \frac{2\varepsilon_a(\rho)}{|c_p| p^{-\sigma}} \right)^{1/2}. \quad (45)$$

*Proof.* The equation  $F_c(\rho) = a$  becomes

$$\sum_{p \in \mathcal{S}_c} |c_p| p^{-\sigma} \zeta_p = |a|.$$

Taking real parts and using  $|1 - \zeta|^2 = 2(1 - \Re \zeta)$  on the unit circle proves (44). Dropping every term but the  $p$ th gives (45).  $\square$

Suppose that  $|a|$  lies in the range of  $A_c$ , and put

$$E_c(a) = A_c^{-1}(|a|).$$

The right-edge theorem gives  $a$ -points with real parts tending to  $E_c(a)$  from the left. The preceding equality immediately makes their ordinates inverse-complete.

**Corollary 13.2** (Fixed-target recovery). *If  $\rho_n = \sigma_n + it_n$  are  $a$ -points and  $\sigma_n \rightarrow E_c(a)$ , then, for every  $p \in \mathcal{S}_c$ ,*

$$\frac{c_p}{|c_p|} = \frac{a}{|a|} \lim_{n \rightarrow \infty} p^{it_n}. \quad (46)$$

*In particular, the following two pieces of data determine the complete coefficient system  $c$ :*

- (i) *edge samples  $(r_j, E_c(r_j))$  for distinct  $r_j$  having one accumulation point in the interior of the edge range;*
- (ii) *one sequence of points of one fixed nonzero target tending to its right edge.*

*Proof.* As  $\sigma_n \rightarrow E_c(a)$ , continuity gives  $\varepsilon_a(\rho_n) \rightarrow 0$ , so (46) follows from (45). The sparse edge samples recover  $A_c$  and hence every  $|c_p|$  by Corollary 5.9; the displayed limit then recovers the phases.  $\square$

For zeros of a unimodular twist, the least-prime term is separated from the tail. Let

$$S = \sigma_e, \quad D(\sigma) = \sum_{p \geq 3} p^{-\sigma} - 2^{-\sigma}, \quad (47)$$

so  $D(S) = 0$  and  $D(\sigma) > 0$  for  $1 < \sigma < S$ . Recall that

$$2^{-S} = \sum_{p \geq 3} p^{-S}.$$

**Corollary 13.3** (Exact zero defect and relative phase recovery). *Let  $F_\omega(s) = \sum_p \omega_p p^{-s}$  with  $|\omega_p| = 1$ , and let  $\rho = \sigma + it$  be a zero. Then*

$$\sum_{p \geq 3} p^{-\sigma} \left| \frac{\omega_p}{\omega_2} + \left(\frac{p}{2}\right)^{it} \right|^2 = 2D(\sigma). \quad (48)$$

Hence

$$\left| \frac{\omega_p}{\omega_2} + \left(\frac{p}{2}\right)^{it} \right| \leq \sqrt{2D(\sigma)p^\sigma} \quad (p \geq 3). \quad (49)$$

If  $\rho_n = \sigma_n + it_n$  and  $\sigma_n \rightarrow S$ , then

$$\frac{\omega_p}{\omega_2} = - \lim_{n \rightarrow \infty} \left(\frac{p}{2}\right)^{it_n} \quad (p \geq 3). \quad (50)$$

Thus one fixed-level zero sequence approaching the edge recovers the whole twist modulo its unavoidable common unimodular factor.

*Proof.* After division of  $F_\omega(\rho) = 0$  by  $\omega_2 2^{-it}$ , the tail has value  $-2^{-\sigma}$ . Expanding the left side of (48) therefore gives

$$2 \sum_{p \geq 3} p^{-\sigma} - 2 \cdot 2^{-\sigma} = 2D(\sigma).$$

The remaining assertions follow by dropping terms and letting  $\sigma \uparrow S$ .  $\square$

## 13.2 Universal rooted profiles

The phase defect controls not merely one coefficient at a time but the entire analytic function rooted at a near-edge point.

**Theorem 13.4** (Rooted profile at a nonzero target). *Fix the coefficient moduli  $(|c_p|)_p$ , let  $a \neq 0$ , and put  $E = E_c(a)$  and  $\alpha = a/|a|$ . The coefficient phases may vary with  $n$ . If  $\rho_n = \sigma_n + it_n$  is an  $a$ -point and  $\sigma_n \rightarrow E$ , then*

$$\bar{\alpha} F_{c(n)}(\rho_n + z) - |a| \longrightarrow H_{c,a}(z) := A_c(E + z) - A_c(E) \quad (51)$$

locally uniformly on  $\Re z > 1 - E$ . Consequently the translated  $a$ -point divisors converge locally vaguely, with multiplicity, to the zero divisor of  $H_{c,a}$ .

Moreover,

$$H_{c,a}(0) = 0, \quad H'_{c,a}(0) = A'_c(E) < 0,$$

and all normalized jets at the rooted points converge to the corresponding jets of  $H_{c,a}$ . If the support contains two distinct primes, then 0 is the only zero of  $H_{c,a}$  on the imaginary axis.

*Proof.* For each supported prime, [Theorem 13.1](#) gives

$$\frac{c_p^{(n)}}{\bar{\alpha} |c_p|} p^{-it_n} \longrightarrow 1.$$

On a compact subset of  $\Re z > 1 - E$ , the summands are dominated, for all large  $n$ , by a fixed convergent series  $Cp^{-1-\delta}$ . Dominated convergence proves (51); convergence of all derivatives follows either in the same way on a slightly smaller half-plane or from Cauchy's formula. Hurwitz's theorem, applied on compact sets whose boundaries avoid the limiting divisor, gives divisor convergence.

The derivative formula is immediate. If  $H_{c,a}(iy) = 0$ , equality in the triangle inequality forces  $p^{-iy} = 1$  for every supported prime. For two distinct primes this is impossible when  $y \neq 0$ , by unique factorization.  $\square$

The zero edge has a different canonical profile because one term opposes the entire tail.

**Theorem 13.5** (Universal rooted zero-edge profile). *Allow both the twist  $\omega^{(n)}$  and the zero to vary with  $n$ . Let*

$$\rho_n = \sigma_n + it_n, \quad \sigma_n \rightarrow S, \quad U_{2,n} = \omega_2^{(n)} 2^{-it_n}.$$

Then

$$G_n(z) := \frac{F_{\omega^{(n)}}(\rho_n + z)}{U_{2,n}} \longrightarrow H_e(z) := 2^{1-(S+z)} - \mathcal{P}(S+z) \quad (52)$$

locally uniformly on

$$\Omega_e = \{z : \Re z > 1 - S\}. \quad (53)$$

The convergence is uniform over all twists and all zeros with  $D(\Re \rho) \leq \varepsilon$  as  $\varepsilon \downarrow 0$ .

Write

$$\kappa_e = \sum_{p \geq 3} \log(p/2) p^{-S} = -D'(S) > 0. \quad (54)$$

Then

$$H_e(0) = 0, \quad H'_e(0) = \kappa_e, \quad \frac{F'_{\omega^{(n)}}(\rho_n)}{U_{2,n}} \longrightarrow \kappa_e, \quad (55)$$

and the analogous assertion holds for every higher derivative.

More quantitatively, let  $K \Subset \Omega_e$  and put  $s_K = \min_{z \in K} \Re(S+z) > 1$ . If  $\varepsilon_n = D(\sigma_n)$ , then

$$\sup_{z \in K} |G_n(z) - H_e(z)| \ll_K \begin{cases} \varepsilon_n^{(s_K-1)/(S-1)}, & 1 < s_K < (S+1)/2, \\ \sqrt{\varepsilon_n \log(1/\varepsilon_n)}, & s_K = (S+1)/2, \\ \sqrt{\varepsilon_n}, & s_K > (S+1)/2. \end{cases} \quad (56)$$

At either transition one may instead use an arbitrarily small loss in the exponent. In particular, an  $O_K(\sqrt{\varepsilon})$  estimate on the whole of (53) is not asserted.

*Proof.* Put

$$r_{p,n} = \frac{\omega_p^{(n)} p^{-it_n}}{\omega_2^{(n)} 2^{-it_n}} \quad (p \geq 3).$$

Then (48) is equivalently

$$\sum_{p \geq 3} p^{-\sigma_n} |r_{p,n} + 1|^2 = 2\varepsilon_n. \quad (57)$$

For each fixed  $p$ , this implies  $r_{p,n} \rightarrow -1$ . Since  $|r_{p,n} + 1| \leq 2$ , dominated convergence gives (52) on every compact subset of  $\Omega_e$ . It also shows that the convergence is uniform in the choice of twist once only the defect is prescribed.

For completeness, the rate follows from a prime split. If  $s = \min_{z \in K} \Re(\sigma_n + z)$  and  $X \geq 3$ , then

$$\sum_{p \geq 3} |r_{p,n} + 1| p^{-s} \leq \sqrt{2\varepsilon_n} \left( \sum_{p \leq X} p^{\sigma_n - 2s} \right)^{1/2} + 2 \sum_{p > X} p^{-s}. \quad (58)$$

Bounding the prime sums by the corresponding integer sums and taking  $X = \varepsilon_n^{-1/(S-1)}$  gives the three cases of (56). Also

$$S - \sigma_n = \frac{\varepsilon_n}{\kappa_e} + O(\varepsilon_n^2),$$

so replacing  $\sigma_n$  by  $S$  contributes only  $O_K(\varepsilon_n)$ . The derivative assertions follow from local uniform convergence or directly from (57).  $\square$

**Corollary 13.6** (Divisor and rooted Palm collapse). *Let*

$$\mathcal{D}_{\omega,\rho} = \sum_{F_\omega(\rho')=0} m_{\rho'} \delta_{\rho'-\rho}$$

*be the zero divisor viewed from the zero  $\rho$ . Uniformly over all twists and roots with  $D(\Re\rho) \leq \varepsilon$ ,*

$$\mathcal{D}_{\omega,\rho} \xrightarrow[\varepsilon \downarrow 0]{\text{vague}} \mathcal{D}_e := \text{div}(H_e) \quad \text{on } \Omega_e. \quad (59)$$

*Consequently, any Palm sampling scheme supported on roots in the layer  $D(\Re\rho) \leq \varepsilon$ , after recentering at its sampled root, converges in law to the point mass at the deterministic divisor  $\mathcal{D}_e$ .*

*Proof.* This is Hurwitz's theorem applied to the uniform compact convergence in Theorem 13.5. The Palm statement is immediate because the convergence is uniform over every admissible root, not only almost sure with respect to a particular sampling measure.  $\square$

The canonical function has a one-sided boundary geometry.

**Proposition 13.7** (Zero-free face of the rooted profile). *The zero at 0 of  $H_e$  is simple, and*

$$H_e(z) \neq 0 \quad (\Re z \geq 0, z \neq 0). \quad (60)$$

*If  $d_{\text{hc}}(\sigma)$  is any valid deterministic planar hard-core radius for zeros near  $S$ , continuous from the left at  $S$ , then*

$$H_e(z) \neq 0 \quad (0 < |z| < d_{\text{hc}}(S)). \quad (61)$$

*Proof.* Simplicity follows from (54). The open half-plane in (60) is zero-free by the definition of the projection edge. If  $H_e(iy) = 0$ , equality in the polygon inequality forces

$$e^{-iy \log(p/2)} = 1 \quad (p \geq 3).$$

Taking  $p = 3, 5$  and using unique factorization gives  $y = 0$ .

If (61) failed, Hurwitz's theorem would produce, near a nonzero limiting root  $z$ , a second zero of  $F_\omega$  at distance strictly smaller than  $d_{\text{hc}}(\sigma_n)$  from the root  $\rho_n$ , a contradiction.  $\square$

### 13.3 Exact boundary attainment and orbit classification

Right-edge projection is phase blind, but actual attainment of the edge is maximally phase rigid.

**Theorem 13.8** (Classification of boundary-attaining phases). *Let  $a = |a|\alpha \neq 0$  and  $E = A_c^{-1}(|a|)$ . Then*

$$F_c(E + it) = a \quad \Longleftrightarrow \quad \frac{c_p}{|c_p|} = \alpha p^{it} \quad \text{for every } p \in \mathcal{S}_c. \quad (62)$$

*If  $\mathcal{S}_c$  contains two distinct primes, the boundary ordinate  $t$  is unique.*

*For a unimodular prime twist,*

$$F_\omega(S + it) = 0 \quad \Longleftrightarrow \quad \frac{\omega_p}{\omega_2} = -\left(\frac{p}{2}\right)^{it} \quad (p \geq 3). \quad (63)$$

*The boundary ordinate is unique. If*

$$\omega_2^* = 1, \quad \omega_p^* = -1 \quad (p \geq 3),$$

*then the complete family of boundary-attaining zero twists is*

$$\omega_p = \kappa \omega_p^* p^{it}, \quad |\kappa| = 1, \quad t \in \mathbb{R}. \quad (64)$$

*This family is dense in the coefficient torus in the product topology and has Haar measure zero.*

*Proof.* At the boundary the sum of the moduli equals the modulus of the sum, so all summands must have the target phase. This proves (62). Equality at the zero edge forces every tail summand to have the phase opposite to the  $p = 2$  summand, which gives (63). Two different ordinates in the zero case would imply simultaneously

$$(3/2)^{i(t-t')} = (5/2)^{i(t-t')} = 1;$$

unique factorization forces  $t = t'$ . The nonzero-target case is the same argument with any two supported primes.

Equation (64) is a rearrangement of (63). Density follows from Kronecker's theorem. Haar nullity already follows after projection to the coordinates 2, 3, 5: on each bounded  $t$ -interval the image of  $(\kappa, t)$  is a two-dimensional Lipschitz image in the three-torus, and the whole image is a countable union of such null sets.  $\square$

### 13.4 An edge-layer spacing law

The fixed hard core becomes arbitrarily large in the vertical direction when both roots are required to lie in a shrinking edge layer.

**Theorem 13.9** (Divergent ordinate exclusion). *There are constants  $\varepsilon_*, c, C > 0$ , independent of the twist, such that whenever  $0 < \varepsilon < \varepsilon_*$  and  $\rho_j = \sigma_j + it_j$  are distinct zeros of the same unimodular twist satisfying*

$$D(\sigma_1) \leq \varepsilon, \quad D(\sigma_2) \leq \varepsilon,$$

one has

$$\boxed{|t_1 - t_2| \geq c \log(1/\varepsilon) - C.} \quad (65)$$

One may take

$$c = \frac{\pi}{(\log(3/2) + \log(5/2)) \log 10}, \quad (66)$$

after enlarging  $C$  and decreasing  $\varepsilon_*$ . In particular, the minimum ordinate spacing in the defect- $\varepsilon$  layer tends to infinity as  $\varepsilon \downarrow 0$ , uniformly over all twists.

*Proof.* The condition  $D(\sigma_j) \leq \varepsilon$  forces  $\sigma_j \rightarrow S$  uniformly as  $\varepsilon \downarrow 0$ . The rooted convergence to a function having a simple zero at 0 gives a uniform constant  $h_0 > 0$  such that  $|t_1 - t_2| \geq h_0$  for all sufficiently small  $\varepsilon$ . Equivalently, one may combine the planar hard core near  $S$  with  $|\sigma_1 - \sigma_2| \rightarrow 0$ .

Put  $h = |t_1 - t_2|$ ,  $a_0 = \log(3/2)$ , and  $b_0 = \log(5/2)$ . Applying (49) at the two roots gives

$$|e^{iha_0} - 1| \leq 2\sqrt{2\varepsilon 3^S}, \quad |e^{ihb_0} - 1| \leq 2\sqrt{2\varepsilon 5^S}. \quad (67)$$

For small  $\varepsilon$ , choose  $m, n \in \mathbb{Z}$  such that

$$|ha_0 - 2\pi m| \leq \pi\sqrt{2 3^S \varepsilon}, \quad |hb_0 - 2\pi n| \leq \pi\sqrt{2 5^S \varepsilon}. \quad (68)$$

The fixed lower bound  $h_0$  ensures that  $(m, n) \neq (0, 0)$  after a further decrease of  $\varepsilon_*$ . Hence

$$|mb_0 - na_0| \leq C_0\sqrt{\varepsilon}, \quad C_0 = \frac{b_0\sqrt{2 3^S} + a_0\sqrt{2 5^S}}{2}. \quad (69)$$

On the other hand,

$$mb_0 - na_0 = \log \left( \frac{(5/2)^m}{(3/2)^n} \right) \neq 0.$$

Writing the rational number in lowest terms, its numerator and denominator are at most  $10^M$ , where  $M = |m| + |n|$ . Thus

$$|mb_0 - na_0| \geq \frac{1}{10^M + 1} \geq \frac{1}{2 \cdot 10^M}. \quad (70)$$

Moreover,

$$M \leq Ah + 1, \quad A = \frac{a_0 + b_0}{2\pi}.$$

Combining (69) and (70), then absorbing fixed terms into  $C$ , gives (65) with (66).  $\square$

*Remark 13.10.* The logarithmic rate is elementary and intentionally conservative. An effective lower bound for the nonzero linear form  $m \log(5/2) - n \log(3/2)$  from the theory of logarithms of algebraic numbers would give a power-law exclusion with a very small effective exponent. No such external strengthening is needed here.



### 13.5 Edge-germ rigidity and stable matching

Two zero sets have the same *edge germ* if they agree in some strip  $S - \eta < \Re s \leq S$ . Equality after an imaginary translation is defined analogously.

**Theorem 13.11** (Edge-germ orbit rigidity). *For two unimodular twists  $\omega, \omega'$  and  $\tau \in \mathbb{R}$ , the following are equivalent:*

- (i) *the zero edge germ of  $F_{\omega'}$  is the translate by  $i\tau$  of the zero edge germ of  $F_{\omega}$ ;*
- (ii) *there exists  $\kappa \in \mathbb{T}$  such that*

$$\omega'_p = \kappa \omega_p p^{i\tau} \quad \text{for every prime } p. \quad (71)$$

When these conditions hold,

$$F_{\omega'}(s) = \kappa F_{\omega}(s - i\tau),$$

so the equality extends from the edge germs to the full zero sets. In particular, an anchored zero edge germ determines the twist modulo one common unimodular factor.

More generally, let  $|a| = |b| \neq 0$ , and let  $c, d$  have the same coefficient moduli. If the  $b$ -point edge germ of  $F_d$  is the translate by  $i\tau$  of the  $a$ -point edge germ of  $F_c$ , then

$$\frac{d_p}{|d_p|} = \frac{b}{a} \frac{c_p}{|c_p|} p^{i\tau} \quad (p \in \mathcal{S}_c). \quad (72)$$

The converse holds and gives equality of the complete translated divisors.

*Proof.* Choose zeros  $\sigma_n + it_n$  in the common germ with  $\sigma_n \rightarrow S$ . Their translates have ordinates  $t_n + \tau$ . Applying (50) to both twists gives

$$\frac{\omega'_p}{\omega'_2} = \frac{\omega_p}{\omega_2} \left(\frac{p}{2}\right)^{i\tau}.$$

Taking  $\kappa = (\omega'_2/\omega_2)2^{-i\tau}$  proves (71). The converse follows by direct substitution. The proof of (72) is identical, using (46); the factor  $b/a$  fixes the global phase.  $\square$

The reconstruction is stable on every finite collection of prime coordinates. The hard core supplies the otherwise missing canonical matching of noisy point clouds.

**Proposition 13.12** (Canonical noisy matching and inverse stability). *Let  $J \Subset (1, S]$  lie in a region where every zero of every twist has planar separation at least  $d_J > 0$ . Suppose two locally finite zero patterns in  $J + i\mathbb{R}$  have Hausdorff distance at most  $\eta < d_J/2$  and ignore points within distance  $\eta$  of the boundary of the observation window. Then proximity within  $\eta$  defines a unique bijective matching.*

*If  $\rho = \sigma + it$  and  $\rho' = \sigma' + it'$  are matched zeros of twists  $\omega$  and  $\omega'$ , respectively, then for every  $p \geq 3$ ,*

$$\begin{aligned} \left| \frac{\omega_p}{\omega_2} - \frac{\omega'_p}{\omega'_2} \right| &\leq \sqrt{2D(\sigma)p^\sigma} + \sqrt{2D(\sigma')p^{\sigma'}} + 2 \left| \sin \frac{(t - t') \log(p/2)}{2} \right| \\ &\leq \sqrt{2D(\sigma)p^\sigma} + \sqrt{2D(\sigma')p^{\sigma'}} + \min\{2, \eta \log(p/2)\}. \end{aligned} \quad (73)$$

*For nonzero targets, after normalizing coefficient phases by the target phase, the same formula holds with the two defects from Theorem 13.1, the weights  $|c_p|p^{-\sigma}$ , and  $\log p$  in place of  $\log(p/2)$ .*

*Proof.* Two candidates for the same matched point would be at mutual distance at most  $2\eta < d_J$ , contrary to separation. Mutual Hausdorff approximation then makes the matching bijective. The first two terms of (73) are (49); the last is the chordal distance between  $(p/2)^{it}$  and  $(p/2)^{it'}$ .  $\square$

There is also a useful forward stability statement. Let  $m_*(\sigma) > 0$  be a deterministic lower bound for  $|F'_\omega(\rho)|$  at a zero with real part  $\sigma$ , and let  $d_{\text{hc}}(\sigma) > 0$  be its hard-core radius.

**Proposition 13.13** (Forward root stability). *Let  $F_\omega(\rho) = 0$ ,  $\sigma = \Re \rho$ , and choose  $R > 0$  so that*

$$R < \min\{d_{\text{hc}}(\sigma), \sigma - 1\}, \quad R\mathcal{P}''(\sigma - R) \leq m_*(\sigma). \quad (74)$$

*For another twist  $\nu$ , put*

$$\delta_R = \sum_p |\omega_p - \nu_p| p^{-(\sigma-R)}.$$

*If  $\delta_R < m_*(\sigma)R/2$ , then  $F_\nu$  has exactly one zero  $\rho_\nu$  in  $|s - \rho| < R$ , counted with multiplicity, and*

$$|\rho_\nu - \rho| \leq \frac{2\delta_R}{m_*(\sigma)}. \quad (75)$$

*Proof.* On  $|z - \rho| = R$ , Taylor's formula gives

$$|F_\omega(z)| \geq m_*(\sigma)R - \frac{1}{2}\mathcal{P}''(\sigma - R)R^2 \geq \frac{1}{2}m_*(\sigma)R.$$

The hard core says that  $\rho$  is the only zero of  $F_\omega$  in the disc. Rouché's theorem gives one zero of  $F_\nu$ . Applying the same Taylor bound at that zero, now with its distance from  $\rho$  in place of  $R$ , proves (75).  $\square$

### 13.6 Phase-readable Delone divisors

The preceding results expose a sharp distinction between anchored and translation-averaged data.

**Corollary 13.14** (Delone geometry and phase readability). *Fix a twist  $\omega$  and an edge strip*

$$S - \eta < \Re s < S$$

*whose closure lies in the deterministic hard-core region. Above a sufficiently large ordinate, its zero divisor is a Delone set in the strip: it is uniformly discrete by the hard core and relatively dense by Corollary 4.3. Its anchored edge germ determines  $\omega$  modulo global phase, and its germ modulo imaginary translation determines the orbit of  $\omega$  modulo global phase and the vertical Kronecker flow.*

*Nevertheless, every translation-averaged finite-order correlation law of these divisors is the same for all unimodular twists. Thus the family is stationary-homometric at every finite order but anchored-rigid at the edge.*

*Proof.* Uniform discreteness and relative density are precisely the hard-core and bounded-window recurrence conclusions. Phase readability is Theorem 13.11; phase blindness of the averaged laws is the joint Kac–Rice universality established earlier.  $\square$

**Remark 13.15** (Logical scope). The exact defect identities, rooted limits, orbit classifications, and spacing theorem are deterministic. They do not require a numerical value for the onset of the hard-core region. Any decimal threshold for that onset should be quoted only after interval certification. Likewise, an edge density describes a limiting frequency after the height limit; without a uniform moving-edge error term it does not determine the first height at which a prescribed phase accuracy occurs.

## 14 Phase distortion and its exact zero-edge cost

The horizontal divisor law is phase blind, whereas the ordinate of a zero contains phase information. In this section we make that contrast quantitative. The natural loss function is not introduced ad hoc: at every zero it is exactly twice the polygonal defect from the outer edge. Combining this identity with the logarithmic edge profile gives a universal small-loss law for every unimodular prime twist.

Let

$$F_\omega(s) = \sum_{p \in \mathbb{P}} \omega_p p^{-s}, \quad |\omega_p| = 1,$$

and let  $\sigma_e$  be the unique solution of

$$\sum_{p \geq 3} p^{-\sigma_e} = 2^{-\sigma_e}.$$

For  $1 < \sigma < \sigma_e$  put

$$\varepsilon_0(\sigma) := \sum_{p \geq 3} p^{-\sigma} - 2^{-\sigma}. \quad (76)$$

Thus  $\varepsilon_0(\sigma) > 0$ ,  $\varepsilon_0(\sigma) \downarrow 0$  as  $\sigma \uparrow \sigma_e$ , and  $\varepsilon'_0(\sigma_e) = D_1(\sigma_e) < 0$  in the notation of [Section 10](#).

**Definition 14.1** (Relative-phase decoder and distortion). If  $\rho = \sigma + it$  is a zero of  $F_\omega$ , use the phase of the coefficient at 2 as reference and set

$$\hat{\vartheta}_p(t) := - \left( \frac{p}{2} \right)^{it}, \quad p \geq 3.$$

The associated weighted chordal distortion is

$$\mathcal{D}_\omega(\rho) := \sum_{p \geq 3} p^{-\sigma} \left| \frac{\omega_p}{\omega_2} - \hat{\vartheta}_p(t) \right|^2 = \sum_{p \geq 3} p^{-\sigma} \left| \frac{\omega_p}{\omega_2} + \left( \frac{p}{2} \right)^{it} \right|^2. \quad (77)$$

**Proposition 14.2** (Exact phase-defect identity). *Every zero  $\rho = \sigma + it$  of every unimodular twist satisfies*

$$\boxed{\mathcal{D}_\omega(\rho) = 2\varepsilon_0(\sigma)}. \quad (78)$$

Consequently, for every  $p \geq 3$ ,

$$\left| \frac{\omega_p}{\omega_2} + \left( \frac{p}{2} \right)^{it} \right| \leq \sqrt{2\varepsilon_0(\sigma)p^\sigma}. \quad (79)$$

In particular, any sequence of zeros with real parts tending to  $\sigma_e$  recovers every relative coefficient phase  $\omega_p/\omega_2$ .

*Proof.* This is [Corollary 13.3](#), rewritten in the notation of (77). The coordinatewise estimate follows by discarding every nonnegative summand except the one indexed by  $p$ .  $\square$

The identity allows the zero divisor itself to be marked by an intrinsic phase-reconstruction error. For sufficiently small  $d > 0$ , let  $\sigma(d)$  be the unique number near  $\sigma_e$  such that

$$2\varepsilon_0(\sigma(d)) = d, \quad (80)$$

and define

$$\mathcal{C}(d) := \int_{\sigma(d)}^{\sigma_e} G(\sigma, 0) \, d\sigma. \quad (81)$$

Here and below an isolated zero on the boundary line has no effect on a vertical density.

**Theorem 14.3** (Universal small-distortion law). *For  $\bullet \in \{\text{mult}, \text{dist}, \text{simp}\}$  let*

$$M_\omega^\bullet(d; T) := \#\{\rho : F_\omega(\rho) = 0, \ 0 < \Im \rho < T, \ \mathcal{D}_\omega(\rho) \leq d\}.$$

*For every fixed sufficiently small  $d > 0$ ,*

$$\lim_{T \rightarrow \infty} \frac{M_\omega^\bullet(d; T)}{T} = \mathcal{C}(d), \quad (82)$$

*and the limit is independent of  $\omega$  and of the counting convention. By the universal rooted edge profile, after decreasing the allowed range of  $d$ , every zero counted here has a twist-independent neighbourhood in which it is the unique zero and is simple.*

*Let*

$$\beta_e = \frac{1}{\sigma_e - 1}, \quad \ell_d = \log \frac{2}{d},$$

*and let  $K_e$  and  $\mathcal{J}_e$  be the constants and rate function of [Theorem 10.4](#). Then*

$$-\log \mathcal{C}(d) = \mathcal{J}_e(\sigma(d)) + O\left(\mathcal{J}_e(\sigma(d))^{1/2}\right), \quad (83)$$

*and hence*

$$-\log \mathcal{C}(d) = K_e \left(\frac{d}{2}\right)^{-\beta_e} \ell_d^{-(1+\beta_e)} \left(1 + (1 + \beta_e) \frac{\log \ell_d}{\ell_d} + O(\ell_d^{-1})\right). \quad (84)$$

*Proof.* By [Proposition 14.2](#), the condition  $\mathcal{D}_\omega(\rho) \leq d$  is precisely  $\Re \rho \in [\sigma(d), \sigma_e]$ . The phase-blind horizontal divisor theorem, including its distinct- and simple-point forms, therefore gives (82).

For the asserted pointwise isolation, choose a disc about 0 whose closure lies in the domain of [Theorem 13.5](#) and contains no other zero of  $H_e$ . Uniform rooted convergence and Rouché's theorem then give, for all sufficiently small  $d$ , exactly one simple zero in that disc—the rooted zero itself—uniformly in the twist.

It remains to check that integration in  $\sigma$  does not alter the exact logarithmic edge scale. Write  $u = \varepsilon_0(\sigma)$ ,  $x = d/2$ ,  $\sigma = \sigma(u)$ , and

$$R(u) := \mathcal{J}_e(\sigma(u)).$$

Since  $\varepsilon'_0(\sigma_e) \neq 0$ , the Jacobian  $|\sigma'(u)|$  is bounded above and below by positive constants for small  $u$ . The exact edge theorem gives, uniformly there,

$$-\log G(\sigma(u), 0) = R(u) + O(R(u)^{1/2}). \quad (85)$$

Moreover  $R$  decreases as  $u$  increases, tends to infinity at zero, and its closed asymptotic implies

$$R\left(x \left(1 - \frac{1}{R(x)}\right)\right) = R(x) + O(1). \quad (86)$$

For the upper bound, monotonicity of  $R$  and (85) give

$$\mathcal{C}(d) \leq Cx \exp\{-R(x) + O(R(x)^{1/2})\}.$$

For the lower bound, integrate only over

$$x \left(1 - \frac{1}{R(x)}\right) \leq u \leq x.$$

Equations (85)–(86) give

$$\mathcal{C}(d) \geq \frac{cx}{R(x)} \exp\{-R(x) + O(R(x)^{1/2})\}.$$

Now  $|\log(x/R(x))| = O(\log R(x) + \log(1/x)) = o(R(x)^{1/2})$ . The two bounds prove (83). Substituting  $x = d/2$  in the closed edge expansion proves (84); the integration error is smaller than every displayed inverse-logarithmic correction.  $\square$

The preceding theorem can be inverted without a probabilistic or mixing hypothesis. It is an inversion of limiting intensity, not a statement about the first zero observed at finite height. Since  $\varepsilon_0$  is strictly decreasing near  $\sigma_e$  and  $G(\sigma, 0) > 0$  immediately to its left,  $\mathcal{C}(d)$  is strictly increasing for all sufficiently small  $d > 0$  and therefore has the local inverse used below.

**Corollary 14.4** (Density–fidelity frontier). *Let  $d(c)$  be the local inverse of  $\mathcal{C}(d)$  near  $c = 0$ , and put  $Y = \log(1/c)$ . Then*

$$d(c) \sim 2 \left( K_e \beta_e^{1+\beta_e} \right)^{1/\beta_e} Y^{-1/\beta_e} (\log Y)^{-(1+\beta_e)/\beta_e}. \quad (87)$$

In particular, since  $1/\beta_e = \sigma_e - 1$  and  $(1 + \beta_e)/\beta_e = \sigma_e$ ,

$$d(c) \asymp [\log(1/c)]^{-(\sigma_e-1)} [\log \log(1/c)]^{-\sigma_e}. \quad (88)$$

*Proof.* Set

$$\varepsilon = AY^{-1/\beta_e} (\log Y)^{-q}.$$

Substitution in the leading term of (84) shows that  $q = (1 + \beta_e)/\beta_e$  and  $A = (K_e \beta_e^{1+\beta_e})^{1/\beta_e}$ . Since  $d = 2\varepsilon$ , this proves the assertion.  $\square$

There is also a useful finite-dimensional consequence. It must be stated as a sufficient certificate, rather than as the optimal cost of learning a finite set of phases.

**Corollary 14.5** (Certified relative-phase recovery through  $Q$ ). *Fix  $0 < \delta < 2$ . If a zero  $\rho = \sigma + it$  satisfies*

$$\mathcal{D}_\omega(\rho) \leq \delta^2 Q^{-\sigma_e}, \quad (89)$$

*then*

$$\left| \frac{\omega_p}{\omega_2} + \left( \frac{p}{2} \right)^{it} \right| \leq \delta \quad (3 \leq p \leq Q).$$

*The vertical intensity of the defect band certified by (89) is*

$$\mathcal{C}_{Q,\delta}^{\text{cert}} := \mathcal{C}(\delta^2 Q^{-\sigma_e}),$$

*and, for fixed  $\delta$  as  $Q \rightarrow \infty$ ,*

$$-\log \mathcal{C}_{Q,\delta}^{\text{cert}} \sim \kappa_{\text{cert}} \delta^{-2\beta_e} \left( \frac{Q}{\log Q} \right)^{1+\beta_e}, \quad \kappa_{\text{cert}} = K_e 2^{\beta_e} \sigma_e^{-(1+\beta_e)}. \quad (90)$$

*Thus, by the prime number theorem,*

$$-\log \mathcal{C}_{Q,\delta}^{\text{cert}} \sim \kappa_{\text{cert}} \delta^{-2\beta_e} \pi(Q)^{1+\beta_e}. \quad (91)$$

*Proof.* For  $p \leq Q$  and  $\sigma \leq \sigma_e$ , Proposition 14.2 gives

$$\left| \frac{\omega_p}{\omega_2} + \left( \frac{p}{2} \right)^{it} \right|^2 \leq \mathcal{D}_\omega(\rho) p^\sigma \leq \delta^2 Q^{-\sigma_e} Q^{\sigma_e} = \delta^2.$$

Substitute  $d = \delta^2 Q^{-\sigma_e}$  into (84). Since  $\sigma_e \beta_e = 1 + \beta_e$  and  $\log(2/d) = \sigma_e \log Q + O_\delta(1)$ , one obtains (90). Equation (91) follows from  $\pi(Q) \sim Q/\log Q$ .  $\square$

*Remark 14.6* (What the cost law does not say). The condition (89) is sufficient, not necessary. The first prime-indexed phases may happen to be recovered accurately while the distortion is carried by larger primes. Thus  $\mathcal{C}_{Q,\delta}^{\text{cert}}$  is the intensity of a canonically certified family, not the optimal sampling rate for finite-dimensional phase recovery.

Likewise,  $1/\mathcal{C}(d)$  is a reciprocal limiting intensity. The limit in (82) first sends  $T \rightarrow \infty$  with  $d$  fixed. It does not locate the first zero of distortion at most  $d$ , and one may not replace  $d$  by a moving threshold  $d(T)$  without a uniform effective divisor law. In particular, the present theorem does not prove an extreme-value law for the rightmost zero. The hard core prevents clustering at very small separation, but it does not turn a mean density into a first-hitting estimate.

### 14.1 The exact identity at a nonzero target

The elementary phase identity has a fully rigorous nonzero-target form. We record it separately because the corresponding logarithmic density cost requires an outer-edge theorem for  $G(\sigma, a)$  that is not part of the zero-edge theorem proved above.

**Proposition 14.7** (Exact distortion at a nonzero  $a$ -point). *Let  $a \neq 0$ , and suppose  $F_\omega(\rho) = a$  with  $\rho = \sigma + it$  and  $\sigma > 1$ . Put*

$$\varepsilon_a(\sigma) = \mathcal{P}(\sigma) - |a|,$$

*and define*

$$\mathcal{D}_{\omega,a}(\rho) := \sum_p p^{-\sigma} \left| \omega_p - \frac{a}{|a|} p^{it} \right|^2.$$

*Then*

$$\boxed{\mathcal{D}_{\omega,a}(\rho) = 2\varepsilon_a(\sigma)}. \tag{92}$$

*In particular,*

$$\left| \omega_p - \frac{a}{|a|} p^{it} \right| \leq \sqrt{2\varepsilon_a(\sigma) p^\sigma} \quad (p \in \mathbb{P}). \tag{93}$$

*Consequently any sequence of  $a$ -points approaching the outer edge  $E(a) = \mathcal{P}^{-1}(|a|)$  recovers all coefficient phases.*

*Proof.* Write  $a = |a|e^{i\alpha}$  and put  $\zeta_p = \omega_p e^{-it \log p - i\alpha}$ . The  $a$ -point equation is

$$\sum_p p^{-\sigma} \zeta_p = |a|.$$

Taking real parts and using  $|1 - \zeta|^2 = 2(1 - \Re \zeta)$  gives

$$\sum_p p^{-\sigma} |1 - \zeta_p|^2 = 2(\mathcal{P}(\sigma) - |a|).$$

Multiplication of each difference by the appropriate unimodular factor gives (92); the coordinate bound follows by discarding all other summands.  $\square$

*Remark 14.8* (Conditional nonzero-target extension). It is natural to expect the preceding small-distortion law at every fixed  $a \neq 0$ . With  $E = E(a)$  and  $\beta_a = (E - 1)^{-1}$ , the required input would be the general outer-edge estimate below, where

$$H_0(E) = \int_0^\infty (x - \log I_0(x)) x^{-1/E-1} dx$$

and

$$K(E) := \frac{E-1}{E} H_0(E)^{1+\beta_a} E^{-\beta_a} (1+\beta_a)^{-(1+\beta_a)} :$$

$$\begin{aligned} -\log G(\sigma, a) &= K(E) \varepsilon_a(\sigma)^{-\beta_a} L_a^{-(1+\beta_a)} \\ &\times \left( 1 + (1+\beta_a) \frac{\log L_a}{L_a} + O(L_a^{-1}) \right), \quad L_a = \log \frac{1}{\varepsilon_a(\sigma)}. \end{aligned}$$

The same PNT–Cramér mechanism should apply after replacing the dominant-prime cancellation by alignment of the full prime sum; on the edge fibre the derivative weight tends to  $|\mathcal{P}'(E)|^2$ . Nevertheless, the zero-edge theorem as stated proves only  $a = 0$ . A complete proof of the full-prime tilted local-density lower bound and of the derivative-weight comparison must be supplied before the displayed nonzero-target expansion, or any cost law derived from it, is promoted to a theorem. No such extension is used in the proved results of this section.

## 15 Uniform small-target surgery and divisor monodromy

The weak escape theorem describes what happens to the horizontal divisor measure as  $a \rightarrow 0$ , while the hard-core theorem controls individual zeros in a right-hand band. Together they give a stronger, pointwise description on every compact hard-core subband: its bounded  $a$ -points are uniformly controlled holomorphic perturbations of the zero divisor, whereas the missing mass is carried by a disjoint logarithmic lattice at infinity. These controlled persistent branches and the escaping branches have different monodromy about the singular target  $a = 0$ .

Throughout this section

$$F_\omega(s) = \sum_p \omega_p p^{-s}, \quad |\omega_p| = 1, \quad \mathcal{P}''(\sigma) = \sum_p (\log p)^2 p^{-\sigma}.$$

For  $\lambda > 0$  put

$$f_\sigma(\lambda) := \sum_{p \geq 3} p^{-\sigma} |1 - \lambda \log(p/2)|, \tag{ST.1}$$

$$q_\lambda(\sigma) := \frac{2^{-\sigma} - f_\sigma(\lambda)}{\lambda}, \tag{ST.2}$$

$$\kappa_\lambda := \lambda^{-1} + \log 2. \tag{ST.3}$$

The hard-core region is the open set

$$\mathcal{H} := \{\sigma > 1 : \mathfrak{m}(\sigma) > 0\}.$$

The preceding hard-core theorem gives  $|F'_\omega(\rho)| \geq \mathfrak{m}(\Re \rho)$  at every zero with real part in  $\mathcal{H}$ . This agrees with the notation in [Theorem 12.10](#); no numerical description of  $\mathcal{H}$  is needed below.

### 15.1 A derivative bound stable under perturbation of the target

The general nonzero-target estimate obtained by treating all prime terms symmetrically is useful for large targets. Near  $a = 0$  one should instead retain the distinguished  $p = 2$  term. This gives the estimate required for a uniform implicit-function theorem.

**Lemma 15.1** (Small-target derivative bound). *Let  $z = \sigma + it$ ,  $\sigma > 1$ , and suppose that  $F_\omega(z) = a$ . Then, for every  $\lambda > 0$ ,*

$$\boxed{|F'_\omega(z)| \geq q_\lambda(\sigma) - \kappa_\lambda|a|}. \quad (\text{ST.4})$$

At  $a = 0$ , taking the supremum over  $\lambda$  recovers the hard-core lower bound  $|F'_\omega(z)| \geq \mathfrak{m}(\sigma)$ .

*Proof.* Write  $U_p = \omega_p p^{-it}$ ,  $a_p = p^{-\sigma}$ , and

$$S = \sum_{p \geq 3} a_p U_p, \quad D = \sum_{p \geq 3} \log(p/2) a_p U_p.$$

Since  $a_2 U_2 + S = a$ ,

$$|S| \geq 2^{-\sigma} - |a|.$$

Moreover

$$F'_\omega(z) = -(\log 2)a - D,$$

and hence

$$S = \sum_{p \geq 3} (1 - \lambda \log(p/2)) a_p U_p - \lambda F'_\omega(z) - \lambda(\log 2)a.$$

The triangle inequality now gives

$$2^{-\sigma} - |a| \leq \mathfrak{f}_\sigma(\lambda) + \lambda |F'_\omega(z)| + \lambda(\log 2)|a|,$$

which is (ST.4). □

The estimate can be made uniform without using a numerically computed threshold. If  $K \Subset \mathcal{H}$  is a compact interval, compactness supplies a finite set  $\Lambda_K \subset (0, \infty)$  such that

$$\mu_K := \inf_{\sigma \in K} \max_{\lambda \in \Lambda_K} q_\lambda(\sigma) > 0. \quad (\text{ST.5})$$

Indeed, choose one positive witness  $q_\lambda(\sigma) > 0$  at each point and take a finite subcover. Set

$$\kappa_K := \max_{\lambda \in \Lambda_K} \kappa_\lambda. \quad (\text{ST.6})$$

**Corollary 15.2** (A small-target hard core). *Let  $K = [\alpha, \beta] \Subset \mathcal{H}$  and let  $0 \leq r < \mu_K / \kappa_K$ . Every  $a$ -point with  $|a| \leq r$  and real part in  $K$  satisfies*

$$|F'_\omega(z)| \geq \mu_K - \kappa_K r > 0. \quad (\text{ST.7})$$

*In particular all such points are simple. If  $z \neq w$  are two such  $a$ -points, then*

$$|z - w| \geq d_{K,r}, \quad (\text{ST.8})$$

*where  $d_{K,r} \in (0, \alpha - 1)$  is the unique solution of*

$$d_{K,r} \mathcal{P}''(\alpha - d_{K,r}) = 2(\mu_K - \kappa_K r). \quad (\text{ST.9})$$

*Thus simplicity and planar hard core persist uniformly under sufficiently small perturbations of the target.*



*Proof.* The first assertion follows from Lemma 15.1, choosing at each  $\sigma$  a maximizing witness in  $\Lambda_K$ . Put  $h = w - z$ . If  $|h| \geq \alpha - 1$  there is nothing to prove. Otherwise the segment from  $z$  to  $w$  lies in  $\Re s > 1$ , and Taylor's formula gives

$$0 = F_\omega(w) - F_\omega(z) = hF'_\omega(z) + h \int_0^1 (F'_\omega(z + \theta h) - F'_\omega(z)) d\theta.$$

Consequently

$$2|F'_\omega(z)| \leq |h| \mathcal{P}''(\Re z - |h|) \leq |h| \mathcal{P}''(\alpha - |h|).$$

The left-hand side of (ST.9), viewed as a function of  $d$ , is strictly increasing from 0 to  $+\infty$  on  $(0, \alpha - 1)$ . Equations (ST.7)–(ST.9) therefore give the claim.  $\square$

## 15.2 Uniform zero- $a$ -point continuation

The next theorem is stated with three nested intervals. This buffer is necessary: an  $O(|a|)$  displacement may cross a prescribed vertical boundary even when the correspondence itself is perfectly regular.

**Theorem 15.3** (Uniform holomorphic continuation of the persistent divisor). *Let*

$$I \Subset J \Subset K \Subset \mathcal{H}$$

*be compact intervals. There are constants  $a_0, C_1, C_2 > 0$ , depending only on these intervals, such that the following assertions hold simultaneously for every unimodular twist  $\omega$  and every  $|a| < a_0$ .*

- (i) *Every zero  $\rho$  of  $F_\omega$  with  $\Re \rho \in J$  has a unique associated  $a$ -point  $\rho(a)$  in a fixed disc about  $\rho$ . The map  $a \mapsto \rho(a)$  is holomorphic on  $|a| < a_0$ , and*

$$|\rho(a) - \rho| \leq C_1|a|, \quad \left| \rho(a) - \rho - \frac{a}{F'_\omega(\rho)} \right| \leq C_2|a|^2. \quad (\text{ST.10})$$

- (ii) *Every  $a$ -point  $z$  with  $\Re z \in I$  is  $\rho(a)$  for a unique zero  $\rho$  with  $\Re \rho \in J$ . Thus, writing*

$$Z_a(L) := \{z : F_\omega(z) = a, \Re z \in L\},$$

*one has the exact buffered identity*

$$Z_a(I) = \{\rho(a) : \rho \in Z_0(J), \Re \rho(a) \in I\}. \quad (\text{ST.11})$$

- (iii) *If  $d_0 > 0$  is a uniform zero hard-core constant on  $K$ , then, after decreasing  $a_0$  if necessary,*

$$|\rho_1(a) - \rho_2(a)| \geq d_0 - 2C_1|a| \geq \frac{d_0}{2} \quad (\text{ST.12})$$

*for distinct zeros  $\rho_1, \rho_2$  in  $J$ .*

*All constants and conclusions are uniform in the height of the points and in the twist.*

*Proof.* We give explicit safe choices. Let

$$b := \min\{\text{dist}(I, \mathbb{R} \setminus J), \text{dist}(J, \mathbb{R} \setminus K)\} > 0,$$

let  $\mu = \mu_K$ ,  $\kappa = \kappa_K$ , and put

$$M := \mathcal{P}''(\min K).$$

Let  $d_0 = d_{K,0}$  in Corollary 15.2, and choose

$$0 < \delta < \frac{1}{3} \min \left\{ b, d_0, \frac{\mu}{M} \right\}. \quad (\text{ST.13})$$

It is enough to take

$$a_0 < \min \left\{ \frac{\mu}{2\kappa}, \frac{\mu^2}{8M}, \frac{\mu\delta}{4}, \frac{\mu d_0}{8} \right\}. \quad (\text{ST.14})$$

Let  $F_\omega(\rho) = 0$  and  $\Re \rho \in J$ . On  $|w| = \delta$ ,

$$|F_\omega(\rho + w) - F'_\omega(\rho)w| \leq \frac{M}{2}\delta^2,$$

whereas  $|F'_\omega(\rho)| \geq \mu$ . Equations (ST.13)–(ST.14) and Rouché's theorem show that  $F_\omega - a$  has exactly one zero in  $B(\rho, \delta)$ . Repeating the comparison on the circle

$$|w| = \frac{2|a|}{\mu}$$

gives

$$|\rho(a) - \rho| \leq \frac{2|a|}{\mu}. \quad (\text{ST.15})$$

The zero is simple by Corollary 15.2; the implicit-function theorem and uniqueness in the fixed disc make  $\rho(a)$  holomorphic throughout the target disc. Taylor's formula once more yields

$$\left| \rho(a) - \rho - \frac{a}{F'_\omega(\rho)} \right| \leq \frac{M}{2\mu} |\rho(a) - \rho|^2 \leq \frac{2M}{\mu^3} |a|^2. \quad (\text{ST.16})$$

Thus one may take  $C_1 = 2/\mu$  and  $C_2 = 2M/\mu^3$  in part (i).

Conversely, suppose that  $F_\omega(z) = a$  and  $\Re z \in I$ . By (ST.7),  $|F'_\omega(z)| \geq \mu/2$ . On the circle

$$|w| = \frac{4|a|}{\mu},$$

the linear term  $F'_\omega(z)w$  has modulus at least  $2|a|$ , while the quadratic remainder has modulus at most

$$\frac{M}{2} \left( \frac{4|a|}{\mu} \right)^2 < |a|.$$

Rouché's theorem, now applied to  $F_\omega(z + w)$ , supplies exactly one zero within  $4|a|/\mu$  of  $z$ . By the buffer and (ST.14) this zero lies in  $J$ . The  $a$ -point  $z$  must be the branch already constructed from that zero, because the fixed  $\delta$ -disc contains only one  $a$ -point. The zero hard core and  $\delta < d_0/3$  make the correspondence injective and prove (ST.11).

Finally (ST.15) and the triangle inequality give

$$|\rho_1(a) - \rho_2(a)| \geq |\rho_1 - \rho_2| - \frac{4|a|}{\mu} \geq d_0 - \frac{4|a|}{\mu}.$$

The last condition in (ST.14) gives (ST.12). None of the estimates involves  $\Im \rho$  or the phases  $\omega_p$ , proving the asserted uniformity.  $\square$

*Remark 15.4* (Sharp scope of the continuation theorem). The buffers in Theorem 15.3 cannot simply be deleted from a statement about raw rectangular counts: a branch can cross either vertical boundary. More importantly, below  $\mathcal{H}$  arbitrary twists may have multiple zeros. There one retains local convergence of divisors with multiplicity, but not a uniform decomposition into simple one-point branches.

### 15.3 The zero-free corridor and two-scale degeneration

We now return to the full prime support. Put

$$A(\sigma) = \sum_p p^{-\sigma}, \quad B(\sigma) = 2^{-\sigma} - \sum_{p \geq 3} p^{-\sigma}. \quad (\text{ST.17})$$

Thus  $B(\sigma_e) = 0$  and  $B(\sigma) > 0$  for  $\sigma > \sigma_e$ . The phase-annulus criterion says that the real part of an  $a$ -point with  $r = |a|$  must satisfy

$$\max\{B(\sigma), 0\} \leq r \leq A(\sigma). \quad (\text{ST.18})$$

**Proposition 15.5** (A separating corridor). *Fix  $\beta > \sigma_e$ . For all sufficiently small  $r > 0$  there are numbers*

$$\beta < \ell(r) < u(r)$$

*such that no  $a$ -point with  $|a| = r$  has real part in  $[\beta, \ell(r))$ , and every  $a$ -point to the right of  $\beta$  lies in  $[\ell(r), u(r)]$ . The endpoints are characterized on the far-right monotone branch by*

$$B(\ell(r)) = r, \quad A(u(r)) = r, \quad (\text{ST.19})$$

and

$$\ell(r), u(r) = \frac{\log(1/r)}{\log 2} + O\left(r^{\log_2 3 - 1}\right). \quad (\text{ST.20})$$

*Consequently the bounded and escaping parts of the  $a$ -point divisor are separated by a genuine zero-free corridor whose length tends to infinity.*

*Proof.* Choose  $M > \beta$  so large that  $B$  is strictly decreasing on  $[M, \infty)$ ; this is possible because

$$B'(\sigma) = -(\log 2)2^{-\sigma} + \sum_{p \geq 3} (\log p)p^{-\sigma} < 0$$

eventually. Since  $B$  is positive on  $[\beta, M]$ , its minimum there is positive. For  $r$  below that minimum,  $B(\sigma) > r$  throughout  $[\beta, M]$  and until the unique far-right solution  $\ell(r)$  of  $B(\sigma) = r$ . Criterion (ST.18) excludes every  $a$ -point in this interval. The function  $A$  is strictly decreasing to zero, so its unique solution  $u(r)$  satisfies  $u(r) > \ell(r)$ ; (ST.18) also excludes points to the right of  $u(r)$ .

Finally

$$\sum_{p \geq 3} p^{-\sigma} = O(3^{-\sigma}) \quad (\sigma \rightarrow \infty).$$

Both equations in (ST.19) therefore have the form  $2^{-\sigma} = r(1 + O(r^{\log_2 3 - 1}))$ , which proves (ST.20).  $\square$

Combining Proposition 15.5 with Proposition 4.8 gives the following divisor-level strengthening of Theorem 4.7. On each compact subband of  $\mathcal{H}$ , the bounded part is the canonical divisor

$$\sum_{F_\omega(\rho)=0} \left[ \rho + \frac{a}{F'_\omega(\rho)} + O(|a|^2) \right], \quad (\text{ST.21})$$

where the statement is interpreted locally and the error is uniform. The points beyond the separating corridor form the disjoint family

$$\rho_{a,k} = \frac{\log(1/|a|)}{\log 2} + i \frac{\arg \omega_2 - \arg a + 2\pi k}{\log 2} + O(|a|^{\log_2 3 - 1}), \quad k \in \mathbb{Z}. \quad (\text{ST.22})$$

Thus weak convergence to an atom at infinity reflects an actual two-scale degeneration of the divisor, not merely a cancellation in an averaged counting formula.

## 15.4 Monodromy of the two families

**Theorem 15.6** (Identity versus shift monodromy). *Fix a sufficiently small  $r > 0$ . Analytically continue (a) any persistent branch furnished by Theorem 15.3 and (b) the far-right escaping family along the positively oriented loop  $a(\theta) = re^{i\theta}$ ,  $0 \leq \theta \leq 2\pi$ .*

- (i) *Every persistent branch furnished by Theorem 15.3 returns to itself.*
- (ii) *With the labeling in (ST.22), continuation of the escaping family acts by the bilateral shift*

$$\boxed{\rho_{a,k} \longmapsto \rho_{a,k-1}}. \quad (\text{ST.23})$$

*The reverse loop gives  $k \mapsto k + 1$ .*

*Hence the controlled persistent sheets extend across  $a = 0$ , whereas the escaping sheet is an infinite cyclic covering of the punctured target disc. No assertion is made here about possible branching of bounded points outside the chosen hard-core subbands.*

*Proof.* A persistent branch is holomorphic on the full disc  $|a| < a_0$ , including its centre, so the monodromy theorem gives (i). The escaping roots are simple by Proposition 4.8, and hence admit analytic continuation along the loop. Increasing the continuous argument of  $a$  by  $2\pi$  changes the leading ordinate in (ST.22) from the one labeled  $k$  to the one labeled  $k - 1$ . The error is uniformly  $o(1)$ , while consecutive leading ordinates are separated by  $2\pi/\log 2$ . For sufficiently small  $r$ , the disjoint Rouché discs defining the labels therefore force the terminal permutation to be exactly  $k \mapsto k - 1$ .  $\square$

**Remark 15.7** (Least-prime recovery in three equivalent forms). For a general infinite support  $S$  whose two least primes are  $p_0 < p_1$ , replace  $\log(p/2)$  by  $\log(p/p_0)$  in (ST.1). Lemma 15.1 becomes

$$|F'_{S,\omega}(z)| \geq \frac{p_0^{-\sigma} - \sum_{p \in S, p > p_0} p^{-\sigma} |1 - \lambda \log(p/p_0)|}{\lambda} - (\lambda^{-1} + \log p_0) |a|. \quad (\text{ST.24})$$

The escaping error exponent is  $\log p_1/\log p_0 - 1$ , the vertical lattice step is  $2\pi/\log p_0$ , its density is  $\log p_0/(2\pi)$ , and the monodromy is again a unit shift. Thus the escape mass, reciprocal lattice spacing, and monodromy translation all encode the same least support prime.

## 15.5 An even smooth expansion behind the quantized jump

The atom at infinity is nonperturbative, but the bounded part has an ordinary smooth expansion. This separates the two effects to every finite order.

**Corollary 15.8** (Quantized jump plus an even smooth jet). *Fix  $\sigma_0 > 1$  and define*

$$C_a(\sigma_0) = \int_{\sigma_0}^{\infty} G(\sigma, a) d\sigma.$$

*For every integer  $N \geq 0$ , as  $a \rightarrow 0$  through nonzero values,*

$$C_a(\sigma_0) = C_0(\sigma_0) + \frac{\log 2}{2\pi} + \sum_{j=1}^N c_j(\sigma_0) |a|^{2j} + O_{N, \sigma_0}(|a|^{2N+2}). \quad (\text{ST.25})$$

*If  $\Delta_a$  denotes the planar Laplacian in the target variable, then*

$$c_j(\sigma_0) = \frac{1}{4^j (j!)^2} \int_{\sigma_0}^{\beta} (\Delta_a^j G)(\sigma, 0) d\sigma, \quad (\text{ST.26})$$

*where any fixed  $\beta > \sigma_e$  lying in the separating corridor for all sufficiently small  $|a|$  may be used. Formula (ST.25) is a  $C^\infty$  asymptotic jet; no convergence of the infinite Taylor series is asserted.*

*Proof.* Choose  $\beta > \max\{\sigma_0, \sigma_e\}$  and then restrict  $|a|$  so that Proposition 15.5 applies. On the 2-dominance side of the remote component the Jessen function is  $J_a(\sigma) = -\sigma \log 2$ , while beyond the outer value-support edge it is  $J_a(\sigma) = \log |a|$ . The Riesz-measure identity is  $J_a''(d\sigma) = 2\pi G(\sigma, a) d\sigma$  (and  $J_a''(\sigma) = 2\pi G(\sigma, a)$  on each smooth piece). Integration across the remote component therefore gives its exact mass

$$\frac{0 - (-\log 2)}{2\pi} = \frac{\log 2}{2\pi}. \quad (\text{ST.27})$$

There is no density in the corridor, and  $G(\sigma, 0) = 0$  for  $\sigma > \sigma_e$ , so

$$C_a(\sigma_0) - \frac{\log 2}{2\pi} = \int_{\sigma_0}^{\beta} G(\sigma, a) d\sigma, \quad C_0(\sigma_0) = \int_{\sigma_0}^{\beta} G(\sigma, 0) d\sigma.$$

The function  $G$  is smooth and invariant under  $a \mapsto e^{i\theta} a$ . Its Taylor polynomial at  $a = 0$  therefore contains only powers of  $|a|^2$ , uniformly for  $\sigma \in [\sigma_0, \beta]$ . For a smooth radial function, the coefficient of  $|a|^{2j}$  is  $(4^j (j!)^2)^{-1} \Delta_a^j G(\sigma, 0)$ . Taylor's theorem followed by integration proves (ST.25)–(ST.26).  $\square$

*Remark 15.9* (What has been strengthened). Theorem 4.7 gives weak convergence of horizontal measures. Theorem 15.3 identifies the bounded divisor on compact subbands of  $\mathcal{H}$  with individual, uniformly separated zero branches; Proposition 15.5 separates those branches from the escaping lattice; Theorem 15.6 distinguishes the two parts topologically; and Corollary 15.8 resolves the remaining bounded mass to every finite perturbative order. None of these statements requires an uncertified decimal value for the hard-core threshold.

## 16 Limitations and next problems

The fixed-level correlation and phase-recovery questions are now resolved at the level stated above. The remaining bottlenecks are different and should not be hidden by those qualitative closures.

1. **Effective height errors and a moving edge.** The divisor and point-process limits use qualitative unique ergodicity of the Kronecker flow. No uniform rate is proved as a test function approaches a diagonal, a hard-core boundary, or the outer edge. In particular, the limiting edge density does not determine the first height at which a point of a prescribed defect occurs. A moving-edge theorem would require effective high-dimensional equidistribution with a relative error meaningful when the expected count is of order one.
2. **Gaussian prefactors and the remaining edge profiles.** The logarithmic zero-edge rate and its integrated small-distortion consequence are exact, but a sharp Gaussian prefactor still requires a tilted local central limit theorem. The exact phase-defect identity also holds at every fixed nonzero target, but the analogous nonzero-target density–fidelity law remains conditional on a full edge-profile theorem for  $G(\sigma, a)$ . Regular internal band edges should have related log-weighted profiles; proving them requires uniform tilted density estimates and conditional nondegeneracy of the derivative weight. Smooth fixed-parameter Kac–Rice densities do not by themselves supply this uniformity.
3. **Explicit local laws and the hard-core transition.** The Palm nearest-neighbour and hole laws are exact finite inclusion–exclusion expressions in the  $K_k$ , not closed elementary functions. The complete region in which the joint jet value  $(0, 0)$  is interior to its support, and hence the exact boundary between positive quadratic coefficient and hard core, remains to be classified analytically. Decimal transition values, double-zero examples, and equal-height examples require reproducible interval certificates before they may be used as theorem statements.
4. **Finite-height phase tomography and optimal sampling.** The defect identity and canonical edge matching give deterministic stability from a supplied near-edge point or germ. What remains unavailable is an effective deterministic bound for how high one must search to find that datum. Reciprocal limiting intensity is only a frequency scale, not a first-occurrence theorem. The stated finite-prime criterion is a sufficient certificate; the optimal intensity for reconstructing a prescribed finite set of phases is not determined. Zero data retain the unavoidable common-rotation ambiguity, and unanchored data additionally retain the vertical-flow ambiguity.
5. **Small targets beyond the hard-core subbands.** Uniform holomorphic continuation and identity monodromy of the persistent divisor are proved on compact subbands where the symbolic derivative margin is positive. Across multiple-zero caustics that labeling can branch, and a global description of the persistent divisor under arbitrary target loops remains open. The escaping lattice and its unit-shift monodromy are controlled only in the small-target regime; no uniform statement is asserted while the target and height vary together.
6. **Crossing  $\Re s = 1$ .** The endpoint transfer uses the special factorization  $e^{\mathcal{P}} = \zeta H$ . It neither counts zeros on  $\Re s = 1$  nor gives a general twisted continuation theorem below that line. The singularities of  $\zeta(ks)$  and the required branch choices lie outside the present almost-periodic framework.
7. **Uniform moving-prime geometry.** The internal bands are classified at each fixed derivative order, and the two-prime lattice theorem fixes the adjacent pair before the order tends to infinity. Uniformity for prime pairs growing with the derivative order, together with interval-certified computations of large birth levels, remains open.

The integrated structural conclusion is

$$\begin{array}{l}
 \text{vertical Haar laws and} \\
 \text{joint Kac–Rice densities} \implies \text{exact simple counts and the full local process,} \\
 \mathbf{m}(\sigma) > 0 \implies \text{uniform simplicity, hard cores, and persistent branches,} \\
 -\log G(\sigma, 0) \sim \mathcal{J}_e(\sigma) \implies \text{the PNT edge profile and the exact fidelity frontier,} \\
 \text{near-edge roots and their rooted germs} \implies \text{relative phases and edge-orbit rigidity,} \\
 a \rightarrow 0 \implies \begin{array}{l} \text{quantized escape of } \log p_0/(2\pi) \\ \text{and identity/shift divisor monodromy,} \end{array} \\
 \begin{array}{l} \text{right-edge moduli+} \\ \text{anchored near-edge ordinates} \end{array} \implies \text{the complete coefficient system,} \\
 \text{component masses and moving lattices} \implies \text{prime-pair and relative-phase recovery.}
 \end{array}$$

## Acknowledgement and status note

The manuscript distinguishes throughout among projection closure, boundary attainment, positive limiting density, deterministic recurrence, and effective first occurrence. It also distinguishes multiplicity, distinct-point, and simple-point counts, and separates self-inclusive from factorial correlations. In particular, the smoothed self-inclusive two-point function is Bohr almost periodic, while deletion of the diagonal leaves an  $AP + C_0$  function rather than, in general, another Bohr almost-periodic function.

The polygonal projection mechanism and multiplicity-weighted Jessen mean theory are classical foundations. The exact edge underlying Conjecture 1 of Belovas, Čepaitytė, and Sabaliauskas is therefore not claimed as a new general phenomenon, and the independent specialization of Camargo is explicitly acknowledged. The claims of this paper concern the specialized density, distinct/simple divisor, local-process, hard-core, edge-profile, phase-recovery, rooted-rigidity, zero-edge rate-distortion, small-target monodromy, and inverse consequences proved from those foundations. The rate-distortion theorem is a limiting-intensity statement at the zero edge: it is neither a deterministic first-hit estimate nor, in its finite-prime corollary, an optimality theorem. The nonzero-target defect identity is unconditional, but its matching edge-cost asymptotic is not asserted without a corresponding nonzero-target edge theorem. The persistent small-target divisor theorem is restricted to compact hard-core subbands. Numerical values are illustrative unless accompanied by an explicit interval certificate, and no historical “first” claim is made beyond the cited targeted search. A full MathSciNet and zbMATH priority review remains appropriate before journal submission.

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